Berry–Esseen bounds for two summands and Bernoulli arrays

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Abstract

We establish sharp and strict Berry–Esseen bounds for small independent arrays and for centered two-point (Bernoulli) summands. First, we prove that the Berry–Esseen constant C_2 for at most two arbitrary independent centered summands satisfies $C_2 < 49/120 = 0.408333\ldots < C_{\mathrm{E}}$, where $C_{\mathrm{E}} = (3+\sqrt{10})/(6\sqrt{2\pi}) \approx 0.409732$ is Esseen's constant. Second, for independent centered two-point summands, we prove the sharp inequality $\Delta \le C_{\mathrm{E}} L$ whenever either (i) there are at most three summands with arbitrary support diameters, or (ii) there are arbitrarily many summands whose nondegenerate members share a common support diameter, extending Schulz's binomial theorem to unequal probabilities. For at most two two-point summands with positive total variance, we prove the uniform strict margin $C_{\mathrm{E}} L - \Delta > 3/80$. Combining these finite-array bounds with a Stein–Ornstein–Uhlenbeck Gaussian comparison yields $\Delta \le C_{\mathrm{E}} L$ for any independent array in which two two-point summands carry at least $22/23$ of the total variance, or two arbitrary summands carry at least $439/440$ of the total variance. The proof for two arbitrary summands uses strong quadratic moment duality to reduce each summand at a fixed threshold and variance to a distribution supported on at most three points, and then rules out all possible threshold configurations by variational contact identities and certified interval enclosures.

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1 Introduction and main results

1.1 Context and problem formulation

Let $X_1, \dots, X_n$ be independent centered real random variables with positive total variance and finite third absolute moments. Define the total variance V, the sum of third absolute moments R, the Lyapunov ratio L, and the Kolmogorov normal-approximation error Δ by

$$V = B^2 = \sum_{i=1}^{n} \mathbb{E}X_i^2 > 0, \qquad R = \sum_{i=1}^{n} \mathbb{E}|X_i|^3 < \infty, \qquad L = \ell = \frac{R}{V^{3/2}},$$
$$\Delta = \sup_{x \in \mathbb{R}} \left| \mathbb{P}\left\{ \frac{\sum_{i=1}^{n} X_i}{\sqrt{V}} \le x \right\} - \Phi(x) \right|, \tag{1.1}$$

where Φ and ϕ denote the standard normal distribution function and density. (We write V and B^2, as well as L and ℓ, interchangeably when convenient.) Set

$$C_{\mathrm{E}} = \frac{3+\sqrt{10}}{6\sqrt{2\pi}} = 0.4097321837\dots, \qquad K_{\mathrm{E}} = \sqrt{2\pi} C_{\mathrm{E}} = \frac{3+\sqrt{10}}{6}, \qquad c_2 = \frac{49}{120}, \qquad t_* = \sqrt{10} - 3. \tag{1.2}$$

By Esseen [4], no universal constant smaller than C_{E} can hold in the inequality $\Delta \le CL$ across all finite independent arrays. Indeed, for $p_{\mathrm{E}} = (4-\sqrt{10})/2$ and $q_{\mathrm{E}} = 1 - p_{\mathrm{E}}$, the centered unit-variance two-point distribution

$$\mu_{\mathrm{E}} = p_{\mathrm{E}} \delta_{\sqrt{q_{\mathrm{E}}/p_{\mathrm{E}}}} + q_{\mathrm{E}} \delta_{-\sqrt{p_{\mathrm{E}}/q_{\mathrm{E}}}} \tag{1.3}$$

achieves $\lim_{n\to\infty} \Delta/L = C_{\mathrm{E}}$ for sums of n independent copies.

While the unrestricted conjecture $\Delta \le C_{\mathrm{E}} L$ remains open for general independent arrays, restricting either the number of summands n or the structure of the summand distributions allows one to establish $\Delta \le C_{\mathrm{E}} L$ (and, for small n, strictly smaller constants). For a single summand ($n = 1$), Bentkus and Kirsa [2, Theorem 1] proved that the optimal constant is $C_1 \approx 0.370352 < C_{\mathrm{E}}$, achieved by a two-point distribution. For n identically distributed two-point summands ($N \sim \mathrm{Bin}(n, p)$ with $0 < p < 1$ and $q = 1 - p$), Schulz [8, Theorem 1] proved the sharp binomial bound

$$\sup_{x \in \mathbb{R}} \left| \mathbb{P}\left\{ \frac{N - np}{\sqrt{npq}} \le x \right\} - \Phi(x) \right| < C_{\mathrm{E}} \frac{p^2 + q^2}{\sqrt{npq}}. \tag{1.4}$$

1.2 Main results

Our first theorem establishes that for at most two arbitrary independent summands, the Berry–Esseen constant C_2 is strictly below $49/120 < C_{\mathrm{E}}$.

Theorem 1.1 (Two arbitrary summands)**.** *Let C_2 be the supremum of Δ/L over all independent centered arrays with at most two summands. Then*

$$C_2 < \frac{49}{120} = 0.408333\dots < C_{\mathrm{E}}.$$

Throughout this paper, a *two-point* (or *centered Bernoulli*) *summand* means a centered random variable supported on at most two points; its support diameter is the distance between its two support points when nondegenerate. Deterministic zero summands are permitted. Our second theorem establishes the sharp Esseen bound $\Delta \le C_{\mathrm{E}} L$ for two fundamental classes of non-identically distributed two-point arrays.

Theorem 1.2 (Two-point arrays)**.** *The sharp inequality $\Delta \le C_{\mathrm{E}} L$ holds whenever either*

(i) the array consists of at most three two-point summands with arbitrary support diameters, or

(ii) every summand is supported on at most two points and all nondegenerate summands share a common support diameter.

In class (ii)*, the constant C_{E} is optimal.*

For at most two two-point summands, we establish a uniform positive gap between $C_{\mathrm{E}} L$ and Δ across all probabilities and support diameters.

Theorem 1.3 (Uniform deficit for two two-point summands)**.** *For every independent array consisting of at most two two-point summands with positive total variance $V > 0$,*

$$C_{\mathrm{E}} L - \Delta > \frac{3}{80}.$$

Combining Theorems 1.1 and 1.3 with a Stein–Ornstein–Uhlenbeck normal comparison for the remaining summands yields the sharp inequality $\Delta \le C_{\mathrm{E}} L$ for arbitrary arrays whose variance is concentrated in two summands.

Corollary 1.4 (Arrays with variance concentrated in two summands)**.** *An arbitrary finite independent array satisfies $\Delta \le C_{\mathrm{E}} L$ whenever it contains either two two-point summands with combined variance at least $\frac{22}{23} V$, or two arbitrary summands with combined variance at least $\frac{439}{440} V$.*

1.3 Open questions and overview of the proofs

Two natural questions directly extend the results of this paper:

(1) *Exact value of C_2:* For two two-point summands with arbitrary diameters, a companion paper proves that the maximal ratio is $C_1 \approx 0.370352$, attained when one summand is zero. Whether $C_2 = C_1$ holds for two *arbitrary* independent summands remains open; Theorem 1.1 suffices to show $C_2 < C_{\mathrm{E}}$.

(2) *Arbitrary diameters for $n \ge 4$ two-point summands and $n \ge 3$ general summands:* Extending Theorem 1.2(i) beyond three two-point summands with unequal diameters, and extending Theorem 1.1 to $n \ge 3$ general summands, are key steps toward the general conjecture $C_{\mathrm{ind}} = C_{\mathrm{E}}$. Meanwhile, Theorem 1.2(ii) supplies the finite-array input used in the companion small-Lyapunov manuscript to prove $\Delta \le C_{\mathrm{E}} L$ for all independent arrays with $L \le \exp(-\exp(97))$.

Let us summarize the methods used in each part of the paper. For any centered sum S of variance $V = B^2$ and $\varepsilon \in \{-1, 1\}$, define the closed and left-continuous distribution functions and the signed deficits by

$$F_S^+(y) = \mathbb{P}(S \le y), \qquad F_S^-(y) = \mathbb{P}(S < y), \qquad H_S^\varepsilon(y) = C_{\mathrm{E}} L + \varepsilon \Phi(y/B) - \varepsilon F_S^\varepsilon(y).$$

By continuity of Φ,

$$\Delta = \sup_{y \in \mathbb{R}} \max\{F_S^+(y) - \Phi(y/B),\ \Phi(y/B) - F_S^-(y)\}, \tag{1.5}$$

and reflection $S \mapsto -S$ gives $H_S^-(y) = H_{-S}^+(-y)$.

- **Two and three two-point summands with arbitrary diameters (Sections 2–3 and Appendix A):** At the two extreme support endpoints of any n-summand two-point array, Proposition 2.2 proves $H_S^{\pm} > 0$ analytically by optimizing over the support diameters via Hölder's inequality and bounding the resulting function of the Bernoulli odds $o_i = p_i/q_i$ via two exact Bernstein polynomial certificates over $\mathbb{Q}(\sqrt{5})$. For $n = 2$ and $n = 3$, the remaining interior atoms are parameterized on compact cubes and certified by outward-rounded Arb interval partitions.

- **Common-diameter two-point arrays (Section 4):** To prove Theorem 1.2(ii), we minimize the deficit $\mathcal{H}_+(p;k)$ over the probability cube $[0,1]^n$ at a fixed mean μ and integer threshold k. Adapting Mattner [6], we vary three probabilities along a curve preserving their elementary symmetric polynomials e_1, e_2; because both $F_k(p)$ and the third-moment sum $\mathcal{B}(p)$ are affine in e_3, any extremal point has at most two distinct interior probabilities $a < b$. A uniform bound on third differences of Poisson-binomial point probabilities (Lemma 4.2) rules out the case where both a and b have multiplicity at least two, reducing the problem to $\mathrm{Bin}(M,p) + \mathrm{Bern}(x)$, which is resolved by Schulz's theorem (when $p = x$), a certified second-derivative curvature bound (when $M \geq 2$ and $p \neq x$), and Theorem 1.3 (when $M = 1$).

- **Two arbitrary summands via moment duality and contact exclusions (Sections 5–6 and Appendix B):** At a fixed threshold z and fixed individual variances v_1, v_2, maximizing $\mathbb{P}(X_1 + X_2 \leq z) - C\sum_{i=1}^2 \mathbb{E}|X_i|^3$ in each marginal law is a linear program over probability measures with three moment constraints $(1, u, u^2)$. By strong quadratic moment duality (Bertsimas and Popescu [3], Winkler [10]; Theorem 5.1), if $C_2 \geq 49/120$, the supremum is attained by two distributions X_1, X_2 each supported on at most three points, and each support lies in the contact set where a quadratic majorant $q_i(u)$ touches $F^+_{X_{3-i}}(z-u) - C|u|^3$. Classifying the possible downsets of $\{(j,k) : x_j + y_k \leq z\}$ on the 3×2 and 3×3 support grids and combining the contact equations with second-order support and mass variations rules out every configuration analytically or by interval arithmetic.

- **Variance-concentration transfer (Section 7):** When two summands carry variance $v \in (0,1)$ and the remaining summands carry variance $1 - v$, convolving the two-summand Kolmogorov or L^1 bound with the Stein–Ornstein–Uhlenbeck zero-bias bound for the remainder yields Corollary 1.4.

2 Two Bernoulli summands with arbitrary diameters

For independent Bernoulli variables $B_i \sim \mathrm{Bern}(p_i)$ with $p_i \in (0,1)$ and $q_i = 1 - p_i$, write

$$X_i = h_i(B_i - p_i), \qquad a_i = p_i q_i, \qquad r_i = p_i^2 + q_i^2, \qquad V = \sum_{i=1}^n a_i h_i^2 = B^2, \qquad R = \sum_{i=1}^n a_i r_i h_i^3 = B^3 \ell.$$

We begin by recording the lower bound on $L = R/V^{3/2}$ obtained by optimizing over the support diameters h_i.

Lemma 2.1 (Diameter lower bound via Hölder's inequality)**.** *Let $0 < p_i < 1$ and $h_i \geq 0$ with $\sum_{i=1}^n a_i h_i^2 > 0$. Then*

$$\frac{\sum_{i=1}^n a_i r_i h_i^3}{(\sum_{i=1}^n a_i h_i^2)^{3/2}} \geq \left(\sum_{i=1}^n \frac{a_i}{r_i^2}\right)^{-1/2}, \tag{2.1}$$

with equality if and only if $r_i h_i$ is constant across all i.

Proof. By Hölder's inequality with exponents $3/2$ and 3,

$$\sum_{i=1}^{n} a_i h_i^2 = \sum_{i=1}^{n} (a_i r_i h_i^3)^{2/3} (a_i/r_i^2)^{1/3} \le \left(\sum_{i=1}^{n} a_i r_i h_i^3 \right)^{2/3} \left(\sum_{i=1}^{n} \frac{a_i}{r_i^2} \right)^{1/3},$$

with equality if and only if $a_i r_i h_i^3$ is proportional to a_i/r_i^2. □

Our next result shows that at the two extreme support endpoints of $S = \sum_{i=1}^{n} h_i(B_i - p_i)$, the sharp Esseen bound holds strictly for every $n \ge 1$ and all choices of $p_i \in (0,1)$ and $h_i > 0$.

Proposition 2.2 (Strict positivity at support endpoints)**.** *For every finite two-point array with $0 < p_i < 1$ and $h_i > 0$, both $H_S^+(y)$ and $H_S^-(y)$ are strictly positive at the minimum and maximum of the support of $S = \sum_{i=1}^{n} h_i(B_i - p_i)$.*

The proof of Proposition 2.2 is given in Appendix A: the two branches where the endpoint atom is excluded follow from an elementary normal-tail bound, while the two branches where the endpoint atom is included reduce via Lemma 2.1 to a sharp inequality for the Bernoulli odds $o_i = p_i/q_i$ established using exact Bernstein polynomials over $\mathbb{Q}(\sqrt{5})$.

For $n = 2$ summands, the two interior atoms of S are covered by the following two compact inequalities.

Lemma 2.3 (Interior-atom inequalities for two two-point summands)**.** *Let $0 < p_1, p_2 < 1$, $q_i = 1 - p_i$, $a_i = p_i q_i$, and $r_i = p_i^2 + q_i^2$. For $0 \le t \le 1$, put $V = a_1 t^2 + a_2$ and $R = a_1 r_1 t^3 + a_2 r_2$. Then*

$$C_{\mathrm{E}} \frac{R}{V^{3/2}} + \Phi\left(\frac{-p_1 t + q_2}{\sqrt{V}} \right) - (1 - p_1 p_2) > \frac{3}{80}. \tag{2.2}$$

Similarly, for $0 \le \theta \le \pi/2$, set $\alpha = \cos\theta$ and $\beta = \sin\theta$. Then

$$C_{\mathrm{E}} \left(\alpha^3 \frac{r_1}{\sqrt{a_1}} + \beta^3 \frac{r_2}{\sqrt{a_2}} \right) + \Phi\left(\alpha \sqrt{\frac{q_1}{p_1}} - \beta \sqrt{\frac{p_2}{q_2}} \right) - q_2 > \frac{3}{80}. \tag{2.3}$$

Method of verification. Inequalities (2.2) and (2.3) (with $\theta = \pi t/2$), together with the endpoint square $[0, 99/100]^2$ in Lemma 2.4 below, are verified on their closed unit domains by a 63,792-leaf partition evaluated in 320-bit Arb ball arithmetic (with minimum certified excess $> 2.5361 \times 10^{-7}$ over $3/80$); the monotonicity bounds used near the singular boundary faces $p_i \in \{0, 1\}$ are recorded in Section 8. □

Lemma 2.4 (Uniform endpoint deficit for at most two summands)**.** *At either support endpoint of a one- or two-summand two-point array with $V > 0$, both $H_S^+(y)$ and $H_S^-(y)$ exceed $3/80$.*

Proof. At the lower support endpoint, set $A = \sum_i a_i/r_i^2$, $s = \sum_i p_i/q_i$, and $\Pi = \prod_i q_i$. Normalizing $V = 1$, Lemma 2.1 and Cauchy–Schwarz give $R \ge A^{-1/2}$ and $m = \sum_i p_i h_i \le \sqrt{s}$, so the closed deficit is bounded below by

$$\frac{C_{\mathrm{E}}}{\sqrt{A}} + \Phi(-\sqrt{s}) - \Pi.$$

On $[0, 99/100]^2$, the interval partition in the supplement certifies that this lower bound exceeds $3/80$. If some $p_i \ge 99/100$, then $\Pi \le 1/100$ while $R \ge 1/\sqrt{2} > 2/3$, giving a deficit exceeding $49/180 - 1/100 > 3/80$. For the strict branch at the lower endpoint, where $F_S^- = 0$, the proof of

Lemma A.1 gives $R \geq c/x$ with $c = 2(\sqrt{2}-1) > 49/60$ and $x = m$. Since $x\Phi(-x) < 1/4$ for all $x > 0$, the strict deficit exceeds

$$\left(C_{\mathrm{E}} - \frac{1}{4c}\right) R > \left(\frac{49}{120} - \frac{15}{49}\right)\frac{2}{3} = \frac{601}{8820} > \frac{3}{80}.$$

Reflection $S \mapsto -S$ establishes the same bounds at the upper endpoint. □

Proof of Theorem 1.3. Omit any deterministic summand. If only one summand remains, both its atoms are support endpoints, which are covered by Lemma 2.4. If two nondegenerate summands remain, relabel and scale so that $h_2 = 1$ and $0 < h_1 = t \leq 1$. For $t < 1$, the four atoms $x_A = \sum_{i=1}^{2}(A_i - p_i)h_i$ $(A \in \{0,1\}^2)$ are ordered as $00 < 10 < 01 < 11$, with distribution values

A	$F_S^-(x_A)$	$F_S^+(x_A)$
00	0	q_1q_2
10	q_1q_2	q_2
01	q_2	$1-p_1p_2$
11	$1-p_1p_2$	1.

Lemma 2.4 covers 00 and 11. For the closed branch at 01, the deficit $H_S^+(x_{01})$ is exactly the left-hand side of (2.2). For the closed branch at 10, setting $\alpha = h_1\sqrt{a_1/V}$ and $\beta = h_2\sqrt{a_2/V}$ (so that $\alpha^2 + \beta^2 = 1$) transforms $H_S^+(x_{10})$ into the left-hand side of (2.3). When $t = 1$, the interior atoms 10 and 01 coalesce, with $F_S^- = q_1q_2$ and $F_S^+ = 1 - p_1p_2$, whose closed deficit is again (2.2). Finally, replacing (B_1, B_2) by $(1-B_1, 1-B_2)$ negates S while preserving V, R, and $0 < h_1 \leq h_2$, so $F_S^-(y) = 1 - F_{-S}^+(-y)$ reduces all strict deficits $H_S^-(x_A)$ to the closed deficits already proved. □

As an immediate consequence of Theorem 1.2(ii), the sharp Esseen bound also holds for arbitrary mixtures of positively and negatively oriented Esseen two-point variables of equal variance.

Corollary 2.5 (Mixtures of the two Esseen orientations)**.** *Fix $c > 0$ and $0 \leq \eta_i \leq 1$ for $1 \leq i \leq n$. If $X_1, \ldots, X_n$ are independent with marginal distributions*

$$\mathcal{L}(X_i) = (1-\eta_i)\mathcal{L}(cZ) + \eta_i\mathcal{L}(-cZ), \qquad Z \sim \mu_{\mathrm{E}},$$

then $\Delta \leq C_{\mathrm{E}}L$.

Proof. Realize each X_i by first drawing an independent sign choice and then drawing from $\mathcal{L}(\pm cZ)$. Conditional on the vector of sign choices, the summands are independent two-point variables with identical variances c^2, identical third absolute moments $c^3(p_{\mathrm{E}}^2 + q_{\mathrm{E}}^2)/\sqrt{p_{\mathrm{E}}q_{\mathrm{E}}}$, and common support diameter $c/\sqrt{p_{\mathrm{E}}q_{\mathrm{E}}}$. Theorem 1.2(ii) gives $|\mathbb{P}(S \leq x \mid \text{signs}) - \Phi(x/\sqrt{V})| \leq C_{\mathrm{E}}L$ for every sign configuration, and averaging over the signs preserves the inequality. □

3 Three Bernoulli summands with arbitrary diameters

We now prove Theorem 1.2(i). Write $X_i = h_i(B_i - p_i)$ with $q_i = 1 - p_i$, $a_i = p_iq_i$, and $r_i = p_i^2 + q_i^2$ for $i = 1, 2, 3$; arrays with at most two active summands are already covered by Theorem 1.3.

Normalize total variance to $B^2 = 1$. By Cantelli's inequality (see Bell [1]), every centered unit-variance law satisfies

$$\Delta < \kappa := \frac{271}{500} = 0.542, \tag{3.1}$$

so any counterexample to $\Delta \leq C_{\mathrm{E}}R$ must satisfy $R < \kappa/C_{\mathrm{E}} < 4/3$.

Furthermore, no summand in a counterexample can have variance $w \le 1/25$. Indeed, suppose a summand X has variance $w \le 1/25$, leaving a two-summand two-point sum T of variance $v = 1-w \ge 24/25$. By Theorem 1.3, $T/\sqrt{v}$ has deficit exceeding $3/80 > 1/30$. Replacing T by a normal variable $N(0,v)$ and comparing $N(0,v)+X$ with $N(0,1)$ by a second-order Taylor expansion of $\Phi(\cdot/\sqrt{v})$ incurs an error bounded by

$$L(v)R_X, \qquad L(v) = \frac{\phi(0)(1+4\phi(0))}{6v^{3/2}} < \frac{1}{5} \qquad (v \ge 24/25),$$

where $R_X = \mathbb{E}|X|^3 \ge w^{3/2}$ and $R_T = R - R_X$. The full deficit therefore exceeds

$$\frac{1}{30} - C_{\mathrm{E}} R_T (v^{-3/2}-1) + (C_{\mathrm{E}} - L(v))R_X > b(w) := \frac{1}{30} - \kappa((1-w)^{-3/2}-1) + \frac{5}{24}w^{3/2}.$$

On $[0,1/25]$, $b'(w) \le -3\kappa/2 + 1/16 < 0$ and $b(1/25) > 1/2000 > 0$. Thus any counterexample must have $w_i = a_i h_i^2 > 1/25$ for each $i = 1,2,3$. Since $R \ge w_i^{3/2}/(2\sqrt{a_i})$, $R \ge w_i h_i/2$, and $h_i \ge 2\sqrt{w_i}$, the bound $R < 4/3$ implies the a priori bounds

$$a_i > 9 \times 10^{-6}, \qquad \frac{2}{5} < h_i < \frac{200}{3}, \qquad p_i \in \left[\frac{1}{125000}, 1 - \frac{1}{125000}\right]. \tag{3.2}$$

Relabel and rescale the raw diameters so that $0 < h_1 \le h_2 \le h_3 = 1$. Depending on whether $h_1 + h_2 \le 1$ or $h_1 + h_2 \ge 1$, the eight atoms $A \in \{0,1\}^3$ are ordered either as

$$000 \le 100 \le 010 \le 001 \le 110 \le 101 \le 011 \le 111$$
$$\text{or} \quad 000 \le 100 \le 010 \le 110 \le 001 \le 101 \le 011 \le 111.$$

Proposition 2.2 covers the extreme endpoints 000 and 111. With $V = \sum_{i=1}^3 a_i h_i^2$, $R_h = \sum_{i=1}^3 a_i r_i h_i^3$, and $z_A = (A-p)\cdot h/\sqrt{V}$, the six interior atoms in both orderings are covered by the eight parameterizations $O = (st, s, 1)$, $T = ((1-s)/2 + st, (1+s)/2, 1)$, and $U = (s/2, s/2 + (1-s)t, 1)$ for $(s,t) \in [0,1]^2$:

Case	$h(s,t)$	A	$F(p)$	Accepted leaves
First	O	100	$q_2 q_3$	244,953
Second	O	010	$q_3(1-p_1p_2)$	373,301
Upper pair	O	101	$1-p_2p_3$	102,405
Last	O	011	$1-p_1p_2p_3$	175,197
Triangle lower	T	001	$1-p_1p_2-p_1p_3-p_2p_3+2p_1p_2p_3$	263,494
Triangle upper	T	110	$q_3+p_3q_1q_2$	62,488
Separated lower	U	110	q_3	20,070
Separated upper	U	001	$q_3+p_3q_1q_2$	70,253

On each of these eight five-dimensional domains, the inequality

$$C_{\mathrm{E}} \frac{R_h}{V^{3/2}} + \Phi(z_A) - F(p) \ge 0 \tag{3.3}$$

is certified by the subdivision and boundary-projection recipes in `certificates/finite/current-finite.zip` (1,312,161 accepted leaves across 2,624,314 nodes, reevaluated at 320-bit Arb precision; see Section 8). At any atom collision, the last configuration in the tied block carries the full closed CDF value $F_S^+(y)$, while every other row has $F(p) \le F_S^+(y)$ so that (3.2) remains valid whenever the row expression is negative. Reflection supplies the strict CDF branches, proving Theorem 1.2(i).

4 Common-diameter Bernoulli arrays

Theorem 4.1 (Common support diameter)**.** *Part* (ii) *of Theorem 1.2 holds: every finite independent array of two-point summands with a common support diameter satisfies* $\Delta \le C_{\mathrm{E}} L$*, and the constant* C_{E} *is optimal.*

After omitting zero summands and dividing by the common support diameter, we have $X_i = B_i - p_i$ where $B_1, \dots, B_n$ are independent with $B_i \sim \mathrm{Bern}(p_i)$, $p_i \in (0,1)$, and $q_i = 1 - p_i$. Put

$$N_p := \sum_{i=1}^{n} B_i, \qquad \mu := \sum_{i=1}^{n} p_i, \qquad V(p) := \sum_{i=1}^{n} p_i q_i, \qquad \mathcal{B}(p) := \sum_{i=1}^{n} p_i q_i (p_i^2 + q_i^2), \tag{4.1}$$

and write $F_j(p) = \mathbb{P}(N_p \le j)$. For an integer k and $V(p) > 0$, the closed and strict deficits are

$$\mathcal{H}_+(p;k) := C_{\mathrm{E}} \frac{\mathcal{B}(p)}{V(p)^{3/2}} + \Phi\left(\frac{k-\mu}{\sqrt{V(p)}}\right) - F_k(p), \tag{4.2}$$

$$\mathcal{H}_-(p;k) := C_{\mathrm{E}} \frac{\mathcal{B}(p)}{V(p)^{3/2}} - \Phi\left(\frac{k-\mu}{\sqrt{V(p)}}\right) + F_{k-1}(p). \tag{4.3}$$

By (1.5), $\Delta \le C_{\mathrm{E}} L$ is equivalent to $\mathcal{H}_\pm(p;k) \ge 0$ for all $k \in \mathbb{Z}$, and complementation $B_i \mapsto 1 - B_i$ gives the exact reflection identity

$$\mathcal{H}_-(p_1, \dots, p_n; k) = \mathcal{H}_+(q_1, \dots, q_n; n-k). \tag{4.4}$$

4.1 Third differences of Poisson-binomial point probabilities

Lemma 4.2 (Poisson-binomial third-difference bound)**.** *Let* $c_j = \mathbb{P}(C = j)$ $(j \in \mathbb{Z})$ *be the point mass function of a finite Poisson-binomial sum* $C = \sum_i B_i$ *of variance* $W = \operatorname{Var} C$*, and let* $\Delta^3 c_j := c_j - 3c_{j-1} + 3c_{j-2} - c_{j-3}$*. Then*

$$\sup_{j \in \mathbb{Z}} |\Delta^3 c_j| < \frac{8 C_{\mathrm{E}}}{(W+1)^{3/2}}. \tag{4.5}$$

Proof. Set $A = 8C_{\mathrm{E}}$; the rational bounds $\sqrt{10} > 79/25$ and $\pi < 22/7$ give $A > 16/5$.

First suppose $0 \le W \le 1/4$. Taking the more probable outcome of each B_i as a reference configuration, let $\varepsilon_i = \min(p_i, q_i)$, $E = \sum_i \varepsilon_i \le 2W \le 1/2$, $R = \sum_i \varepsilon_i/(1-\varepsilon_i) \in [W, 1]$, and let J count the number of coordinates that depart from the reference configuration. Then $P_0 = \mathbb{P}(J = 0) \ge \max(1 - E, e^{-R}) \ge 1/2$ and $P_1 = \mathbb{P}(J = 1) = P_0 R \ge W e^{-W}$. If m is the sum of the reference configuration, then at $j \in \{m+1, m+2\}$ the event $J = 0$ contributes ± 3 to $\Delta^3 c_j$ while every $J = 1$ outcome contributes with the opposite sign and magnitude at least 1, so that

$$|\Delta^3 c_j| \le 3P_0 - P_1 + 3\mathbb{P}(J \ge 2) = 3 - 4P_1 \le 3 - 4We^{-W} \tag{4.6}$$

for $j \in \{m+1, m+2\}$, whereas at all other j the $J = 0$ contribution has magnitude at most 1 and

$$|\Delta^3 c_j| \le P_0 + 3(1 - P_0) \le 1 + 4W \le 2 < A(5/4)^{-3/2}. \tag{4.7}$$

In (4.6), using $e^{-W} \ge 1 - W$ and $(1+W)^{-3/2} \ge h(W) := 1 - \frac{3}{2}W + \frac{15}{8}W^2 - \frac{35}{16}W^3$, the difference $P(W) = Ah(W) - (3 - 4W + 4W^2)$ satisfies $P'(W) < 3h'(W) + 4 - 8W = -\frac{1}{2} + \frac{13}{4}W - \frac{315}{16}W^2 < 0$ on $[0, 1/4]$ and $P(1/4) = \frac{725}{1024}A - \frac{9}{4} > \frac{1}{64} > 0$.

Now suppose $W \geq 1/4$. Fourier inversion and $|1-p+pe^{it}| \leq \exp\{-2p(1-p)\sin^2(t/2)\}$ give

$$|\Delta^3 c_j| \leq I(W) := \frac{16}{\pi}\int_0^{\pi/2} \sin^3 x\, e^{-2W\sin^2 x}\, dx. \tag{4.8}$$

Writing $J_r(W) = \int_0^{\pi/2} \sin^r x\, e^{-2W\sin^2 x}\, dx$, integration by parts of $(\sin^4 x \cos x\, e^{-2W\sin^2 x})'$ gives $(5+4W)J_5 = 4J_3 + 4WJ_7$, so $J_5/J_3 \geq 4/(5+4W)$ and

$$\frac{d}{dW}\log\{I(W)(1+W)^{3/2}\} = -2\frac{J_5}{J_3} + \frac{3}{2(1+W)} \leq -\frac{8}{5+4W} + \frac{3}{2(1+W)} < 0.$$

At $W = 1/4$, $e^{-u} \leq 1-u+u^2/2$ gives $I(1/4) \leq \frac{16}{\pi}(J_3(0) - \frac{1}{2}J_5(0) + \frac{1}{8}J_7(0)) = \frac{256}{35\pi} < A(5/4)^{-3/2}$. □

4.2 Reduction to a binomial sum plus one Bernoulli summand

Proposition 4.3 (Reduction along elementary-symmetric probability fibers)**.** *Fix $n \geq 1$, $\mu \in (0,n)$, $k \in \mathbb{Z}$, and either deficit in (4.2)–(4.3). If that deficit is negative somewhere on $\mathcal{P}_{n,\mu} := \{p \in [0,1]^n : \sum_{i=1}^n p_i = \mu\}$, then after deleting deterministic coordinates in $\{0,1\}$ and possibly applying (4.4), a negative global minimum on a face of minimal dimension has the form*

$$N_p \sim \mathrm{Bin}(M,p) + \mathrm{Bern}(x), \qquad M \geq 1, \quad 0 < p, x < 1, \quad p \neq x. \tag{4.9}$$

Proof. Since $\mathcal{B}(p) \geq V(p)/2$, the deficit tends to $+\infty$ as $V(p) \to 0$, so it attains its minimum on every closed face of $\mathcal{P}_{n,\mu}$. Choose a face of smallest dimension on which the minimum is negative, and among the minimizers on that face choose one maximizing $\sum_i p_i^3$. Deleting coordinates equal to 0 or 1 (replacing (μ,k) by $(\mu-1,k-1)$ when $p_i = 1$) preserves the deficit, so minimal dimension implies $p_i \in (0,1)$ for all remaining coordinates.

Following Mattner [6, pp. 5–6], for any three coordinates p_1, p_2, p_3 let e_1, e_2, e_3 be their elementary symmetric polynomials:

$$\prod_{i=1}^3 \{1 + p_i(w-1)\} = 1 + e_1(w-1) + e_2(w-1)^2 + e_3(w-1)^3. \tag{4.10}$$

Holding e_1, e_2 fixed preserves both their mean e_1 and their variance $V_3 = e_1 - e_1^2 + 2e_2$, while making both $F_k(p)$ and their third-moment sum

$$\mathcal{B}_3 = e_1 - 3e_1^2 + 4e_1^3 - 2e_1^4 + (6 - 12e_1 + 8e_1^2)e_2 - 4e_2^2 + 4(3-2e_1)e_3 \tag{4.11}$$

affine in e_3. If $p_1, p_2, p_3 \in (0,1)$ were distinct, e_3 could be varied in both directions, contradicting either minimality of the deficit (if the slope is nonzero) or maximality of $p_1^3+p_2^3+p_3^3 = e_1^3 - 3e_1e_2 + 3e_3$ (if the slope is zero). Thus at most two distinct values $0 < a < b < 1$ occur, with multiplicities $r, s \geq 1$ (since $r = 0$ or $s = 0$ is ruled out by Schulz's theorem (1.4)).

Suppose $r, s \geq 2$. After applying (4.4) if necessary so the deficit is closed, let d_j be the point mass function of the $n-3$ coordinates outside a chosen triple. The slope of $\mathcal{H}_+$ with respect to e_3 is

$$L(e_1; d) = 4\gamma(3 - 2e_1) + \Delta^2 d_k, \qquad \gamma := \frac{C_{\mathrm{E}}}{V^{3/2}}, \tag{4.12}$$

where $\Delta^2 d_k = d_k - 2d_{k-1} + d_{k-2}$. Because (a,a,b) maximizes e_3 and (a,b,b) minimizes e_3 on their respective (e_1,e_2) fibers, minimality of $\mathcal{H}_+$ requires

$$L_{aab} \leq 0 \leq L_{abb}. \tag{4.13}$$

Letting c_j denote the point mass function of the $n-4$ coordinates obtained by deleting two copies of a and two copies of b, we have $d^{(aab)} = c * \mathrm{Bern}(b)$ and $d^{(abb)} = c * \mathrm{Bern}(a)$, so subtraction gives

$$0 \le L_{abb} - L_{aab} = (b-a)\{-8\gamma + \Delta^3 c_k\}, \tag{4.14}$$

which contradicts Lemma 4.2 since $V \le \mathrm{Var}(C) + 1$. Hence $\min(r, s) = 1$. □

4.3 Excluding the one-exception family

Lemma 4.4 (Curvature inequality on the bounded variance domain)**.** *Let $0 < p, x < 1$, $q = 1-p$, $pq \le W \le 3$, $0 \le f \le 1$, and $A_0 := W - (1-f)q^2 \ge 0$. Set*

$$B_0 := p^2 + (1-f)(W + 2pq), \qquad V := x(1-x) + W + pq,$$

$$t_x := 1 - 2x, \qquad t_p := 1 - 2p, \qquad y := \frac{f(W+pq)}{1-qf}, \qquad z := \frac{y-x}{\sqrt{V}}.$$

Then

$$\frac{2C_{\mathrm{E}}}{\sqrt{V}}\{4W + 2 + t_x t_p\} + z\phi(z)(2+z^2) - 5V\frac{fB_0}{(2-f)B_0 + A_0} > 0. \tag{4.15}$$

Proof. For $V \le 1/5$, set $S_2 = t_x^2 + t_x t_p + t_p^2 \ge (t_x^2 + t_p^2)/2 \ge 1 - 2V \ge 1/3$. Since $4V + S_2 = 4W + 2 + t_x t_p$, the arithmetic–geometric mean inequality gives $C_{\mathrm{E}}(8\sqrt{V} + 2S_2/\sqrt{V}) \ge 8C_{\mathrm{E}}/\sqrt{3}$. Because $z\phi(z)(2+z^2) \ge -4/(e\sqrt{\pi})$ and $fB_0/[(2-f)B_0 + A_0] \le 1$, we have for $V \le 1/5$

$$\frac{8C_{\mathrm{E}}}{\sqrt{3}} - \frac{4}{e\sqrt{\pi}} - 5V \ge \frac{8C_{\mathrm{E}}}{\sqrt{3}} - \frac{4}{e\sqrt{\pi}} - 1 > \frac{24}{13} - \frac{16}{19} - 1 = \frac{1}{247} > 0. \tag{4.16}$$

For $V \ge 1/5$, using the lower bound

$$z\phi(z)(2+z^2) \ge \sqrt{\frac{2}{\pi}}\min\{z, 0\}, \tag{4.17}$$

the domain is covered by three explicit maps from $[0,1]^4$ (with fourth coordinate x):

(i) $p = (1+s_p)/2$, $W = pq + s_W(3-pq)$, $f = s_f$ (covering $p \ge 1/2$);

(ii) $p = s_p/2$, $W = q^2 + s_W(3-q^2)$, $f = s_f$ (covering $p \le 1/2$, $W \ge q^2$);

(iii) $p = s_p/2$, $W = pq + s_W(q^2 - pq)$, $f = 1 - (1-s_f)W/q^2$ (covering $p \le 1/2$, $W \le q^2$, $A_0 \ge 0$).

The verifier `verify/structure/check.py` (`--arithmetic --case common-gap`) certifies positivity of (4.15) at the smaller coefficient $199/500 < C_{\mathrm{E}}$ across 275,283 accepted boxes (from 550,563 processed boxes) at 192-bit Arb precision. □

Proposition 4.5 (Nonnegativity for $\mathrm{Bin}(M,p) + \mathrm{Bern}(x)$)**.** *For every $M \ge 1$, $0 < p, x < 1$ with $p \ne x$, and $k \in \mathbb{Z}$, both deficits of $\mathrm{Bin}(M,p) + \mathrm{Bern}(x)$ are nonnegative.*

Proof. By (4.4) and Proposition 4.3, if a negative deficit occurred, $\mathcal{H}_+$ would have a negative global minimum at some $\mathrm{Bin}(M,p) + \mathrm{Bern}(x)$ with $p \ne x$. When $M = 1$, this is excluded by Theorem 1.3, so assume $M \ge 2$, put $m = M - 1$, and let $D \sim \mathrm{Bin}(m,p)$, $d_j = \mathbb{P}(D = j)$, and $W = mpq$.

Varying the exceptional coordinate x and one copy of p as $\bar{p} + d, \bar{p} - d$ with $u = d^2$ makes the closed CDF affine in u:

$$\mathcal{F}_k(u) = \mathcal{F}_k(0) + (d_{k-1} - d_k)u. \tag{4.18}$$

With $t_x = 1-2x$, $t_p = 1-2p$, $\xi = d_{k-1} - d_k$, $S_2 = t_x^2 + t_x t_p + t_p^2$, $V = x(1-x) + Mpq$, and $z = (k - x - Mp)/\sqrt{V}$, we have $V' = -2$, $\mathcal{B}' = -2S_2$, and $\mathcal{B}'' = -8$ with respect to u. Eliminating $\mathcal{B}$ at the stationary point $\mathcal{H}_+'(u) = 0$ yields the exact second derivative

$$V^2 \mathcal{H}_+'' = 5V\xi - C_{\mathrm{E}} \left(8\sqrt{V} + \frac{2S_2}{\sqrt{V}} \right) - z\phi(z)(2 + z^2). \tag{4.19}$$

By the binomial recurrence,

$$\xi = \begin{cases} 0, & k < 0 \text{ or } k \geq m+2, \\ -d_0, & k = 0, \\ d_{k-1} \dfrac{k - (m+1)p}{kq}, & 1 \leq k \leq m+1. \end{cases} \tag{4.20}$$

If $\xi \leq 0$, then $V^2 \mathcal{H}_+'' \leq -(8C_{\mathrm{E}}/\sqrt{3} - 4/(e\sqrt{\pi})) < 0$. If $\xi > 0$, put $y = k - (m+1)p > 0$, $f = y/(kq) \in (0, 1]$, and $j = k - 1$, so that $d_{j+1} = (1-f)d_j$, $d_{j-1} = r_0 d_j$, and $d_{j-1} + d_j + d_{j+1} \leq 1$ give

$$\xi = f d_j \leq \frac{f}{2 - f + r_0}, \tag{4.21}$$

with

$$r_0 = \frac{A_0}{B_0}, \qquad A_0 = W - (1-f)q^2 \geq 0, \qquad B_0 = p^2 + (1-f)(W + 2pq), \tag{4.22}$$

and

$$y = \frac{f(W + pq)}{1 - qf}, \qquad C_{\mathrm{E}} \left(8\sqrt{V} + \frac{2S_2}{\sqrt{V}} \right) = \frac{2C_{\mathrm{E}}}{\sqrt{V}} \{ 4W + 2 + t_x t_p \}. \tag{4.23}$$

For $W \leq 3$, Lemma 4.4 implies $V^2 \mathcal{H}_+'' < 0$. For $W \geq 3$, Fourier inversion gives

$$|\xi| = |d_{k-1} - d_k| \leq \frac{4}{\pi} \int_0^{\pi/2} \sin u \, e^{-2W \sin^2 u} \, du < \frac{2}{\pi W}, \tag{4.24}$$

so $-V^2 \mathcal{H}_+'' \geq 8C_{\mathrm{E}}\sqrt{V} - 4/(e\sqrt{\pi}) - 10V/(\pi W) \geq 44/45 > 0$ on $V \in [W, W + 1/2]$. In every case $\mathcal{H}_+'' < 0$, contradicting local minimality. □

Proof of Theorem 4.1. Combine Propositions 4.3 and 4.5; optimality of C_{E} follows from Esseen's i.i.d. family (1.3). □

5 Moment duality and the three-point reduction

When all summands except X_i are held fixed, the distribution function of S at a fixed threshold y is an expectation $\mathbb{E}h(X_i)$. Maximizing the Berry–Esseen deficit at fixed summand variance $s = \mathbb{E}X_i^2$ is therefore a linear optimization problem over probability measures with prescribed first and second moments, governed by the classical extreme-point and duality theorems of Winkler [10, Theorem 2.1] and Bertsimas and Popescu [3, Theorem 2.2].

Theorem 5.1 (Strong quadratic moment duality)**.** *Let $s > 0$ and let $h : \mathbb{R} \to \mathbb{R}$ be upper semicontinuous and satisfy*

$$-K_-(1 + |u|^3) \le h(u) \le K_+ - \lambda|u|^3$$

for some constants $K_-, K_+ < \infty$ and $\lambda > 0$. Then

$$\max_{P:\ \int dP=1,\ \int u\,dP=0,\ \int u^2\,dP=s,\ \int |u|^3 dP<\infty} \int h\,dP = \min_{\substack{q(u)=A+Bu+Cu^2 \\ q\ge h\ \text{on}\ \mathbb{R}}} (A + Cs). \tag{5.1}$$

Both extrema are attained. Every extreme point of the set of primal maximizers is supported on at most three points, and every primal maximizer is supported within the contact set $\{u \in \mathbb{R} : q(u) = h(u)\}$ of every optimal quadratic majorant q.

Proof. The upper bound $h(u) \le K_+ - \lambda|u|^3$ forces any maximizing sequence to have uniformly bounded third absolute moments, which ensures tightness and uniform integrability of u and u^2. By upper semicontinuity of h and Fatou's lemma, the set of primal maximizers is nonempty, convex, and weakly compact, and forms an extremal face of the feasible moment set. If an extreme feasible measure P charged four disjoint Borel sets $A_1, \dots, A_4$, their moment vectors $(P(A_j), \int_{A_j} u\,dP, \int_{A_j} u^2\,dP) \in \mathbb{R}^3$ would be linearly dependent, yielding a nonzero bounded density perturbation $(1 \pm \varepsilon \sum_{j=1}^4 c_j \mathbf{1}_{A_j})P$ preserving the three moments. Thus every extreme maximizer has at most three atoms.

For strong duality, the moment-hypograph cone

$$\mathcal{D} = \left\{ \left(\mu(\mathbb{R}), \int u\,d\mu, \int u^2\,d\mu, z \right) : \mu \ge 0,\ \int |u|^3\,d\mu < \infty,\ z \le \int h\,d\mu \right\} \subset \mathbb{R}^4$$

is convex and closed (by the same cubic-coercivity compactness argument). At its boundary point $p = (1, 0, s, M)$, where M is the primal maximum, a nonzero supporting hyperplane $am_0 + bm_1 + cm_2 + dz \le 0$ has $d > 0$ because $(1, 0, s)$ lies in the interior of the moment cone $\{(m_0, m_1, m_2) : m_0 > 0, m_2 \ge m_1^2/m_0\}$. Normalizing $d = 1$ and testing against point masses δ_u shows that $q(u) = -a - bu - cu^2 \ge h(u)$ on $\mathbb{R}$ with $q(0) + q''(0)s/2 = -a - cs = M$, establishing (5.1) and complementary slackness $\int (q - h)\,dP = 0$. □

Theorem 5.2 (Three-point reduction)**.** *Any independent centered array satisfying $\Delta > cL$ for some $c > 0$ can be replaced by an independent centered array with the same number of summands and the same individual variances, each supported on at most three points, satisfying $\Delta > cL$.*

Proof. After reflection if needed, fix a threshold y with $F_S^+(y) - \Phi(y/B) > cL$. Holding all summands except X_i fixed with $T_i = S - X_i$, apply Theorem 5.1 to the upper semicontinuous function $h_i(u) = F_{T_i}^+(y - u) - cB^{-3}|u|^3$ at variance $s_i = \mathbb{E}X_i^2$ to replace X_i by a three-point extreme maximizer without decreasing $F_S^+(y) - \Phi(y/B) - cB^{-3} \sum_j \mathbb{E}|X_j|^3$. □

6 Proof of Theorem 1.1: two arbitrary summands

Normalize total variance to $B^2 = 1$. By reflection, C_2 is the supremum of D/R, where $D = \mathbb{P}(X_1 + X_2 \le z) - \Phi(z) > 0$ and $R = \mathbb{E}|X_1|^3 + \mathbb{E}|X_2|^3$.

Lemma 6.1 (Attainment by three-point marginals)**.** *If $C_2 \ge 49/120$, then C_2 is attained at some threshold $z \in \mathbb{R}$ by two nondegenerate centered distributions X_1, X_2, each supported on at most three points.*

Proof. Along a maximizing sequence with $D_j/R_j \to C_2 \geq 49/120$, we have $1/\sqrt{2} \leq v_{1,j}^{3/2} + v_{2,j}^{3/2} \leq R_j \leq D_j/c \leq 1/c$ (for any fixed $c \in (463/1250, 49/120)$), while Chebyshev's inequality and $1-\Phi(z_j)$ keep the thresholds z_j bounded. Passing to a weak limit preserves total variance one (by uniform integrability of squares) and yields $D \geq C_2 r \geq C_2 R$, so equality $D/R = C_2$ holds. Because $C_1 < 463/1250 < 49/120$ by Bentkus and Kirsa [2, Theorem 1], both limiting marginals have positive variance. Replacing each marginal by a three-point extreme maximizer from Theorem 5.1 preserves $D - C_2 R = 0$ and shifts z to a sum atom $x_j + y_k = z$. □

At an attained maximum with $C = C_2 \in [49/120, 227/500)$ (where the upper bound follows from Bell [1]), set $s_i = \mathbb{E}X_i^2$, $\beta_i = \mathbb{E}|X_i|^3$, $M_i = \mathbb{E}[X_i|X_i|]$, and

$$A = \frac{z\phi(z) + 3D}{2}, \qquad K = \frac{2A}{3C} = R + \frac{z\phi(z)}{3C}.$$

Lemma 6.2 (Variational contact majorants)**.** *For $i = 1, 2$, the quadratic polynomial*

$$q_i(u) = F + As_i - C\beta_i - (\phi(z) + 3CM_i)u - Au^2$$

majorizes $F^+_{X_{3-i}}(z-u) - C|u|^3$ on $\mathbb{R}$, with equality at every support point of X_i. Moreover, $f_i(u) = q_i(u) + C|u|^3$ satisfies $f_i'(u) \leq 0$ at every support point $u \in \operatorname{supp} X_i$, with $f_i'(u) = 0$ whenever $z - u \notin \operatorname{supp} X_{3-i}$.

Proof. Perturbing $\mathcal{L}(X_i)$ to $(1-t)\mathcal{L}(X_i) + t\delta_u$, recentering by tu, and shifting the threshold to $z - tu$ gives right derivatives $F^+_{X_{3-i}}(z-u) - F$, $u^2 - s_i$, and $|u|^3 - \beta_i - 3uM_i$ for the event probability, variance, and third-moment sum at $t = 0$. Nonpositivity of the right derivative of $D(t)V(t)^{3/2}/R(t)$ yields $F^+_{X_{3-i}}(z-u) - C|u|^3 \leq q_i(u)$, and integrating against $\mathcal{L}(X_i)$ gives equality on $\operatorname{supp} X_i$. Because $u \mapsto F^+_{X_{3-i}}(z-u)$ is nonincreasing and is constant near u when $z - u \notin \operatorname{supp} X_{3-i}$, the smooth majorant f_i satisfies $f_i'(u) \leq 0$ at every contact point and $f_i'(u) = 0$ away from threshold representations. □

From Cantelli's inequality ($D < 11/20$), $|z\phi(z)| < 1/\sqrt{2\pi e} < 49/200$, and $R \geq 1/\sqrt{2}$, we obtain the a priori bounds

$$\frac{1}{2} < K < \frac{8}{5}, \tag{6.1}$$

and comparing a dominant summand of variance $v \geq 187/200$ against a Gaussian via the Bentkus–Kirša bound $C_1 < 463/1250$ and $L(v) < 1/5$ yields

$$\frac{13}{200} < s_i < \frac{187}{200} \qquad (i = 1, 2). \tag{6.2}$$

Furthermore, any centered three-point law on $\{-a, c, b\}$ ($a, b > 0$, $-a < c < b$) is a mixture of two centered two-point laws sharing an extreme atom, whose conditional variances differ by $d = \min\{b(a+c), a(b-c)\}$ (we set $d = 0$ for a two-point law). When $d_1 + d_2 \leq B^2/3$, conditioning on the mixture components and applying Theorem 1.3 bounds the ratio strictly below 49/120:

Lemma 6.3 (Exclusion of small mixture-variance widths)**.** *If two centered distributions supported on at most three points have mixture-variance widths $d_1 + d_2 \leq B^2/3$, then*

$$C_{\mathrm{E}} L - \Delta > \frac{4877}{1056500}, \qquad \frac{\Delta}{L} < \frac{572623}{577500} C_{\mathrm{E}} < \frac{49}{120}.$$

Proof. Normalize $B^2 = 1$ and draw the two mixture labels independently. Each conditional array consists of two two-point summands with total variance $W \in [2/3, 4/3]$ of mean 1, and its conditional third absolute moment is affine in its variance with nonnegative slope. Because $u \mapsto u^{-3/2}$ is decreasing, $\mathbb{E}[(U_i - \mathbb{E}U_i)W^{-3/2}] \le 0$, so the average of C_{E} times the conditional Lyapunov ratio exceeds $C_{\mathrm{E}}R$ by at most $C_{\mathrm{E}}R(\mathbb{E}W^{-3/2} - 1)$, while the normal CDF convexity error is at most $(7/50)\mathbb{E}[-\log W]$ (since for $N(v) = \varepsilon\Phi(y/\sqrt{v})$, $N''(v) = \varepsilon\phi'''(y/\sqrt{v})/(4v^2) \le |\phi'''(y/\sqrt{v})|/(4v^2) < 7/(50v^2)$). Over all centered random variables W of mean 1 and support width at most 1/3, a 256-cell rational midpoint-derivative check verifies $\mathbb{E}W^{-3/2} - 1 < 113/2000$ and $\mathbb{E}[-\log W] < 1/70$. Combining the conditional margin $3/80$ from Theorem 1.3 with $\Delta < \kappa = 271/500$ gives the stated bounds. □

Proposition 6.4 (Exclusion of all three-point contact configurations)**.** *No pair of nondegenerate centered distributions supported on at most three points can attain a global maximum of Δ/L with $49/120 \le C < 227/500$.*

Proof. By Theorem 1.3, at least one summand must be a nondegenerate three-point distribution, and Lemma 6.3 gives $d_1 + d_2 > 1/3$. Ordering the support points of X_1 and X_2, the event $\{(j,k) : x_j + y_k \le z\}$ is a downset encoded by the number of included pairs in each row of X_1.

- *One three-point and one two-point summand (3×2 grid):* Among the ten downsets $000, 100, 110, 111, 200, 210, 211, 220, 221, 222$, the contact conditions of Lemma 6.2 immediately rule out $000, 100, 200, 111, 222$. Appendix B rules out $211, 220, 221$ (unique representation), the middle/low representation of 210, and the 211 collision analytically, and rules out the remaining four configurations (`lower110`, `low210`, `collision210`, `collision221`) by certified interval covers.

- *Two three-point summands (3×3 grid):* For ordered support points $x_{i,0} < x_{i,1} < x_{i,2}$ with adjacent gaps $u_i = x_{i,1} - x_{i,0}$ and $v_i = x_{i,2} - x_{i,1}$, set $\alpha_i = v_i/(u_i + v_i)$ and $\beta_i = u_i/(u_i + v_i)$, and let
$$\eta_i = \frac{\alpha_i\delta_{x_{i,0}} - \delta_{x_{i,1}} + \beta_i\delta_{x_{i,2}}}{u_i v_i}$$
be the signed measure on $\operatorname{supp} X_i$ having zero total mass, zero first moment, and unit second moment ($\int x^2\, d\eta_i = 1$). The opposite joint perturbation $(P_1 + t\eta_1) \otimes (P_2 - t\eta_2)$ preserves total variance and makes R affine in t, so maximality requires $g = \iint \mathbf{1}_{\{x+y\le z\}}\, d\eta_1 d\eta_2 \ge 0$. This sign condition immediately rules out $110, 200, 210, 211, 310, 311, 332$, while the $g = 0$ shapes $000, 111, 222, 300, 330, 333$ and 100 are ruled out by Lemma 6.2. Appendix B rules out all remaining unique-representation configurations and $322, 331$ collisions analytically, and rules out the remaining collision configurations (`three_owner321`, `adjacent_owner321`, `endpoint_owner321`, `tt221_negative`, `tt221_positive`) by certified interval covers, including strict exclusion at the boundary $C = 49/120$.

Together with Lemma 6.1, this completes the proof of Theorem 1.1. □

7 Transfer to larger arrays with two dominant summands

Normalize the total variance of $(X_1, \dots, X_n)$ to $B^2 = 1$. For $0 < v \le 1$, set
$$K(v) = \frac{\phi(0)}{2} \int_0^1 \frac{s^2\, ds}{[1 - (1-v)s^2]^{3/2}}.$$

Lemma 7.1 (Gaussian remainder comparison)**.** *If T is a sum of independent centered random variables with total variance $1-v$ and third absolute moment sum r, and $G_v \sim N(0,v)$ is independent of T, then $d_{\mathrm{K}}(T+G_v, N(0,1)) \le K(v)r$.*

Proof. Put $a = 1-v > 0$. By Goldstein [5, Lemma 1.1] and Tyurin [9, Theorem 3 and Corollary 1], the zero-biased distribution T^* satisfies $W_1(T,T^*) \le r/(2a)$. For $h(u) = \Phi((x-u)/\sqrt{v})$, the Stein equation $af'(u) - uf(u) = h(u) - \mathbb{E}h(\sqrt{a}Z)$ has solution $f(u) = -\int_0^1 \mathbb{E}h'(su + \sqrt{a(1-s^2)}Z)\,ds$, with $\|f''\|_\infty \le \phi(0)\int_0^1 s^2(1-as^2)^{-3/2}\,ds$, so $|\mathbb{E}h(T) - \Phi(x)| = a|\mathbb{E}f'(T) - \mathbb{E}f'(T^*)| \le aW_1(T,T^*)\|f''\|_\infty \le K(v)r$. □

Partitioning the array into two disjoint blocks H_1, H_2 of combined variance $v = \mathrm{Var}(H_1 + H_2) > 0$ and original third-moment sum b, plus an independent remainder T of third-moment sum $r = R - b$, let $e = \mathbb{E}|H_1|^3 + \mathbb{E}|H_2|^3 - b \ge 0$ be the increase in third absolute moments under grouping (nonnegativity follows from the identity

$$\mathbb{E}|X+Y|^3 - \mathbb{E}|X|^3 - \mathbb{E}|Y|^3 = 2\mathbb{E}\Big[\mathbf{1}_{\{XY<0\}} \min(|X|,|Y|)^2\{3\max(|X|,|Y|) - \min(|X|,|Y|)\}\Big] \ge 0 \tag{7.1}$$

for independent centered X, Y). Conditioning on T, applying Theorem 1.1 to $H_1 + H_2$, and then applying Lemma 7.1 gives

$$\Delta \le \frac{c_2(b+e)}{v^{3/2}} + K(v)r, \tag{7.2}$$

so that $\Delta \le C_{\mathrm{E}}R$ holds whenever

$$\frac{c_2}{v^{3/2}}\left(\frac{b}{R} + \frac{e}{R}\right) + K(v)\left(1 - \frac{b}{R}\right) \le C_{\mathrm{E}}. \tag{7.3}$$

In particular, since $K(v) < \frac{2}{75} + \frac{1}{25v^{3/2}}$ by convexity,

$$v \ge \frac{11}{16}, \qquad b \le r, \qquad e \le \frac{R}{500} \quad \Longrightarrow \quad \Delta < C_{\mathrm{E}}R. \tag{7.4}$$

Proof of Corollary 1.4. For two arbitrary summands ($e = 0$), (7.2) implies $\Delta \le C_{\mathrm{E}}R$ whenever $c_2/v^{3/2} \le C_{\mathrm{E}}$ and $K(v) \le C_{\mathrm{E}}$, both of which hold for $v \ge 439/440$ since $(49/120)^2 < (40973/100000)^2(439/440)^3$. For two two-point summands, Theorem 1.3 gives $\Delta < C_{\mathrm{E}}b/v^{3/2} - 3/80 + K(v)r$; if $\Delta > C_{\mathrm{E}}R$, then $C_{\mathrm{E}}R < \kappa = 271/500$ and $K(v) < C_{\mathrm{E}}$ give $\Delta - C_{\mathrm{E}}R < \kappa(v^{-3/2} - 1) - 3/80 \le 0$ for $v \ge 22/23$ since $(22/23)^3 > (1084/1159)^2$. □

7.1 Signed convolution refinements

Retaining the signed error $f_H(x) = F_H(x) - \Phi(x/\sqrt{v})$ of a two-summand partial sum $H = X_1 + X_2$ under convolution with T gives the sharper bound

$$\Delta(H+T) \le \mathcal{N}_T(f_H) + K(v)r, \qquad \mathcal{N}_T(f_H) = \sup_{x\in\mathbb{R}} |\mathbb{E}f_H(x-T)|. \tag{7.5}$$

For example, for three independent Rademacher summands of size $1/\sqrt{3}$ ($v = 2/3$), $\|f_H\|_\infty = 1/4$ exceeds $C_{\mathrm{E}}/\sqrt{3}$, whereas $\mathcal{N}_T(f_H) = \frac{1}{4} - \frac{1}{2}\Phi(-\sqrt{2})$ and $\mathcal{N}_T(f_H) + K(2/3)/(3\sqrt{3}) < C_{\mathrm{E}}/\sqrt{3}$. Similarly, convolving f_H with an explicit independent normal $G_a \sim N(0,a)$ and using Both $\|f_H\|_\infty \le c_2 b/v^{3/2}$ and $\|f_H\|_1 \le b/v$ [5, 9] yields

$$d_{\mathrm{K}}(H + G_a, G_{v+a}) \le \frac{c_2 b}{v^{3/2}}\left\{2\Phi\left(\frac{\sqrt{v}}{4c_2\sqrt{a}}\right) - 1\right\}. \tag{7.6}$$

8 Overview of the computational certificates

The computational certificates supporting Theorems 1.1, 1.2, and 1.3 are stored in `certificates/finite/` and verified by `verify/finite/check.py` (structural tree and exact algebra checks) and `verify/finite/current/check.py--arithmetic` (full Arb interval reevaluation):

(1) **Two-Bernoulli margin (`margin`, 63,792 leaves at 320 bits):** Covers the two unit cubes in Lemma 2.3 and the endpoint square $[0, 99/100]^2$ in Lemma 2.4, using exact interval monotonicity of $p(1-p)$, $a/(1-2a)^2$, and $(1-2a)/\sqrt{a}$.

(2) **Three-Bernoulli arbitrary diameters (`bernoulli_three`, 1,312,161 accepted leaves at 320 bits):** Covers the eight five-dimensional domains in Section 3 using midpoint bisections, derivative-sign boundary projections under (3.2), and mean-value enclosures.

(3) **Two-summand contact exclusions (Appendix B):** Covers `lower110` (15,081 leaves, plus 427 leaves at $C = 49/120$), `low210` (113,322 leaves), `collision221` (113,154 leaves, plus 34,084 leaves at $C = 49/120$), `collision210` (154,848 leaves), `tt221_negative` (68,963 leaves), `tt221_positive` (90,233 leaves), and the complete Cartesian grids for 321 (`three_owner321`, `adjacent_owner321`, `endpoint_owner321`).

(4) **Common-diameter curvature and Bernstein odds tables:** Reconstructs the 275,283-leaf cover for Lemma 4.4 and verifies the exact $\mathbb{Q}(\sqrt{5})$ Bernstein tables for Proposition A.2 in SymPy [7].

A Proof of the support-endpoint inequalities

In this appendix we prove Proposition 2.2. Set $V = \sum_{i=1}^n p_i q_i h_i^2$ and $\mathcal{B}_h = \sum_{i=1}^n p_i q_i (p_i^2 + q_i^2) h_i^3$.

Lemma A.1 (Atom-free branches at the support endpoints)**.** *For any $n \geq 1$, $p_i \in (0,1)$, $q_i = 1-p_i$, $h_i > 0$, $u = \sum_{i=1}^n q_i h_i$, and $m = \sum_{i=1}^n p_i h_i$,*

$$C_{\mathrm{E}} \frac{\mathcal{B}_h}{V^{3/2}} > \max\left\{1 - \Phi\left(\frac{u}{\sqrt{V}}\right), \Phi\left(-\frac{m}{\sqrt{V}}\right)\right\}. \tag{A.1}$$

Consequently, the closed deficit at the upper support endpoint and the strict deficit at the lower support endpoint are strictly positive.

Proof. By complementation $B_i \mapsto 1 - B_i$, it suffices to prove the upper-endpoint bound. Normalize $V = 1$ and put $x_i = q_i h_i$, $v_i = p_i q_i h_i^2$, $b_i = p_i q_i (p_i^2 + q_i^2) h_i^3$, and $x = \sum_{i=1}^n x_i$. With $c = 2(\sqrt{2} - 1) > 49/60$, the identity $(p_i^2 + q_i^2)/p_i - c = (\sqrt{2} p_i - 1)^2 / p_i \geq 0$ and Cauchy–Schwarz give

$$\mathcal{B}_h = \sum_{i=1}^n \frac{p_i^2 + q_i^2}{p_i} \frac{v_i^2}{x_i} \geq c \sum_{i=1}^n \frac{v_i^2}{x_i} \geq \frac{c}{x}. \tag{A.2}$$

On the other hand, for every $x > 0$,

$$x\{1 - \Phi(x)\} < C_{\mathrm{E}} c. \tag{A.3}$$

Indeed, for $0 < x \leq 2/3$, $x\{1 - \Phi(x)\} \leq 1/3 < 2401/7200 < C_{\mathrm{E}} c$, while for $x \geq 2/3$, Mills' ratio gives $x\{1 - \Phi(x)\} < \phi(x) \leq \phi(2/3) = \phi(0) e^{-2/9} < \phi(0) c < C_{\mathrm{E}} c$. Combining (A.2) and (A.3) proves (A.1). □

Proposition A.2 (Sharp envelope for Bernoulli odds)**.** *Let $o_1, \dots, o_n \geq 0$, not all zero, and set*

$$s = \sum_{i=1}^{n} o_i, \qquad \Pi = \prod_{i=1}^{n} (1+o_i)^{-1}, \qquad g(o) = \frac{o(1+o)^2}{(1+o^2)^2}, \qquad \mathcal{A} = \sum_{i=1}^{n} g(o_i). \tag{A.4}$$

Then

$$\Pi^2 \mathcal{A} \leq \frac{1}{3} \qquad (0 < s \leq 1), \tag{A.5}$$

$$\Pi^2 \mathcal{A} \leq \frac{s}{(1+s/2)^2(1+s^2/4)^2} \qquad \left(1 \leq s \leq \frac{16}{9}\right), \tag{A.6}$$

$$\Pi^2 \mathcal{A} \leq \frac{944784}{6076225} \qquad \left(s \geq \frac{16}{9}\right). \tag{A.7}$$

Moreover, along any nonzero ray $v \in [0,\infty)^n$, $\mathcal{F}_v(t) = \Pi(tv)^2 \mathcal{A}(tv)$ satisfies $\frac{d}{d\log t} \log \mathcal{F}_v(t) < -3/100$ for $t \sum_i v_i \geq 16/9$, and equality in (A.6) holds if and only if two odds equal $s/2$ and all others vanish.

Proof. To analyze the intermediate range $1 \leq s \leq 16/9$, isolate two coordinates $x = o_1, y = o_2$ with sum $a = x + y$ and product $k = xy \in (0, a^2/4]$. Writing $H = \sum_{j=3}^{n} g(o_j)$, $\Pi_0 = \prod_{j=3}^{n} (1+o_j)^{-1}$, $z_a(k) = (1+a+k)^{-2}$, and $R_a(k) = z_a(k)\{g(x)+g(y)\}$, we have

$$\Pi^2 \mathcal{A} = \Pi_0^2 \{R_a(k) + H z_a(k)\}. \tag{A.8}$$

With $D = 1 + a^2 - 2k + k^2$, we have $R_a(k) = N(a,k)/[D^2(1+a+k)^2]$, where $N(a,k) = a + 2a^2 + a^3 + (a^3 - a - 4)k + (2a^2 - a + 8)k^2 + (a-4)k^3$. At the boundary endpoints $k = 0$ (merging $(x,y) \mapsto (a,0)$) and $k = a^2/4$ (equal split $x = y = a/2$), the values $\mathcal{C}_1(a) = a/(1+a^2)^2$ and $\mathcal{C}_2(a) = a/[(1+a/2)^2(1+a^2/4)^2]$ cross at $a_* = 3 - \sqrt{5}$.

- For $0 \leq a \leq a_*$, converting the polynomial quotient $\mathcal{S}(a,k) = [aD^2(1+a+k)^2 - (1+a^2)^2 N(a,k)]/[k(1+a)^2]$ to the tensor Bernstein basis of degree $(60, 20)$ on $(a/a_*, 4k/a^2) \in [0,1]^2$ yields 1280 strictly positive coefficients and a single zero at the corner $(a,k) = (a_*, a_*^2/4)$, proving $R_a(k) \leq R_a(0)$ (and hence $R_a(k) + H z_a(k) \leq R_a(0) + H z_a(0)$ since $z_a(k)$ decreases in k).
- For $a_* \leq a \leq 16/9$, parametrizing $z_a(k) = \theta z_a(0) + (1-\theta) z_a(a^2/4)$ by $\theta \in [0,1]$, all 70 tensor Bernstein coefficients of the polynomial quotient $\mathcal{L}(a,k)$ on $[a_*, 16/9] \times [0, a^2/4]$ are strictly positive, proving the strict chord inequality

$$R_a(k) < \theta R_a(0) + (1-\theta) R_a(a^2/4) \qquad \left(0 < k < \frac{a^2}{4}\right). \tag{A.9}$$

Consequently, at any maximizer of $\Pi^2 \mathcal{A}$ for $s \in [1, 16/9]$, no pair of positive odds can have sum $\leq a_*$, and every pair in the strict chord range must be equal, so all $d \in \{1,2,3,4\}$ positive odds are equal to s/d. Comparing the four functions $\mathcal{C}_d(s) = s/[(1+(s/d)^2)^2(1+s/d)^{2(d-1)}]$ on $[1, 16/9]$ shows $\mathcal{C}_2(s) > \mathcal{C}_d(s)$ for $d \in \{1,3,4\}$, proving (A.6).

For $0 < s \leq 1$, expanding $\Pi^2 \mathcal{A} = \sum_i o_i/[(1+o_i^2)^2 \prod_{j \neq i}(1+o_j)^2]$ and applying the elementary inequality

$$(1+x^2)^2(1+y)^2 \geq 3(x+y) \qquad (x > 0,\ y \geq 0) \tag{A.10}$$

with $x = o_i$ and $y = s - o_i \le \prod_{j\neq i}(1+o_j) - 1$ bounds each term by $o_i/(3s)$, proving (A.5). Finally, for $s \ge 16/9$, using $e(o) := og'(o)/g(o) < 5/4$ and subadditivity of $o \mapsto o/(1+o)$ gives

$$\frac{d}{d\log t}\log \mathcal{F}_v(t) = \frac{\sum_i g(o_i)e(o_i)}{\mathcal{A}} - 2\sum_i \frac{o_i}{1+o_i} < \frac{5}{4} - \frac{2s}{1+s} \le -\frac{3}{100},$$

so scaling back to $s = 16/9$ yields (A.7). The Bernstein conversions use

$$b_\beta = \sum_{\alpha\le\beta} a_\alpha \prod_j \frac{\binom{\beta_j}{\alpha_j}}{\binom{d_j}{\alpha_j}} \tag{A.11}$$

and are checked in `verify/finite/support_odds/`. □

Lemma A.3 (Atom-bearing branches at the support endpoints)**.** *Under the hypotheses of Lemma A.1, the closed deficit at the lower support endpoint and the strict deficit at the upper support endpoint are strictly positive.*

Proof. Normalizing $V = 1$ and setting $o_i = p_i/q_i$, $w_i = p_i q_i h_i^2$, $x_0 = \sum_i \sqrt{o_i w_i}$, and $\Pi = \prod_i (1+o_i)^{-1}$, Lemma 2.1 and Cauchy–Schwarz give $\mathcal{B}_h \ge \mathcal{A}^{-1/2}$ and $x_0 \le \sqrt{s}$. With $Q(x) = 1-\Phi(x)$, it therefore suffices to show

$$\frac{C_{\mathrm{E}}}{\sqrt{\mathcal{A}}} + Q(\sqrt{s}) > \Pi. \tag{A.12}$$

First note that

$$f(s) := (1+s)Q(\sqrt{s}), \qquad f'(s) < 0 \quad (s > 0), \tag{A.13}$$

since $Q(x)/\phi(x) < (1+x^2)/(2x)$ for all $x > 0$. On $0 < s \le 1$, (A.5) gives $C_{\mathrm{E}}/(\Pi\sqrt{\mathcal{A}}) > 7/10$, while $Q(\sqrt{s})/\Pi \ge (1+s)Q(\sqrt{s}) \ge 2Q(1) > 47/150 > 3/10$, whose sum exceeds 1. On $1 \le s \le 16/9$, (A.6) with $u = s/2 \ge 1/2$ gives

$$\frac{C_{\mathrm{E}}}{\Pi\sqrt{\mathcal{A}}} \ge C_{\mathrm{E}}\frac{(1+u)(1+u^2)}{\sqrt{2u}} > \frac{49}{64}, \tag{A.14}$$

while (A.13) gives

$$\frac{Q(\sqrt{s})}{\Pi} \ge (1+s)Q(\sqrt{s}) \ge \frac{25}{9}Q\left(\frac{4}{3}\right) > \frac{15}{64}, \tag{A.15}$$

which again sum to more than 1. Finally, for $s \ge 16/9$, (A.7) gives $\Pi^2\mathcal{A} \le 944784/6076225 < (49/120)^2 < C_{\mathrm{E}}^2$, so $C_{\mathrm{E}}/\sqrt{\mathcal{A}} > \Pi$ already holds. Together with Lemma A.1, this completes the proof of Proposition 2.2. □

B Contact calculations for two three-point summands

Throughout this appendix, suppose that two nondegenerate independent centered distributions X_1, X_2 attain the two-summand supremum $C = C_2 \ge 49/120$. By Theorem 1.1 of Bell [1], $C < 227/500$. By Lemma 6.1 and Theorem 1.3, each marginal is supported on at most three points and at least one is a nondegenerate three-point distribution (a *triple*, having three distinct atoms of positive mass).

We initially normalize total variance to $B^2 = 1$ and write F for the selected closed event probability, $D = F - \Phi(z)$, and $R = \mathbb{E}|X_1|^3 + \mathbb{E}|X_2|^3$ (so that $R > 1/\sqrt{2}$ by strict convexity of $u \mapsto u^{3/2}$). A *representation* of the threshold z is a pair of support points $(x_j, y_k) \in \operatorname{supp} X_1 \times \operatorname{supp} X_2$ with $x_j + y_k = z$.

Proposition 6.4 rules out the zero-event, largest-point, and negative-mixed-derivative downsets. The table below indexes the remaining downsets and threshold-representation patterns. Transposition interchanges the two marginal distributions, and the certificate identifiers refer to the domains in `certificates/finite/current-results.json` described in Section 8.

Marginals and event	Representations	Argument or certificate
Triple/pair 211	Unique	Section B.5
Triple/pair 220	Unique middle/high	Section B.9
Triple/pair 221	Unique middle/high	Section B.10
Triple/pair 210	Unique middle/low	Section B.11
Triple/pair 210	Unique low/high	Section B.12; `low210`
Triple/pair 110	Unique	Section B.13; `lower110`
Triple/pair 211	Two	Section B.14
Triple/pair 221	Two, upper-gap collision	Section B.15; `collision221`
Triple/pair 210	Two, lower-gap collision	Sections B.16–B.18; `collision210`
Two triples 220	Unique	Sections B.6–B.8
Two triples 221, 320	Unique	Section B.19
Two triples 321	Unique middle/middle	Section B.20
Two triples 322, 331	Two	Section B.21
Two triples 321	Three	Sections B.23–B.27; `three_owner321`
Two triples 321	Two adjacent	Sections B.28–B.29; `adjacent_owner321` and its completion
Two triples 321	Two endpoints	Sections B.28, B.30; `endpoint_owner321`
Two triples 221, 320	Two	Sections B.31–B.37; `tt221_negative`, `tt221_positive`

Changes of scale The discrepancy D and ratio C are unchanged by a common positive rescaling. In any raw scale, let V be the total variance, R_0 the sum of the centered third absolute moments, and o the centered threshold. Then

$$z = \frac{o}{\sqrt{V}}, \qquad R_{\text{normalized}} = \frac{R_0}{V^{3/2}}, \qquad C = \frac{DV^{3/2}}{R_0}.$$

Two choices of raw scale are used:

Scale	Definition and conversion
Contact scale	Divide the variance-one supports by $K = R_{\text{normalized}} + z\phi(z)/(3C)$. Then $V = K^{-2}$, $\tau = CK^3 = C/V^{3/2}$, and the contact slope is $\sigma = \phi(z)/(CK^2)$. After division by τ, the cubic majorant has quadratic coefficient $-3/2$.
Unit-span scale	Translate the supports to start at zero and divide by the chosen outer support span. Center these raw laws when computing V, R_0, o.

Each family states its scale. In particular the three-representation calculations use the unit-span scale, while the collided 221 systems use the contact scale.

B.1 Variance derivatives at a triple

The total variance is one in this calculation.

Let $x_1 < x_2 < x_3$ be the support of a genuine triple. Its second divided difference, normalized by $[x_1, x_2, x_3]x^2 = 1$, satisfies

$$[x_1, x_2, x_3]|x|^3 \le \frac{15D + \phi'''(z)}{12C}. \tag{B.1}$$

Indeed, take a nonzero signed measure η on these points with zero mass and first moment. Put $v = \int x^2\, d\eta$ and $r = \int |x|^3\, d\eta$. The Vandermonde determinant gives $v \neq 0$, and $r/v = [x_1, x_2, x_3]|x|^3$. Both signs of t are allowed in $P + t\eta$. The total variance becomes $V(t) = 1 + tv$, the sum of third absolute moments becomes $R + tr$, and the event probability becomes $g(t) = F + tg'$. Maximality gives

$$J(t) = V(t)^{3/2}\{g(t) - \Phi(z/\sqrt{V(t)})\} - C(R + tr) \leq 0, \qquad J(0) = 0.$$

With $A = (3D + z\phi(z))/2$, differentiation gives

$$g' + Av - Cr = 0, \qquad J''(0) = v^2 \left\{ 3C\frac{r}{v} - \frac{15D + \phi'''(z)}{4} \right\} \leq 0.$$

This proves (B.1).

For any three real knots of diameter h,

$$[x_1, x_2, x_3]|x|^3 \geq h/2. \tag{B.2}$$

On one side of zero the divided difference is the sum of the absolute values. Otherwise scale h to one and reflect to write the knots as $-a, c, 1 - a$, where $0 < a < 1$ and $0 \leq c \leq 1 - a$. The divided difference q obeys

$$q = 1 + c - 2a + \frac{2a^3}{a + c}, \qquad (a + c)(q - 1/2) = 2a(a - 1/2)^2 + c(c + 1/2 - a).$$

The right side is nonnegative for $a \leq 1/2$. For $a > 1/2$, put $d = a - 1/2$ and rewrite it as $(c - d/2)^2 + (2a - 1/4)d^2 \geq 0$. Equality requires the symmetric knots $(-h/2, 0, h/2)$.

B.2 A joint curvature and diameter bound

For every centered variance-one law,

$$15\{F(z) - \Phi(z)\} + \phi'''(z) < 79/10. \tag{B.3}$$

For $z \geq 0$ use $F(z) \leq 1$. On $[0, \sqrt{3}]$ the derivative of $15(1 - \Phi(z)) + \phi'''(z)$ is $(z^4 - 6z^2 - 12)\phi(z) < 0$; its value at zero is $15/2$. Beyond $\sqrt{3}$ the third derivative is nonpositive, giving the same upper bound.

For $z = -x < 0$, Cantelli gives $D \leq d(x) = (1+x^2)^{-1} - (1 - \Phi(x))$. The ratio $2x/\{(1+x^2)^2\phi(x)\}$ increases, since its logarithmic derivative is $(x^2 - 1)^2/\{x(1 + x^2)\}$. At $x = 1/2$ the ratio exceeds one; moreover $1 - \Phi(1/2) > 3/10$. Hence d decreases on $[1/2, \infty)$ and $d(1/2) < 1/2$. On $[1/2, \sqrt{3}]$, $\phi'''(-x) \leq 0$, so the quantity in (B.3) is less than $15/2$. For $x \geq \sqrt{3}$, $d(x) < 1/4$ and Fourier inversion gives $\|\phi'''\|_\infty \leq 2/\pi < 2/3$, yielding $53/12$.

It remains to consider $0 \leq x \leq 1/2$. The bounds $1 - \Phi(x) \geq 1/2 - \phi(0)x$, $\phi(x) \geq \phi(0)(1 - x^2/2)$, and $\phi(0) < 2/5$ give the upper bound

$$U(x) = \frac{15}{1 + x^2} - \frac{15}{2} + \frac{24}{5}x + x^3 - \frac{x^5}{5}.$$

Here

$$10(1 + x^2)\{79/10 - U(x)\} = 4 - 48x + 154x^2 - 58x^3 - 8x^5 + 2x^7 =: p(x).$$

On each interval $[i/64, (i + 1)/64]$, $0 \leq i < 32$, expand $p((i + t)/64)$ in the degree-seven Bernstein basis. Direct rational conversion gives a positive coefficient minimum $11964025693/3848290697216$.

For completeness this assertion means the following finite rational identity and comparison: if $p(x) = \sum_{j=0}^{7} a_j x^j$, compute

$$c_{i,k} = \sum_{j=k}^{7} a_j \binom{j}{k} \frac{i^{j-k}}{64^j}, \qquad b_{i,l} = \sum_{k=0}^{l} c_{i,k} \frac{\binom{l}{k}}{\binom{7}{k}} \quad (0 \le i < 32,\ 0 \le l \le 7).$$

The minimum of these 256 fractions is the stated number. The nonnegative Bernstein basis functions sum to one, proving positivity.

Combining (B.1)–(B.3) with $C \ge 49/120$ shows that every normalized triple diameter is less than $158/49$. In raw coordinates of total variance V and triple diameter A_i this is

$$24964V > 2401A_i^2. \tag{B.4}$$

If the two raw spans equal one, it follows that

$$V > 2401/24964, \qquad R \ge V^{3/2}/\sqrt{2} > \frac{49^3}{158^3\sqrt{2}} > 1/48.$$

The last strict comparison follows on squaring from $49^6\,48^2 > 2\,158^6$.

B.3 Threshold and density bounds

For every centered variance-one law with positive closed-CDF discrepancy

$$D = F(z) - \Phi(z) \ge d_0 := 49/(120\sqrt{2}),$$

one necessarily has

$$\boxed{-5/4 < z < 3/5, \qquad \phi(z) > 9/50.} \tag{B.3.1}$$

Since any two-summand maximizer with $C \ge 49/120$ satisfies $R_{\text{normalized}} \ge 1/\sqrt{2}$, its discrepancy obeys $D = CR_{\text{normalized}} \ge d_0$ and therefore satisfies (B.3.1).

The negative threshold For $z = -s < 0$, Cantelli's inequality, including the closed event, gives

$$D \le g(s) := \frac{1}{1+s^2} - \Phi(-s).$$

The ratio

$$H(s) = \frac{2s}{(1+s^2)^2\phi(s)}$$

satisfies

$$\frac{d}{ds}\log H(s) = \frac{(s^2-1)^2}{s(1+s^2)} \ge 0.$$

Also $2s/(1+s^2)^2 = 640/1681 > 1/4$ at $s = 5/4$, while $\phi(5/4) < \phi(1) < 1/4$. The last inequality follows from $2\pi e > 16$, using $\pi > 3$ and $e > 8/3$. Thus $H(s) > 1$ and $g'(s) < 0$ for every $s \ge 5/4$.

Taylor's formula with the signed remainder gives, for $u \ge 0$, $\exp(-u) \le 1 - u + u^2/2 - u^3/6 + u^4/24$. At $s = 5/4$, let

$$I = s - s^3/6 + s^5/40 - s^7/336 + s^9/3456.$$

Integrating the polynomial and using $\phi(0) < 2/5$ gives

$$\Phi(-s) > 1/2 - (2/5)I,$$
$$g(5/4) < 16/41 - 1/2 + (2/5)I = \frac{37147949759}{130006646784} =: B_-.$$

Both B_- and d_0 are positive, and the exact comparison is

$$d_0^2 - B_-^2 = \frac{727351123347544172903}{422543205200493438566400} > 0.$$

Consequently $D \geq d_0$ excludes every $z \leq -5/4$.

The positive threshold The classical $\pi < 22/7$ implies $\phi(0) > 79/200$, because

$$7/44 - (79/200)^2 = 1349/440000 > 0.$$

For $x \geq 0$, $\exp(-x^2/2) \geq 1 - x^2/2$. Hence at $s = 3/5$,

$$1 - \Phi(s) < 1/2 - (79/200)(s - s^3/6) = 13861/50000 =: B_+.$$

The exact difference $d_0^2 - B_+^2 = 146635361/22500000000$ is positive. Since $F(z) \leq 1$, all $z \geq 3/5$ have $D \leq 1 - \Phi(3/5) < d_0$.

A density bound and the raw contact slope By (B.3.1), $|z| < 5/4$, so $\phi(z) > \phi(5/4)$. Put $x = 25/32$. The degree-five Taylor polynomial is a lower bound for $\exp(-x)$ for all $x \geq 0$, giving

$$\phi(5/4) > \frac{79}{200}\sum_{j=0}^{5}\frac{(-25/32)^j}{j!} = \frac{29108943797}{161061273600} > 9/50.$$

The final exact slack is $117914549/161061273600 > 0$.

By the contact-scale definition, $\sigma = \phi(z)/(CK^2)$. Since $C < 227/500$ and $K < 8/5$,

$$\sigma > \frac{9/50}{(227/500)(8/5)^2} > \frac{9}{200}.$$

Thus

$$\boxed{\sigma > 9/200.} \tag{B.3.2}$$

B.4 The raw contact scale

Here supports are divided by K, and V, R, o denote their raw total variance, sum of third absolute moments, and centered threshold.

The collided triple/pair 210 and two-triple 221 contact charts satisfy

$$C = V^{3/2}/G, \quad D = R/G, \quad z = o/\sqrt{V}, \quad \sigma\sqrt{V} = G\phi(z), \quad 3(V - R) = \sigma o.$$

Consequently

$$\boxed{C/\sqrt{V} = D + z\phi(z)/3.} \tag{B.4.1}$$

The threshold result Section B.3 gives $-5/4 < z < 3/5$. On $0 \le z < 3/5$, $z\phi(z)$ increases, and $\phi(3/5) < 1/3$ as verified below. For negative z, the product is negative. Thus in all these contact charts $z\phi(z) < 1/5$. Combining (B.4.1) with $D < 11/20$ and $C \ge 49/120$ proves

$$\frac{1}{\sqrt{V}} < \frac{\frac{11}{20} + \frac{1}{15}}{49/120} = \frac{74}{49}, \qquad \boxed{V > 2401/5476.} \tag{B.4.2}$$

If the actual representation satisfies $o \le 0$, then $z \le 0$ and the stronger bound is

$$\frac{1}{\sqrt{V}} < \frac{11/20}{49/120} = \frac{66}{49}, \qquad \boxed{V > 2401/4356.} \tag{B.4.3}$$

The entire negative 210 chart has $o = c - ph < 0$, so (B.4.3) holds on that whole candidate class. In another chart it can be used when the enclosing interval proves $o \le 0$ for every surviving candidate. It cannot be applied to a box that still permits both signs without an additional valid case split.

These scale formulas are specific to the chart normalization fixing the quadratic coefficient of the cubic contact potential. They are not asserted for the unit-span three-representation 321 chart, which uses a different raw scale.

For the elementary density endpoint, the classical Archimedean lower bound $\pi > 223/71$ gives $\phi(0) < 399/1000$, because

$$(399/1000)^2 - 71/446 = 1823/223000000 > 0.$$

The degree-four Taylor upper bound for $\exp(-x)$ at $x = 9/50$ now gives

$$\phi(3/5) < \frac{399}{1000}\sum_{j=0}^{4}\frac{(-9/50)^j}{j!} = \frac{16663671213}{50000000000} < 1/3.$$

B.5 One triple: contact identities and threshold shapes

In this subsection the support coordinates have total variance one; factors of K are retained explicitly.

Derivative identities for the Bernoulli law Write the Bernoulli support as $-u, v$, with $u, v > 0$, and set

$$h_B = u + v, \qquad K_B = \frac{u^2 + v^2}{u + v}.$$

Its variance is uv, its third absolute moment is uvK_B, and its signed second moment is $M_B = uv(v - u)/(u + v)$. Its dual derivative is

$$f_B'(x) = 3C\{x|x| - Kx - M_B\} - \phi(z). \tag{B.5.1}$$

The nonselected point is a smooth minimum. Applying (B.5.1) there gives

Bernoulli representation	exact identity	consequence
$-u$	$\phi(z) = 3Cv(K_B - K)$	$K_B > K, \quad 2v > K$
v	$\phi(z) = 3Cu(K - K_B)$	$K_B < K, \quad 2u > K.$

(B.5.2)

The second derivative gives the last inequalities. Equality would be a stationary inflection rather than a local minimum. Also

$$K_B \geq \frac{h_B}{2}, \qquad K_B < h_B. \tag{B.5.3}$$

Exhaustive grid and three elementary eliminations List the number of included Bernoulli support points in each successive triple row. The ten possible downsets are

$$000, 100, 110, 111, 200, 210, 211, 220, 221, 222.$$

A unique selected representation must be a maximal included cell. A genuine triple cannot have its largest support point as the selected representation, by the two smooth-minimum argument in the proof of Proposition 6.4. This excludes 111 and 222; 000 has negative signed discrepancy.

Shapes 100 and 200 have a selected lowest triple point, while both nonselected triple points have deletion-CDF value zero. They are thus two equal zero minima. The cubic identity gives them as $-K, K$, with

$$f_T(x) = C|x|^3 - \frac{3}{2}CKx^2 + \frac{1}{2}CK^3.$$

The selected triple point is $-a$, with $a > K$, and let its mass be l. All event probability comes from that triple point, so

$$F = l f_T(-a) = Cl(a-K)^2(a+K/2) < Cla^3 \leq CR.$$

This contradicts $F - \Phi(z) = CR > 0$, excluding both 100 and 200.

For shape 211, the only possible representation is triple-low/Bernoulli-high. The two nonselected triple points have the same deletion level, hence are the symmetric smooth minima $-K, K$. Write the selected triple point as $-a$, with $a > K$. The included triple-high/Bernoulli-low cell is strictly below the representation, giving

$$K - u < -a + v, \qquad h_B > a + K > 2K.$$

But (B.5.2)–(B.5.3), for a high Bernoulli representation, require $h_B/2 \leq K_B < K$. This is a contradiction. Thus 211 is also excluded.

Exactly five representation-specific configurations remain: 110 with the middle/low representation, the two representations of 210, and 220 or 221 with the middle/high representation.

Exact data for the five remaining configurations When the selected representation uses the middle triple point, write the triple support as $-a, c, b$. Its two nonselected minima satisfy

$$\frac{K}{2} < a < K < b, \qquad a^2 + b^2 = K(a+b). \tag{B.5.4}$$

Let its masses be l, m, r, put $q_B = v/(u+v)$, and let $u_T = a+c$, $v_T = b-c$. The actual deletion levels and strict geometry are:

Shape and representation	Triple levels	Bernoulli levels	Gap condition
110, middle/low	$(q_B, q_B, 0)$	$(l+m, 0)$	$h_B > u_T$
210, middle/low	$(1, q_B, 0)$	$(l+m, l)$	$h_B < u_T$
220, middle/high	$(1, 1, 0)$	$(l+m, l+m)$	$h_B < v_T$
221, middle/high	$(1, 1, q_B)$	$(1, l+m)$	$h_B > v_T$

In the middle/low 210 case, (B.5.2)–(B.5.3) give

$$a + c > h_B > K_B > K, \qquad c > K - a > 0. \tag{B.5.5}$$

For the low/high representation of 210, instead write the triple support as $-t, -a, b$, where $t > a$ and its two nonselected minima $-a, b$ satisfy (B.5.4). Its deletion levels are $(1, q_B, 0)$, the Bernoulli deletion levels are $(l + m, l)$, and its exact strict geometry is

$$t - a < h_B < t + b, \qquad K_B < K. \tag{B.5.6}$$

Every table entry follows directly from the indicated threshold cell; none replaces the actual convolution probabilities by free parameters.

B.6 Two triples: symmetry of a unique-representation rectangle

Common shape coordinates Use the full-law cubic dual from Lemma 6.2, and normalize the total variance to one. Put

$$K = \frac{2A}{3C} = R + \frac{z\phi(z)}{3C} > 0.$$

For coordinate i, write its support as $K(-a_i, c_i, b_i)$, where $-a_i, b_i$ are the two nonselected smooth minima and c_i is the selected middle point. These are dimensionless support coordinates. The minimum equations and the strictly decreasing contact levels give

$$\frac{1}{2} < a_i < 1 < b_i, \qquad a_i^2 + b_i^2 = a_i + b_i, \qquad b_i = \frac{1 + \sqrt{1 + 4a_i - 4a_i^2}}{2}. \tag{B.6.1}$$

Set

$$h_i = a_i + b_i, \quad u_i = a_i + c_i, \quad v_i = b_i - c_i,$$

$$D(a) = \frac{1}{2}(b - a)(a^2 + 4ab + b^2), \qquad H(a, c) = 3\{a(1 - a) + c - c|c|\}. \tag{B.6.2}$$

Thus $CK^3 D(a_i)$ is the difference of the two minimum contact levels, and $CK^2 H(a_i, c_i)$ is the negative slope at the representation. Let m_i be the middle mass and q_i the combined mass of the first two support points. The unique threshold representation and centering imply

$$\sigma = \frac{\phi(z)}{CK^2} > 0, \qquad m_i = \frac{\sigma}{H(a_i, c_i)}, \qquad q_i = \frac{b_i}{h_i} + \frac{\sigma u_i}{h_i H(a_i, c_i)}. \tag{B.6.3}$$

For shape 220, the dual equals q_j at the first and middle support points of coordinate i, and equals zero at its last support point. Consequently

$$f_i(c_i K) = f_i(-a_i K), \qquad q_j = CK^3 D(a_i). \tag{B.6.4}$$

In particular,

$$q_1 D(a_1) = q_2 D(a_2). \tag{B.6.5}$$

The contact equations force identical laws For each a in $(1/2, 1)$, equation $f(c) = f(-a)$ selects one middle point on the descending part of the cubic between its local maximum and its positive minimum. Its normalized location is a strictly increasing function of a.

For $1/2 < a \le 3/4$, factorization on the negative half-line gives

$$c = 2a - 3/2, \qquad H = (a + c)^2. \tag{B.6.6}$$

For $3/4 \le a < 1$, the middle point is nonnegative and satisfies $c < a$. Indeed, the linear coefficient of the dual is negative, so $f(Ka) - f(-Ka) = -6CK^3a^2(1-a) < 0$, whereas f decreases strictly along this positive part of the descending branch. Set $k = c/a$, so $0 \le k < 1$. The equal-height equation is exactly

$$a = \frac{3(1+k)^2}{2(k^3 + 3k + 2)}, \qquad c = ka. \tag{B.6.7}$$

Its derivative is

$$a'(k) = \frac{3(1+k)(1-k)(k^2+4k+1)}{2(k^3+3k+2)^2} > 0 \quad (0 \le k < 1). \tag{B.6.8}$$

We claim that, for every fixed $\sigma > 0$, the expression in (B.6.3) obeys

$$a \longmapsto q(a)D(a) = D(a)\frac{b}{a+b} + \sigma\frac{D(a)(a+c)}{(a+b)H(a,c)} \quad \text{strictly decreasing on } (1/2, 1). \tag{B.6.9}$$

First, direct differentiation using (B.6.1) gives $D'(a) = -3(2a-1)(a+b) < 0$. Also b decreases with a, so $b/(a+b)$ strictly decreases. Hence the first term in (B.6.9) strictly decreases. For the second term, put

$$T(a) = \frac{D(a)(a+c)}{(a+b)H(a,c)}.$$

On the nonpositive branch, (B.6.6) gives $T = D/[(a+b)(a+c)]$. Here D decreases, while $a+b$ and $a+c$ increase, so T decreases.

On the positive branch, the cubic factors at its positive minimum:

$$D(a) = (b-c)^2(c+2b-3/2), \qquad H(a,c) = 3(b-c)(b+c-1).$$

Thus the exact function to check is

$$T(k) = \frac{(b-c)(a+c)(c+2b-3/2)}{3(a+b)(b+c-1)}, \tag{B.6.10}$$

with (B.6.1), (B.6.7). The denominator stays positive through $k = 1$, so (B.6.10) and its derivative extend continuously to the closed interval $[0, 1]$. The fixed certificate `rectangle`, checked by the rules in Section 8, proves

$$\boxed{T'(k) < -1/4 \quad \text{for every } 0 \le k \le 1.} \tag{B.6.11}$$

The two formulas for T agree at $a = 3/4$, so (B.6.9) follows. Equation (B.6.5) now gives $a_1 = a_2$. Equations (B.6.1), (B.6.6)–(B.6.8), and (B.6.3) then give identical supports, middle masses, and outer masses for the two coordinates. In particular $c_1 = c_2 = c$.

B.7 Two triples: nonpositive rectangle representations

Here $-a, c, b$ are the support points in the variance-one scale, with K retained explicitly.

The equal-height contacts Write the discrepancy as $D = CR$, the selected threshold as z, and set

$$K = \frac{2A}{3C} = R + \frac{z\phi(z)}{3C}.$$

For each coordinate, write its two nonselected smooth minima as $-a, b$ and its selected middle atom as c, of mass m. The cubic dual has derivative

$$f'(u) = 3Cu|u| - 3CKu - L.$$

The two minimum equations imply

$$a^2 + b^2 = K(a + b), \qquad L = 3Ca(K - a). \tag{B.7.1}$$

The decreasing deletion CDF has strictly larger value at $-a$ than at b in shape 220. The minimum-height difference is

$$f(-a) - f(b) = \frac{C}{2}(b - a)(a^2 + 4ab + b^2) > 0.$$

Thus $b > a$. The strict local-minimum conditions and (B.7.1) consequently give

$$K/2 < a < K < b, \qquad b = \frac{K + \sqrt{K^2 + 4Ka - 4a^2}}{2}. \tag{B.7.2}$$

In this rectangle, the deletion CDF has the same value at $-a$ and c, so $f(c) = f(-a)$. Suppose $c \le 0$. On the negative half-line, direct factorization gives

$$f(c) - f(-a) = C(c + a)^2(2a - 3K/2 - c).$$

Since the support points are distinct, this yields

$$c = 2a - 3K/2, \qquad K/2 < a \le 3K/4, \qquad -f'(c) = C(a + c)^2. \tag{B.7.3}$$

The unique-representation support derivative is $mf'(c) = -\phi(z)$. Therefore

$$m = \frac{\phi(z)}{C(a + c)^2}. \tag{B.7.4}$$

A variance loss forced by the selected mass For a centered law on $-a, c, b$ with middle mass m, its variance is

$$v = ab - m(a + c)(b - c) = ab - \frac{\phi(z)}{C}\frac{b - c}{a + c}. \tag{B.7.5}$$

This follows by solving the two remaining masses, or by averaging the identity $x^2 + (a-b)x - ab = 0$ at the two outer support points.

On $K/2 < a \le 3K/4$, the function ab in (B.7.2) increases with a. Its maximum is

$$\frac{3(2+\sqrt{7})}{16}K^2 < \frac{7}{8}K^2.$$

Indeed, after division by K, the derivative of $xb(x)$ is positive for $1/2 \le x \le 3/4$, while $\sqrt{7} < 8/3$ proves the displayed bound. Also $b > 9K/8$, since the minimum of b in this range is $(2+\sqrt{7})K/4$ and $\sqrt{7} > 5/2$. Equation (B.7.3) gives $a + c \le 3K/4$, so

$$\frac{b-c}{a+c} > \frac{3}{2}.$$

Consequently every such coordinate satisfies

$$\boxed{v < \frac{7}{8}K^2 - \frac{3}{2}\frac{\phi(z)}{C}.} \tag{B.7.6}$$

If both selected middle points are nonpositive, their sum is $z = -t$ with $0 \le t < K$, by (B.7.3). Adding (B.7.6) over the two coordinates gives

$$1 < \frac{7}{4}K^2 - 3\frac{\phi(t)}{C}. \tag{B.7.7}$$

The right side is always less than one The proved global upper bound gives $C < 1/2$, and $C \ge 49/120 > 2/5$. The public variance-only discrepancy bound, with $\kappa = 271/500$, gives $CR < \kappa$. Hence

$$K < \frac{\kappa - t\phi(t)/3}{C}. \tag{B.7.8}$$

First, (B.7.8) and $t < K$ imply $t < \kappa/(2/5) < 7/5$ and

$$t\{2/5 + \phi(t)/3\} < \kappa.$$

The left side increases on $[0, 7/5]$: its derivative is $2/5+(1-t^2)\phi(t)/3 > 2/5-(24/25)(2/5)/3 > 0$. At $t = 6/5$ it exceeds

$$\frac{6}{5}\left(\frac{2}{5} + \frac{19}{300}\right) = \frac{139}{250} > \frac{271}{500},$$

using $\phi(6/5) > 19/100$. Thus $t < 6/5$.

Put $B(t) = \kappa - t\phi(t)/3$. Since $\sup_{t\ge 0} t\phi(t) = \phi(1) < 1/4$, we have $B(t) > 9/20$. For $2/5 \le C \le 1/2$, the derivative in C of

$$\frac{7B(t)^2}{4C^2} - \frac{3\phi(t)}{C}$$

has the sign of $3C\phi(t) - (7/2)B(t)^2$, which is negative because $\phi(t) < 2/5$, $C \le 1/2$, and $B(t) > 9/20$. Equations (B.7.7)–(B.7.8) therefore imply $1 < U(t)$, where

$$U(t) = \frac{7B(t)^2}{4(2/5)^2} - \frac{3\phi(t)}{2/5}, \qquad 0 \le t \le 6/5.$$

This scalar function has no interior maximum. In fact, with $c_0 = 2/5$,

$$\frac{U'(t)}{\phi(t)} = \frac{7B(t)(t^2-1)}{6c_0^2} + \frac{3t}{c_0},$$

whose derivative is

$$\frac{7}{6c_0^2}\left\{\frac{(t^2-1)^2\phi(t)}{3}+2tB(t)\right\}+\frac{3}{c_0}>0.$$

Thus U' changes sign at most once, from negative to positive. At the two endpoints, the elementary bounds $\phi(0)>39/100$ and $\phi(6/5)>19/100$ give

$$U(0)<\frac{46087}{160000},\qquad U(6/5)<\frac{152023}{160000}<1.$$

For the second comparison, the expression defining U is decreasing in $\phi(t)$ at fixed t, so replacing $\phi(6/5)$ by $19/100$ gives an upper bound. This contradicts $1<U(t)$ and completes the exclusion.

The numerical-looking Gaussian comparisons above require no floating point arithmetic. For example $\pi<22/7$ gives $\phi(0)>199/500$. At $x=18/25$, the Taylor bound

$$e^x\le 1+x+x^2/2+x^3/6+\frac{x^4}{24(1-x/5)}<25/12$$

gives $\phi(6/5)>(199/500)(12/25)>19/100$. Also $\pi e>8$ gives $\phi(1)<1/4$, and $\pi>25/8$ gives $\phi(0)<2/5$.

B.8 Two triples: positive rectangle representations

Return to the contact-scale coordinates $(-a,c,b)$ of Section B.6.

Positive representations Suppose $c>0$ in the identical normalized laws obtained above. Omit the coordinate index. The rectangle condition is

$$b-a>2c,\qquad \text{equivalently } v=b-c>u=a+c. \tag{B.8.12}$$

The inequality is strict because a collision between a corner sum and the selected middle-middle sum would give a second representation of the threshold.

Before division by K, each coordinate has variance $1/2$. Writing its middle mass as m, centering and the exact variance formula give

$$\frac{1}{2}=K^2(ab-muv),\qquad q=\frac{b+mu}{h}.$$

Eliminating m yields

$$q=1+\frac{ac}{hv}-\frac{1}{2hvK^2}. \tag{B.8.13}$$

By (B.6.1), $h\le 2$, $ab\le 1$, and $b<(1+\sqrt{2})/2<5/4$. Also $v<b$, while (B.8.12) gives $hv>2u^2$. Therefore

$$hv<5/2,\qquad \frac{ac}{hv}<\frac{ac}{2(a+c)^2}\le\frac{1}{8},$$

and (B.8.13) implies

$$q<Q(K):=\frac{9}{8}-\frac{1}{5K^2},\qquad K\ge 1/\sqrt{2}. \tag{B.8.14}$$

The selected threshold is $z = 2Kc > 0$, and the event probability is q^2. Since $CR = CK - z\phi(z)/3$, its ratio identity would require

$$0 = q^2 - \Phi(z) - CK + z\phi(z)/3 < q^2 - \frac{1}{2} - \frac{2}{5}K. \tag{B.8.15}$$

Here $C \geq 49/120 > 2/5$, and $\Phi(z) \geq 1/2 + z\phi(z)$ for $z \geq 0$. For $K \geq 5/4$, the right side of (B.8.15) is at most $1/2 - (2/5)K \leq 0$, since $q \leq 1$, giving a contradiction.

For $1/\sqrt{2} \leq K \leq 5/4$, define

$$G(K) = Q(K)^2 - 1/2 - (2/5)K.$$

Its second derivative is

$$G''(K) = \frac{4}{5K^4}\left(\frac{1}{K^2} - \frac{27}{8}\right) < 0.$$

Also $G'(5/4) = 654/78125 > 0$. Thus G increases throughout this interval and

$$G(K) \leq G(5/4) = \left(\frac{997}{1000}\right)^2 - 1 = -\frac{5991}{10^6} < 0.$$

Equations (B.8.14)–(B.8.15) again give a contradiction. This excludes positive representations of the 220 event; the only numerical input is (B.6.11).

B.9 One triple: the rectangle

Normalized contacts and exact variance Normalize total variance to one, and use the dimensionless supports

$$T: \quad K(-a, c, b), \qquad B: \quad K(-d, e), \qquad K = R + \frac{z\phi(z)}{3C}.$$

The representation is the middle triple point and the high Bernoulli point, so $z = K(c + e)$. The nonselected triple points are smooth minima with deletion levels $1, 0$; the representation has level 1. Consequently

$$\frac{1}{2} < a < 1 < b, \quad a^2 + b^2 = a + b, \quad f_T(Kc) = f_T(-Ka), \quad -\frac{1}{2} < c < 1. \tag{B.9.1}$$

Both Bernoulli deletion levels are the same, namely $q = \mathbb{P}(T \leq Kc)$. Its low point is a smooth minimum, and its high point has negative derivative. The selected Bernoulli point identity gives

$$d > \frac{1}{2}, \quad \frac{d^2 + e^2}{d + e} < 1.$$

Set $k = e/d > 0$. Factoring the equal-height equation first, without yet restricting k, gives

$$d = d(k) = \frac{3(1 + k)^2}{2(k^3 + 3k + 2)}, \qquad e = kd(k), \qquad k > 0. \tag{B.9.2}$$

Substitution into the strict Bernoulli representation inequality then gives

$$1 - \frac{d^2 + e^2}{d + e} = \frac{(1 - k)(k^2 + 4k + 1)}{2(k^3 + 3k + 2)} > 0.$$

Hence $k < 1$. Differentiating (B.9.2) on $(0,1)$ shows $d'(k) > 0$, while $d(0) = 3/4$ and $d(1) = 1$. Consequently $3/4 < d < 1$ and $0 < e < d$.

This is the equal-height equation factored in Section B.6.

Let m, r be the triple's middle and high masses. Centering gives

$$r = \frac{a - m(a+c)}{a+b} > 0, \qquad \mathrm{Var}(T) = K^2\{ac + r(a+b)(b-c)\}.$$

Writing $h = a + b$, exact total variance is therefore

$$\boxed{1 = K^2\{ac + de + rh(b-c)\}.} \tag{B.9.3}$$

The full event is simply $T \le Kc$, so $F = 1 - r$.

The selected standardized threshold is positive If $c > 0$ this is immediate. Suppose $c \le 0$. The triple equal-height factorization yields

$$\frac{1}{2} < a \le \frac{3}{4}, \qquad c = 2a - \frac{3}{2}, \qquad H_T = -f_T'(Kc)/(CK^2) = (a+c)^2.$$

Put $\sigma = \phi(z)/(CK^2)$. The unique-representation derivative identity is $mH_T = \sigma$. Since the high mass in (B.9.3) is positive,

$$\sigma < a(a+c) \le \frac{9}{16}. \tag{B.9.4}$$

For the Bernoulli law, its smooth low-point derivative instead gives

$$\sigma = 3d\left(1 - \frac{d^2+e^2}{d+e}\right) = \frac{9(1+k)^2(1-k)(k^2+4k+1)}{4(k^3+3k+2)^2}. \tag{B.9.5}$$

We claim that (B.9.4)–(B.9.5) imply $e > 1/2$. The functions $d(k)$ and $e(k) = kd(k)$ strictly increase on $(0,1)$, by differentiating (B.9.2), and

$$e(11/20) = \frac{10571}{20354} > \frac{1}{2}.$$

If $e \le 1/2$, then $0 < k < 11/20$. The difference between the numerator in (B.9.5) and the value $9/16$ has the sign of

$$kB(k), \qquad B(k) = 8 + 7k - 20k^2 - 26k^3 - 4k^4 - k^5.$$

The polynomial B is strictly concave on $[0, 11/20]$, because $B'' = -40 - 156k - 48k^2 - 20k^3 < 0$, and its two endpoint values are

$$B(0) = 8, \qquad B(11/20) = \frac{3385269}{3200000} > 0.$$

Thus $\sigma > 9/16$, contradicting (B.9.4). Hence $e > 1/2$. Since $c > -1/2$, we have proved in both sign cases that

$$\boxed{z = K(c+e) > 0.} \tag{B.9.6}$$

Elementary uniform bounds For every centered coordinate, its third absolute moment is at least its variance to the power 3/2. With two coordinates and total variance one, this gives $R \geq 1/\sqrt{2} > 7/10$. Also $C \geq 49/120$, so $C^2 > 1/6$ and $CR > 343/1200$. Since $F \leq 1$,

$$\Phi(z) < \frac{857}{1200}.$$

The elementary bounds $\phi(0) > 39/100$ and $\exp(-x) \geq 1 - x$ give

$$\Phi(3/5) > \frac{1}{2} + \frac{39}{100}\left\{\frac{3}{5} - \frac{(3/5)^3}{6}\right\} = \frac{17999}{25000} > \frac{857}{1200}.$$

Consequently $0 < z < 3/5$. We will also use

$$\phi(z) > \phi(3/5) > \frac{33}{100}. \tag{B.9.7}$$

For a rational verification of the last bound, $\pi < 22/7$ implies $\phi(0) > 199/500$; the cubic alternating exponential lower bound gives

$$e^{-9/50} > 1 - \frac{9}{50} + \frac{(9/50)^2}{2} - \frac{(9/50)^3}{6} = \frac{208807}{250000}.$$

Their product is $41552593/125000000 > 33/100$.

Set $N(z) = \Phi(z) - z\phi(z)/3$. The ratio identity is

$$1 - r = N(z) + CK,$$

and the decreasing positive Gaussian density gives $N(z) \geq 1/2 + (2/3)z\phi(z)$. Thus

$$K < \frac{1}{2C} < \frac{5}{4}, \qquad r < \frac{1}{2} - CK - \frac{2}{3}z\phi(z). \tag{B.9.8}$$

Variance contradiction We need only the following support estimates:

$$ac + de < c + e + \frac{1}{4}, \qquad h(b - c) < 3. \tag{B.9.9}$$

For the first, $d < 1$ and $e > 0$ give $de < e$. If $c > 0$, also $ac < c$; if $c \leq 0$, then $ac = c + (1-a)|c| < c + 1/4$, by (B.9.1).

For the second, when $c > 0$ we have $a > 3/4$, and

$$h(b - c) < hb \leq \frac{17 + 7\sqrt{7}}{16} < \frac{9}{4} < 3.$$

Here $(hb)' = h(3-4a)/(2b-1) \leq 0$, so the bound is its value at $a = 3/4$. When $c \leq 0$, substitute $c = 2a - 3/2$. Direct differentiation gives

$$\{h(b - c)\}' = \frac{2(a^2 - b^2 - 6ab + 3b)}{2b - 1} < 0 \qquad (1/2 < a \leq 3/4),$$

because $a^2 < b^2$ and $6ab \geq 3b$. Its upper endpoint at $a = 1/2$ is $(1 + \sqrt{2}/2)^2 = 3/2 + \sqrt{2} < 3$. This proves (B.9.9).

Insert (B.9.8)–(B.9.9) into the exact variance (B.9.3):

$$\begin{aligned} 1 &< Kz + \frac{K^2}{4} + 3K^2 r \\ &< \frac{K^2}{4} + 3K^2(1/2 - CK) + zK\{1 - 2K\phi(z)\}. \end{aligned} \tag{B.9.10}$$

Each term has an elementary upper bound. Equations (B.9.7)–(B.9.8) give

$$\frac{K^2}{4} < \frac{25}{64}, \qquad 3K^2(1/2 - CK) \le \frac{1}{18C^2} < \frac{1}{3}.$$

For the last term in (B.9.10), if its brace is nonpositive the term is already nonpositive. Otherwise, using $z < 3/5$ and $\phi(z) > 33/100$,

$$zK\{1 - 2K\phi(z)\} < \frac{3}{5} \max_{x \ge 0} \left(x - \frac{33}{50} x^2 \right) = \frac{5}{22}.$$

Therefore (B.9.10) yields the impossible strict inequality

$$\boxed{1 < \frac{25}{64} + \frac{1}{3} + \frac{5}{22} = \frac{2009}{2112} < 1.}$$

This completes the exclusion of all unique-representation 220 configurations.

B.10 One triple: the upper shape

We exclude a maximizing pair with a unique middle/high representation and event 221. Use the contact scale:

$$T: \quad K(-a, c, b), \qquad B: \quad K(-d, e), \qquad K = R + \frac{z\phi(z)}{3C}.$$

The selected representation is middle/high, with $z = K(c + e)$. Let r be the triple's high mass, let $p = d/(d + e)$ be the Bernoulli high mass, and put $\varepsilon = rp$. The actual event probability is

$$F = 1 - rp = 1 - \varepsilon. \tag{B.10.1}$$

Contact signs force a positive threshold The triple deletion levels are $(1, 1, 1 - p)$, so the two nonselected smooth minima and the selected middle point satisfy

$$\frac{1}{2} < a < 1 < b, \quad a^2 + b^2 = a + b, \quad f_T(Kc) = f_T(-Ka), \quad -\frac{1}{2} < c < 1. \tag{B.10.2}$$

The Bernoulli low point is a smooth minimum, and its high representation has negative derivative. The representation identity gives

$$d > \frac{1}{2}, \qquad \frac{d^2 + e^2}{d + e} < 1. \tag{B.10.3}$$

To show that $d < 1$, note that the Bernoulli deletion levels are 1 and $1 - r$. Set $t = e/d > 0$ and $P(t) = t^3 + 3t + 2$. Subtracting the two contact heights and using the smooth derivative equation at $-Kd$ gives

$$-r = f_B(Ke) - f_B(-Kd) = CK^3 d^2 P(t) \left\{ d - \frac{3(1 + t)^2}{2P(t)} \right\} < 0. \tag{B.10.4}$$

Since

$$2P(t) - 3(1+t)^2 = (t-1)^2(2t+1) \geq 0,$$

equation (B.10.4) gives $d < 1$. The strict grid geometry for 221 is $d + e > b - c$. Consequently

$$c + e > b - d > 0, \qquad z > 0. \tag{B.10.5}$$

Moreover (B.10.3) and $d > 1/2$ imply

$$\frac{1+t^2}{1+t} < 2, \qquad 0 < t < 1 + \sqrt{2} < \frac{17}{7}. \tag{B.10.6}$$

Universal scalar bounds and exact variance The elementary Gaussian comparisons in Section B.9, apply here. They only use $F \leq 1$, $R \geq 1/\sqrt{2}$, $C \geq 49/120$, and $z > 0$, all valid here. Thus

$$0 < z < \frac{3}{5}, \qquad \phi(z) > \frac{33}{100}, \qquad C^2 > \frac{1}{6}. \tag{B.10.7}$$

Writing $N(z) = \Phi(z) - z\phi(z)/3$, the ratio identity (B.10.1) is

$$1 - \varepsilon = N(z) + CK.$$

Since $N(z) \geq 1/2 + (2/3)z\phi(z)$,

$$\varepsilon < \frac{1}{2} - CK - \frac{2}{3}z\phi(z). \tag{B.10.8}$$

Centering gives the exact variance formula

$$1 = K^2\{ac + de + r(a+b)(b-c)\}. \tag{B.10.9}$$

The equal-height triple contact in (B.10.2) gives the support estimate $(a+b)(b-c) < 3$, proved separately on the two sign branches in Section B.9. Also

$$ac + de < c + e + \delta, \qquad \delta = \frac{1}{4} - (1-d)e. \tag{B.10.10}$$

Indeed, $(a-1)c < 1/4$: it is negative for positive c, and is less than $1/4$ when $-1/2 < c \leq 0$ and $1/2 < a < 1$. Let

$$L = 3(1+t), \qquad p = \frac{1}{1+t}.$$

Then (B.10.1), (B.10.9)–(B.10.10) imply

$$1 < Kz + \delta K^2 + LK^2\varepsilon. \tag{B.10.11}$$

A sharper intercept when the high Bernoulli mass is small For $0 < t \le 1$, simply use $\delta < 1/4$. For $1 \le t < 17/7$, the slope condition (B.10.3) gives

$$\frac{1}{2} < d < d_* := \frac{1+t}{1+t^2} \le 1.$$

The function $d(1-d)$ decreases on $(1/2, 1)$. Hence

$$(1-d)e = td(1-d) > td_*(1-d_*) = \frac{t^2(t^2-1)}{(1+t^2)^2} \ge \frac{5}{12}(t-1). \tag{B.10.12}$$

The last inequality is a direct polynomial comparison. After dividing by $t-1$ when $t > 1$, its numerator is

$$P_0(t) = -5t^4 + 12t^3 + 2t^2 - 5.$$

Its derivative $4t(-5t^2+9t+1)$ changes sign at most once on the positive axis, from positive to negative. Thus its minimum on $[1, 17/7]$ is at an endpoint, and

$$P_0(1) = 4, \qquad P_0(17/7) = \frac{11404}{2401} > 0.$$

The assertion also holds at $t = 1$ directly. We have proved

$$\delta < \bar{\delta}(t) := \begin{cases} 1/4, & 0 < t \le 1, \\ 2/3 - 5t/12, & 1 \le t < 17/7. \end{cases} \tag{B.10.13}$$

Maximization of a cubic variance bound Insert (B.10.8), (B.10.13) into (B.10.11), and put $A = \bar{\delta}(t) + L/2 > 0$. We obtain

$$1 < AK^2 - LCK^3 + zK\left\{1 - \frac{2}{3}LK\phi(z)\right\}. \tag{B.10.14}$$

The first two terms have the exact maximum

$$\max_{K \ge 0}(AK^2 - LCK^3) = \frac{4A^3}{27L^2C^2} < \frac{8A^3}{9L^2}, \tag{B.10.15}$$

using (B.10.7). For the remaining term, a nonpositive brace gives a nonpositive contribution. When the brace is positive, (B.10.7) gives

$$zK\left\{1 - \frac{2}{3}LK\phi(z)\right\} < \frac{3}{5}\max_{x \ge 0}\left(x - \frac{11}{50}Lx^2\right) = \frac{15}{22L}. \tag{B.10.16}$$

It remains to check the two explicit upper functions in (B.10.15)–(B.10.16). Write $x = 1 + t$. For $0 < t \le 1$, we have $1 < x \le 2$, $L = 3x$, and $A = (6x+1)/4$. The bound equals

$$H_0(x) = \frac{(6x+1)^3}{648x^2} + \frac{5}{22x} = \frac{x}{3} + \frac{1}{6} + \frac{101}{396x} + \frac{1}{648x^2}.$$

This function is strictly convex. Its maximum on $[1, 2]$ is at an endpoint, with the larger value at $x = 2$:

$$H_0(x) \le H_0(2) = \frac{27407}{28512} < 1. \tag{B.10.17}$$

For $1 \le t < 17/7$, we have $2 \le x < 24/7$ and $A = (13/12)(x+1)$. The bound is

$$H_1(x) = \frac{2197}{17496}\frac{(x+1)^3}{x^2} + \frac{5}{22x}.$$

Put $c_0 = 2197/17496$. Then

$$H_1(x) = c_0 x + 3c_0 + \frac{3c_0 + 5/22}{x} + \frac{c_0}{x^2},$$

again strictly convex. Its two endpoint values are

$$H_1(2) = \frac{27407}{28512} < 1, \qquad H_1(24/7) = \frac{771397337}{775982592} < 1. \tag{B.10.18}$$

The second endpoint is larger, and its positive deficit is $4585255/775982592$. Convexity bounds the whole interval by those endpoints. Equations (B.10.14)–(B.10.18) force $1 < 1$, a contradiction. This excludes every unique-representation 221 configuration .

B.11 One triple: the middle representation of the triangular event

Consider shape 210 with the middle-triple/low-Bernoulli representation.

Normalize total variance to one and write the dimensionless supports as

$$T: \quad K(-a, c, b), \qquad B: \quad K(-d, e), \qquad K = R + \frac{z\phi(z)}{3C} > 0.$$

Let the triple masses be l, m, r, all positive. The selected threshold is $z = K(c-d)$. The actual deletion levels are

$$T: \quad (1, q_B, 0), \qquad B: \quad (l+m, l), \qquad q_B = \frac{e}{d+e}. \tag{B.11.1}$$

The strict representation geometry is

$$d + e < a + c. \tag{B.11.2}$$

The positive Bernoulli minimum must be at least K The nonselected Bernoulli point Ke is a smooth minimum, so $e > 1/2$. Suppose $e < 1$. Its cubic dual then also has a smooth negative minimum at $-K\alpha$, where

$$\alpha = \frac{1 + \sqrt{1 + 4e - 4e^2}}{2} > 1, \qquad \alpha^2 + e^2 = \alpha + e.$$

The difference of the minimum heights is exactly

$$f_B(-K\alpha) - f_B(Ke) = \frac{CK^3}{2}(e - \alpha)(\alpha^2 + 4\alpha e + e^2) < 0. \tag{B.11.3}$$

But $f_B(Ke) = l$ by (B.11.1). The deletion threshold at the new negative minimum is

$$z + K\alpha = K(c - d + \alpha) > K(-a + e + \alpha) > -Ka,$$

using (B.11.2). Its deletion CDF is therefore at least l. Equation (B.11.3) contradicts the global cubic majorant at this point. Hence

$$\boxed{e \geq 1.} \tag{B.11.4}$$

The nonselected triple minima have levels $1, 0$, so

$$\frac{1}{2} < a < 1 < b, \qquad a^2 + b^2 = a + b. \tag{B.11.5}$$

Equations (B.11.2), (B.11.4)–(B.11.5) give

$$c - d > e - a > 0, \qquad z > 0,$$

and in particular $c > 0$.

Contact height bounds the negative triple minimum Define the dimensionless minimum-height gap

$$D(a) = \frac{1}{2}(b - a)(a^2 + 4ab + b^2), \qquad b = \frac{1 + \sqrt{1 + 4a - 4a^2}}{2}.$$

The levels $1, 0$ in (B.11.1) give the exact identity

$$CK^3 D(a) = 1. \tag{B.11.6}$$

The scalar comparisons in Section B.9 give

$$0 < z < \frac{3}{5}, \qquad \phi(z) > \frac{33}{100}, \qquad C^2 > \frac{1}{6}, \tag{B.11.7}$$

because $R \geq 1/\sqrt{2}$, $F \leq 1$, and $C \geq 49/120$. Also $F = N(z) + CK$, where $N(z) = \Phi(z) - z\phi(z)/3 > 1/2$; hence $CK < 1/2$. Together with (B.11.6),

$$1 = CK^3 D(a) < \frac{D(a)}{8C^2}, \qquad D(a) > 8C^2 > \frac{4}{3}. \tag{B.11.8}$$

Direct differentiation gives

$$D'(a) = -3(2a - 1)(a + b) < 0.$$

At $a = 2/3$, its value is

$$D(2/3) = \frac{71 + 17\sqrt{17}}{108} < \frac{191}{144} < \frac{4}{3},$$

using $\sqrt{17} < 17/4$. Therefore (B.11.8) forces

$$\boxed{a < \frac{2}{3}.} \tag{B.11.9}$$

The selected triple gap is bounded away from one

$$D(a) < D(1/2) = \frac{3}{4} + \frac{\sqrt{2}}{2} < \frac{3}{2}.$$

Equation (B.11.6) consequently gives

$$K^3 > \frac{1000}{681} > \left(\frac{25}{22}\right)^3. \tag{B.11.10}$$

The second strict difference is the positive rational 7375/7251288. Put $\sigma = \phi(z)/(CK^2)$. From (B.11.6)–(B.11.8), (B.11.10),

$$\sigma = \phi(z)D(a)K > \frac{33}{100}\frac{4}{3}\frac{25}{22} = \frac{1}{2}. \tag{B.11.11}$$

The Bernoulli low-representation identity is

$$\sigma = 3e\left\{\frac{d^2+e^2}{d+e} - 1\right\} > 0.$$

Let $u = a + c$. Since the Bernoulli third-moment slope is strictly smaller than its support diameter, and (B.11.2) gives $d + e < u$,

$$\sigma < 3u(u-1).$$

Combining this with (B.11.11), and noting $3(8/7)(1/7) = 24/49 < 1/2$, we obtain

$$\boxed{u > \frac{8}{7}.} \tag{B.11.12}$$

The actual variance Hessian gives the opposite bound Let $\gamma = [-Ka, Kc, Kb]|x|^3$ be the third-moment slope in the centered triple's variance direction. Because $c > 0$, its normalized value is

$$g(a,u) := \frac{\gamma}{K} = b + u - 2a + \frac{2a^3}{(a+b)u}. \tag{B.11.13}$$

On $1/2 < a \le 2/3$, $u \ge 8/7$, this function increases with u and decreases with a. For the first derivative,

$$g_u = 1 - \frac{2a^3}{(a+b)u^2} > 0,$$

since $a + b > 2a$ and $u > a$. For the second, write $h = a + b$. Here $b' < 0$, $h' > 0$, and

$$g_a = b' - 2 + \frac{2}{u}\left(\frac{3a^2}{h} - \frac{a^3h'}{h^2}\right) < -2 + \frac{6a^2}{uh} < -2 + \frac{14}{9} < 0,$$

using $a \le 2/3$, $h > 3/2$, $u \ge 8/7$. Equations (B.11.9), (B.11.12) therefore imply

$$\frac{\gamma}{K} > g(2/3, 8/7) = \frac{499 + 35\sqrt{17}}{504} > \frac{71}{56} > \frac{5}{4}. \tag{B.11.14}$$

The strict lower bound uses only $\sqrt{17} > 4$.

On the other hand, both signs of the centered triple variance direction are admissible, because all three masses are positive. The actual second-order condition derived in Section B.1 is

$$3C\gamma - \frac{15CR + \phi'''(z)}{4} \le 0.$$

Use $R = K - z\phi(z)/(3C)$ and $\phi'''(z) = (3z - z^3)\phi(z)$ to rewrite it as

$$12C\gamma \le 15CK - z(z^2+2)\phi(z) < 15CK,$$

because $z > 0$. Thus $\gamma/K < 5/4$, contradicting (B.11.14). This excludes the middle/low 210 representation .

B.12 One triple: the low representation of the triangular event

For the low-triple/high-Bernoulli representation of 210, divide support coordinates by $K = R + z\phi(z)/(3C) > 0$.

Exact contact reduction Write

$$T: \quad -(a+w), -a, b, \qquad B: \quad -d, e, \qquad w > 0,$$

with triple masses ℓ, m, r. The triple's two nonselected minima have levels $q, 0$, while its selected lowest point has level 1. The Bernoulli masses at $-d, e$ are q, p, respectively, and its deletion levels are $\ell + m, \ell$. The representation geometry is

$$w < d + e < a + b + w. \tag{B.12.1}$$

The smooth triple minima give

$$\frac{1}{2} < a < 1 < b, \quad a^2 + b^2 = a + b, \quad b = \frac{1 + \sqrt{1 + 4a - 4a^2}}{2}.$$

Put

$$h = a + b, \quad D = \frac{1}{2}(b - a)(a^2 + 4ab + b^2), \quad P = w^2(3a - 3/2 + w), \quad \tau = CK^3.$$

The minimum-height gap is $q = \tau D$. Factoring the negative cubic at its smooth minimum gives the other level gap as $p = \tau P$. Therefore

$$\boxed{\tau = \frac{1}{D+P}, \qquad p = \frac{P}{D+P}, \qquad q = \frac{D}{D+P}.} \tag{B.12.2}$$

The Bernoulli negative smooth minimum has $d > 1/2$. Its high-representation derivative gives $k_B = (d^2 + e^2)/(d + e) < 1$. The strict level difference $f_B(e) - f_B(-d) = -m < 0$ also gives $d < 1$, by the exact factorization in Section B.10.

Thus, after choosing a, w, d, every quantity is determined. Write

$$x = d/p = d + e, \quad e = qx, \quad r_B = p^2 + q^2, \quad k_B = r_B x,$$

$$\ell = \frac{d(1 - k_B)}{w(2a - 1 + w)}, \quad m = \frac{b - \ell(h + w)}{h}, \quad r = 1 - \ell - m. \tag{B.12.3}$$

The first formula follows by equating the two unique-representation slope identities:

$$\sigma = 3\ell w(2a - 1 + w) = 3d(1 - k_B), \qquad \sigma = \frac{\phi(z)}{CK^2}.$$

The other two formulas in (B.12.3) are centering and total mass.

It remains to impose the exact Bernoulli height difference. Put

$$M = q^3 + 3p^2 q + 2p^3.$$

Using its smooth derivative at $-d$, the difference is

$$\boxed{m = \tau x^2(3/2 - Mx).} \tag{B.12.4}$$

For fixed a, w, this is a cubic equation in x. In particular the CDF levels and coordinate probabilities have not been replaced by independent optimization parameters.

Variance and two necessary ratio inequalities The dimensionless variances are

$$V_T = ab + \ell w(h + w), \qquad V_B = pqx^2, \qquad V_0 = V_T + V_B.$$

Total variance one gives $K = V_0^{-1/2}$, hence

$$C = \tau V_0^{3/2}, \qquad \boxed{\tau^2 V_0^3 < (227/500)^2.} \tag{B.12.5}$$

The dimensionless Bernoulli third absolute moment is $R_B = pqx^3(p^2 + q^2)$. The triple dual has constant term τB_0, where $B_0 = b^2(2b - 3/2)$, since it vanishes at its positive smooth minimum b. Averaging that dual over the centered triple and using $F - CR = \Phi(z) > 0$ gives the further necessary inequality

$$\boxed{B_0 - \frac{3}{2}V_T - R_B > 0.} \tag{B.12.6}$$

Proof of the compact closed domain Since $V_T > ab$ and $R_B > 0$, equation (B.12.6) implies

$$0 < B_0 - \frac{3}{2}ab = \frac{b}{2}(b^2 - 3a^2).$$

If $a \geq 3/4$, then $b < 5/4$ gives $b^2 < 25/16 < 27/16 \leq 3a^2$, a contradiction. Thus $a < 3/4$. The strictly decreasing height function satisfies

$$1 < D(3/4) < D(a) < D(1/2) < 3/2.$$

Here $D(3/4) = (17 + 7\sqrt{7})/32 > 1$, and $D(1/2) = 3/4 + \sqrt{2}/2 < 3/2$.

The inequalities $d > 1/2$ and $k_B < 1$ give $e/d < 1 + \sqrt{2} < 3$, hence $q < 3/4$. Equation (B.12.2) implies $P > D/3 > 1/3$. If $w \leq 1/2$, then $P = w^2(3a - 3/2 + w) \leq (1/4)(3/4 + 1/2) = 5/16$, a contradiction. Thus $w > 1/2$.

Conversely, (B.12.1) and $d = px < 1$ give $pw < 1$, or $(w - 1)P < D$. If $w \geq 3/2$, then $P > w^3$ and $(w - 1)P \geq (1/2)(27/8) = 27/16 > 3/2 > D$, again a contradiction. Thus $w < 3/2$.

Every candidate therefore lies in the interior of the single closed box

$$\boxed{(a, w, d) \in [1/2, 3/4] \times [1/2, 3/2] \times [1/2, 1].} \tag{B.12.7}$$

Keeping all its faces makes the computational cover larger than needed. The fixed certificate `low210` proves that no point of (B.12.7) satisfies all the necessary contact, probability, variance, and ratio conditions above. Thus this representation cannot occur at a maximizing pair. The enclosure and partition rules are given in Section 8.

B.13 One triple: the lower event

For shape 110, the unique selected representation is the triple's middle point and the Bernoulli's low point.

Two equal-height charts Divide all support points by $K = R + z\phi(z)/(3C) > 0$, and write

$$T: \quad -a, c, b, \qquad B: \quad -d, e.$$

The triple's deletion levels are $(q, q, 0)$, where $q = e/(d+e)$. Thus its nonselected minima and selected point satisfy

$$\frac{1}{2} < a < 1 < b, \quad a^2 + b^2 = a + b, \quad f_T(c) = f_T(-a).$$

Both possible representation signs have exact closed parameter charts $0 \le k \le 1$:

chart	a	c
negative	$1/2 + k/4$	$2a - 3/2$
positive	$3(1+k)^2/[2(k^3+3k+2)]$	$ka.$

(B.13.1)

The two charts share the $c = 0$ boundary and together cover the entire equal-height representation range.

Put

$$b = \frac{1 + \sqrt{1 + 4a - 4a^2}}{2}, \quad h = a + b, \quad u = a + c, \quad v = b - c,$$

$$D = \frac{1}{2}(b-a)(a^2 + 4ab + b^2), \quad H = 3\{a(1-a) + c - c|c|\}.$$

On the negative chart the stable exact formula is $H = u^2$. On the positive chart it is $H = 3[a(1-a) + c(1-c)]$. In either case $0 < H \le 3/2$ at a genuine contact.

The middle mass replaces the singular derivative quotient Let the triple middle mass be the independent parameter m, with $0 < m < 1$. Its other masses and its representation derivative parameter are

$$\ell = \frac{b - mv}{h}, \qquad r = \frac{a - mu}{h}, \qquad \sigma = mH = \frac{\phi(z)}{CK^2} > 0. \tag{B.13.2}$$

The Bernoulli's nonselected positive point is a smooth minimum at height zero. Its entire cubic dual majorizes a nonnegative CDF. If $1/2 < e < 1$, the second smooth minimum is at the negative point

$$-\alpha = -\frac{1 + \sqrt{1 + 4e - 4e^2}}{2}, \qquad \alpha > 1,$$

and its height is strictly below the positive minimum's height zero. This is impossible for the global majorant. Hence $e \ge 1$.

The low Bernoulli representation identity is

$$\sigma = 3e\left\{\frac{d^2 + e^2}{d+e} - 1\right\}. \tag{B.13.3}$$

Since $\sigma < 3/2$ and $e \ge 1$, its third-moment slope is less than $3/2$, so its diameter is less than three. Consequently $d < 2$ and

$$0 < t := d/e < 2.$$

Set $f = (t^2+1)/(t+1)$. Solving the positive quadratic root in (B.13.3) determines the Bernoulli support and probabilities without dividing by H:

$$\boxed{e = \frac{1+\sqrt{1+(4/3)f\sigma}}{2f}, \quad d = te, \quad q = \frac{1}{1+t}, \quad k_B = ef.} \tag{B.13.4}$$

Every actual candidate therefore lies in one of the two closed cubes

$$\boxed{(k, m, t) \in [0,1] \times [0,1] \times [0,2].} \tag{B.13.5}$$

The use of m as a parameter avoids artificial singularities where the representation derivative H vanishes on a chart face.

Exact height, ratio, and normal-density equations The strict 110 representation geometry is $d + e > u$. The minimum-level difference gives

$$\tau := CK^3 = q/D.$$

For the Bernoulli dual, whose positive smooth minimum has height zero, the low-representation contact value is the dimensionless cubic

$$L = e^3(t^3 + 3t + 2) - \frac{3}{2}e^2(1+t)^2.$$

Its deletion-CDF value is $\ell + m = 1 - r$. Thus the exact remaining height equation, cleared of its positive denominator, is

$$\boxed{(1-r)D - qL = 0.} \tag{B.13.6}$$

The dimensionless variances and Bernoulli third absolute moment are

$$V_T = ab - muv, \quad V_B = de, \quad V_0 = V_T + V_B, \quad R_B = de\,k_B.$$

Total variance one gives $K = V_0^{-1/2}$, $z = (c-d)/\sqrt{V_0}$, and

$$C = \frac{qV_0^{3/2}}{D}.$$

The established upper and lower ratio bounds require

$$\boxed{(49/120)^2 D^2 \le q^2 V_0^3 < (227/500)^2 D^2.} \tag{B.13.7}$$

As in the low/high 210 reduction, averaging the triple cubic and using $F - CR = \Phi(z) > 0$ supplies the strict necessary inequality

$$\boxed{b^2(2b - 3/2) - \frac{3}{2}V_T - R_B > 0.} \tag{B.13.8}$$

Finally the definition of σ in (B.13.2) is essential: pure algebraic contacts alone can have ratios below 227/500. Its exact squared form is

$$\boxed{\sigma^2 q^2 V_0 - \frac{D^2}{2\pi}\exp\left(-\frac{(c-d)^2}{V_0}\right) = 0.} \tag{B.13.9}$$

Squaring is harmless as a necessary condition. It only enlarges the candidate set, while all actual quantities are positive. Equation (B.13.9) is evaluated only on boxes with a strictly positive variance enclosure. Other necessary constraints remain available on wider boxes. The fixed certificate `lower110` proves that neither closed chart in (B.13.5) contains a point satisfying all these necessary conditions, including the ratio floor $C = 49/120$. Section 8 gives the enclosure and partition rules. This excludes the unique 110 representation.

B.14 One triple: collision across the full span

The collided 211 event has its two maximal cells at the selected threshold.

Normalize total variance to one. Write R for the sum of third absolute moments, z for the threshold, $\phi_0 = \phi(0)$, and

$$K = R + \frac{z\phi(z)}{3C} > 0. \tag{B.14.1}$$

The global cubic dual is the one in Lemma 6.2. It majorizes the decreasing deletion CDF, touches it at every support point, and has nonpositive derivative at every contact. At a nonselected support point its contact is a smooth local minimum. The two threshold representations use the actual marginal masses.

Collision geometry and the sole smooth triple contact The selected fiber of the collided 211 event is exactly $(triple - low, Bernoulli - high)$ and $(triple - high, Bernoulli - low)$. The Bernoulli diameter therefore equals the triple's total diameter. In units of K, write the supports as

$$T/K: \quad -t, -a, b, \qquad B/K: \quad -d, e, \qquad t > a, \quad b, d, e > 0.$$

Set

$$h = t + b = d + e, \qquad u = t - a > 0, \qquad p = \frac{d}{h}, \quad q = \frac{e}{h}, \qquad \tau = CK^3. \tag{B.14.2}$$

The Bernoulli probabilities at its low and high atoms are respectively q, p. The triple deletion levels are exactly $(1, q, q)$.

The middle triple atom is nonselected and is therefore a smooth minimum. It must be negative. If it were a positive minimum, the cubic dual would have positive derivative at the still larger triple contact; zero cannot be a local minimum because its quadratic coefficient is $-3CK/2 < 0$. The negative minimum can thus be written $-Ka$, with $a > 1/2$.

After dividing its dual values by τ and subtracting a constant, the triple dual is

$$g(x) = |x|^3 - \frac{3}{2}x^2 - 3a(1-a)x. \tag{B.14.3}$$

The equality of the two contact levels at $-a, b$ gives $g(b) = g(-a)$. Put $k = b/a > 0$ and $P_0 = k^3 + 3k + 2$. Direct factorization gives

$$a = \frac{3(1+k)^2}{2P_0}. \tag{B.14.4}$$

Moreover,

$$\frac{g'(b)}{3} = a\{a(1+k^2) - (1+k)\} = \frac{a(1+k)(k-1)(k^2+4k+1)}{2P_0}.$$

The contact derivative is nonpositive, so $0 < k \le 1$. The identities

$$2P_0 - 3(1+k)^2 = (k-1)^2(2k+1),$$

$$a - \frac{3}{4} = \frac{3k(1+2k-k^2)}{4P_0} > 0$$

therefore prove

$$\boxed{\frac{3}{4} < a \le 1, \qquad 0 < b \le 1.} \tag{B.14.5}$$

The boundary $a = b = 1$ is retained. A selected contact may have zero derivative at a collision, so it cannot be discarded as a smooth-representation contradiction.

The remaining triple contact level difference is $1 - q = p$. Factoring the negative cubic at its minimum gives

$$\boxed{p = \tau u^2(t + 2a - 3/2).} \tag{B.14.6}$$

Every factor is positive, including $t + 2a - 3/2 > t$ by (B.14.5).

Combine the first and second variance derivatives Let η be the centered signed three-point direction whose variance derivative is one, as in Proposition 6.4, The support and threshold stay fixed when the triple law is changed to $P_T + s\eta$; both signs of sufficiently small s are admissible. Since the support stays fixed, the collided closed event presents no differentiability issue.

Let $\gamma = \int |x|^3 \eta(dx)$ and let L be the corresponding CDF derivative. The levels $(1, q, q)$ and gaps $Ku, K(a+b)$ give

$$\boxed{L = \frac{p}{K^2 uh} > 0.} \tag{B.14.7}$$

Along this direction the signed excess at price C is exactly

$$J(s) = F + sL - \Phi\left(\frac{z}{\sqrt{1+s}}\right) - C(R + s\gamma)(1+s)^{-3/2}.$$

At the ratio maximum, $J'(0) = 0$ and $J''(0) \le 0$. Differentiation gives

$$C\gamma = L + \frac{3}{2}CR + \frac{1}{2}z\phi(z),$$

$$3C\gamma - \frac{15CR + \phi'''(z)}{4} \le 0.$$

Using $\phi'''(z) = (3z - z^3)\phi(z)$ and eliminating γ yields the exact necessary condition

$$\boxed{12L + 3CR + (z^3 + 3z)\phi(z) \le 0.} \tag{B.14.8}$$

In particular $z = -s < 0$, since L, C, R are positive. Equation (B.14.1) now reads $CR = CK + s\phi(s)/3$, so (B.14.8) becomes

$$12L + 3CK \le s(s^2+2)\phi(s) \le 2s\phi_0. \tag{B.14.9}$$

The last inequality is elementary: the derivative of $(s^2+2)\phi(s)$ is $-s^3\phi(s)$, so that function decreases on $[0, \infty)$ from $2\phi_0$.

Geometry makes the positive CDF derivative too large The threshold is the high-triple/low-Bernoulli sum, hence

$$s = K(d - b) < Kt. \tag{B.14.10}$$

Here $d < h = t + b$ because the Bernoulli law is nondegenerate. Since $L > 0$, (B.14.9) gives $3CK < 2s\phi_0$. Combining this with (B.14.10) and $C \geq 49/120 > \phi_0$ gives

$$\boxed{t > \frac{s}{K} > \frac{3C}{2\phi_0} > \frac{3}{2}.} \tag{B.14.11}$$

Substitute the exact probability identity (B.14.6) into (B.14.7). Equations (B.14.5), (B.14.10), and (B.14.11) imply

$$\begin{aligned} L &= CK\,\frac{(t-a)(t+2a-3/2)}{t+b} \\ &> CK\,\frac{t(t-1)}{t+1} \\ &> Cs\,\frac{t-1}{t+1} \\ &> \frac{Cs}{5}. \end{aligned} \tag{B.14.12}$$

The first strict comparison uses $a \leq 1$, $b \leq 1$, and $2a - 3/2 > 0$; the final one uses $t > 3/2$. Inserting (B.14.12) into (B.14.9) yields

$$\frac{12Cs}{5} < 2s\phi_0, \qquad \text{hence} \qquad C < \frac{5}{6}\phi_0.$$

The identity (B.14.8) is also a general necessary condition for any genuine triple at a full-law ratio maximum, with its actual unit-variance CDF derivative L. In particular $L \geq 0$ forces a negative selected threshold. Its successful use here depends on the exact collided 211 geometry and the actual Bernoulli probability in (B.14.6)–(B.14.7).

B.15 One triple: collision across the upper gap

Normalize variance to one and write

$$K = R + z\phi(z)/(3C) > 0, \qquad \tau = CK^3, \qquad \sigma = \phi(z)/(CK^2) > 0.$$

All following support variables are divided by K.

The exact compact support and probability charts The selected fiber is $(triple - middle, Bernoulli - high)$ and $(triple - high, Bernoulli - low)$. Write

$$T:\ -a, c, b, \qquad B:\ -d, e, \qquad a, b, d, e > 0, \quad -a < c < b.$$

The collision forces

$$h = b - c = d + e, \qquad p = d/h, \quad q = e/h, \tag{B.15.1}$$

where q, p are the low and high Bernoulli probabilities. The triple deletion levels are $(1, 1, q)$; its low point is the sole nonselected contact and is a smooth negative minimum. Thus $a > 1/2$ and, up to a constant, the divided triple dual is

$$g(x) = |x|^3 - \frac{3}{2}x^2 - 3a(1-a)x. \tag{B.15.2}$$

The middle contact satisfies $g(c) = g(-a)$ and $g'(c) \le 0$. The two equal-height charts in Section B.13 apply:

	a	c
negative, $0 \le k \le 1$	$1/2 + k/4$	$(k-1)/2$
positive, $0 \le k \le 1$	$3(1+k)^2/[2(k^3+3k+2)]$	$ka.$

(B.15.3)

At a genuine configuration $a < 1$: the common endpoint $a = c = 1$ would put the still larger contact b to the right of the positive minimum. The highest triple contact has $g'(b) \le 0$, so

$$0 < b \le B(a) := \frac{1 + \sqrt{1 + 4a(1-a)}}{2}.$$

Use $b = wB(a)$, $0 \le w \le 1$, retaining the unnecessary $b \le c$ region and rejecting it by the strict gap condition $h > 0$.

Let the triple middle mass m be an independent parameter. Its low and high masses, and the useful contact derivatives, are

$$A = a + b, \quad u = a + c, \qquad \ell = \frac{b - mh}{A}, \qquad r = \frac{a - mu}{A}, \tag{B.15.4}$$

$$H_c = -g'(c) = 3\{a(1-a) + c - c|c|\}, \qquad H_b = -g'(b) = 3\{a(1-a) + b - b^2\} \ge 0.$$

On the negative chart use the exact identity $H_c = u^2$. Averaging the dual derivative gives

$$\boxed{\sigma = mH_c + rH_b.} \tag{B.15.5}$$

Thus every candidate lies in one of the two full closed four-cubes

$$\boxed{(k, w, p, m) \in [0, 1]^4.} \tag{B.15.6}$$

The denominator A is bounded below by $1/2$ throughout these cubes, including extraneous boundary points. No small-gap or probability division is needed in this parameterization.

Actual contact and collision-preserving support equations Put

$$U = g(-a) - g(b) = a^2(3/2 - 2a) - b^3 + \frac{3}{2}b^2 + 3a(1-a)b.$$

The triple contact levels give

$$\boxed{p = \tau U, \qquad U > 0.} \tag{B.15.7}$$

For the Bernoulli law, set $k_B = h(p^2 + q^2)$. Its divided derivative at the high support point is

$$f'_B(Ke)/(CK^2) = 3e(k_B - 1) - \sigma.$$

$$m f_T'(Kc) = p f_B'(Ke).$$

Substitute (B.15.5) and the displayed derivative. The exact necessary collision equation is

$$\boxed{qmH_c - prH_b - 3pqh(1 - k_B) = 0.} \tag{B.15.8}$$

This support motion preserves the collision and uses the actual marginal masses.

The Bernoulli deletion levels are $(1, \ell + m) = (1, 1 - r)$. Its signed second moment is $M_B = de(e - d)/h$, and its divided derivative everywhere is $3[x|x| - x - M_B] - \sigma$. Integrating between $-d, e$ gives its exact value difference

$$f_B(Ke) - f_B(-Kd) = \tau\{J - \sigma h\} = -r,$$

$$J = (e - d)\{(e - d)^2 - 3h/2\}.$$

Using (B.15.7) clears the denominator:

$$\boxed{p(J - \sigma h) + rU = 0.} \tag{B.15.9}$$

Variance, normal comparisons, and exact bounds Use the following dimensionless moments:

$$V_T = ab - muh, \qquad V_B = de, \qquad V_0 = V_T + V_B,$$

$$R_T = \ell a^3 + m|c|^3 + rb^3, \qquad R_B = de\, k_B, \qquad R_0 = R_T + R_B.$$

Normalization and (B.15.7) give

$$K = V_0^{-1/2}, \qquad z = \frac{o}{\sqrt{V_0}}, \quad o = c + e, \qquad C = \frac{pV_0^{3/2}}{U}. \tag{B.15.10}$$

The definition of K and (B.15.5) imply the further necessary polynomial identity

$$\boxed{3(V_0 - R_0) - \sigma o = 0.} \tag{B.15.11}$$

The bounds $1/2 < K < 8/5$ in (6.1) also give the strict necessary bound

$$\boxed{25/64 < V_0 < 4.} \tag{B.15.11a}$$

This holds before any contact division and removes zero-variance limits.

The strict ratio interval gives

$$(49/120)^2 U^2 \le p^2 V_0^3 < (227/500)^2 U^2. \tag{B.15.12}$$

These denominator-cleared tests are available even when a box's variance enclosure crosses zero. On boxes with a strictly positive variance enclosure, also impose the unsquared density and actual CDF equations

$$\boxed{\sigma p\sqrt{V_0} - U\phi(z) = 0,} \tag{B.15.13}$$

$$\boxed{(1-pr)U - pR_0 - U\Phi(z) = 0.} \tag{B.15.14}$$

The true event probability is $F = 1 - pr$. A simple necessary CDF positivity test without transcendental evaluation follows by averaging the triple dual, which equals one at $-a$:

$$\boxed{U - p\{a^2(3/2 - 2a) + (3/2)V_T + R_B\} > 0.} \tag{B.15.15}$$

At the maximizing pair, each marginal variance lies strictly between 13/200 and 187/200 of the total variance, by (6.2). Hence

$$\boxed{V_T > \frac{13}{200}V_0, \qquad V_B > \frac{13}{200}V_0.} \tag{B.15.16}$$

Finally, the triple's actual unit-variance CDF derivative is $L = -p/(K^2hA)$. The first/second variance identity (B.14.8) of Section B.14 applies without changing the closed event. Substitute (B.15.10), (B.15.13), and multiply by the positive quantity $UhAV_0/p$. The necessary second-order condition becomes the polynomial inequality

$$\boxed{-12V_0^2U + 3R_0hAV_0 + \sigma hAo(o^2 + 3V_0) \le 0.} \tag{B.15.17}$$

The fixed certificate `collision221` proves that the two closed support charts above contain no point satisfying their contact equations, probability conditions, and necessary inequalities. Its strict-floor checks include $C = 49/120$, as described in Section 8. Thus a collision across the upper gap cannot occur at a maximizing pair.

B.16 One triple: collision across the lower gap

Use the contact scale: the support coordinates below are divided by K.

$$K = R + z\phi(z)/(3C), \quad \tau = CK^3, \quad \sigma = \phi(z)/(CK^2).$$

The bounds in (6.1) give

$$\boxed{1/2 < K < 8/5, \qquad 1/20 < \tau < 2.} \tag{B.16.1}$$

The lower bound for τ uses $C > 2/5$; the upper bound uses $C < 227/500$ and $(227/500)(8/5)^3 < 2$.

Compact support charts from the zero minimum Divide all support points by K and write

$$T: \ -t, c, b, \qquad B: \ -d, e, \quad t, b, d, e > 0, \quad -t < c < b.$$

The selected fiber is $(triple - low, Bernoulli - high)$ and $(triple - middle, Bernoulli - low)$, so

$$h = t + c = d + e, \qquad p = d/h, \quad q = e/h. \tag{B.16.2}$$

The low/high Bernoulli probabilities are q, p; the triple deletion levels are $(1, q, 0)$. Its high contact is nonselected and is a smooth positive minimum of height zero. The whole dual is nonnegative, so $b \ge 1$. Indeed a positive minimum with $1/2 < b < 1$ has a strictly lower negative local minimum, contradicting the nonnegative global majorant.

The triple dual divided by τ is exactly

$$g_b(x) = |x|^3 - \frac{3}{2}x^2 - 3b(b-1)x + B_0, \qquad B_0 = b^2(2b - 3/2). \tag{B.16.3}$$

It vanishes at b. Put

$$G = g_b(-t) = t^3 - \frac{3}{2}t^2 + 3b(b-1)t + B_0, \qquad G_c = g_b(c).$$

The contact values give

$$\boxed{\tau = 1/G, \quad q = G_c/G, \quad p = 1 - q, \quad 1/2 < G < 20.} \tag{B.16.4}$$

The height bounds in (B.16.4) follow from (B.16.1). They also bound the supports. If $b \geq 5/2$, then $g_b(-t)$ increases for $t \geq 0$, since its derivative is $3[(t - 1/2)^2 + b(b-1) - 1/4] > 0$. Hence $G \geq B_0 \geq 175/8 > 20$, impossible. Since $b \geq 1$,

$$G \geq t^3 - \frac{3}{2}t^2 + \frac{1}{2}.$$

For $t \geq 7/2$, the right side is increasing and is at least 25, also impossible. Thus

$$1 \leq b < 5/2, \qquad 0 < t < 7/2. \tag{B.16.5}$$

Use independent parameters $(s_b, s_t, w, m) \in [0,1]^4$ and maps

$$b = 1 + \frac{3}{2}s_b, \qquad t = \frac{7}{2}s_t,$$

$$c = -t(1-w) \quad \text{on the negative chart}, \qquad c = wb \quad \text{on the positive chart.} \tag{B.16.6}$$

Here m is the triple middle mass. Both closed cubes are retained, including all extra boundary points and the shared $c = 0$ face.

Let $A = t + b$, $v = b - c$. Centering determines

$$\ell = \frac{b - mv}{A}, \qquad r = \frac{t - mh}{A}. \tag{B.16.7}$$

The denominator $A \geq 1$ throughout the whole parameter domain.

Exact collision and Bernoulli contact equations Define the nonnegative selected-contact slopes

$$H_\ell = 3\{t^2 - t + b(b-1)\}, \qquad H_m = 3\{b(b-1) + c - c|c|\}.$$

Averaging the triple dual derivative gives

$$\boxed{\sigma = \ell H_\ell + m H_m > 0.} \tag{B.16.8}$$

Set $d = ph$, $e = qh$, and $k_B = h(p^2 + q^2)$. Move the raw low triple point and the raw high Bernoulli point by opposite amounts, holding all other raw support points and all probabilities fixed. Both selected raw sums and the closed event stay fixed. Recenter the same two marginals and move the threshold by their total mean shift. The two-sided ratio derivative gives

$$\ell f_T'(-Kt) = p f_B'(Ke).$$

Using the Bernoulli derivative $f_B'(Ke)/(CK^2) = 3e(k_B - 1) - \sigma$ and (B.16.8), this is

$$\boxed{q\ell H_\ell - pmH_m - 3pqh(1 - k_B) = 0.} \tag{B.16.9}$$

The Bernoulli deletion levels are $(\ell + m, \ell)$. Integrating its divided derivative, as in the collided 221 reduction, gives

$$\tau\{J - \sigma h\} = -m, \qquad J = (e - d)\{(e - d)^2 - 3h/2\}.$$

Since $\tau = 1/G$, its denominator-cleared contact equation is

$$\boxed{J - \sigma h + mG = 0.} \tag{B.16.10}$$

Normalization, actual CDF, and second-order constraints Write the dimensionless variance and sum of third absolute moments as

$$V_T = tb - mhv, \quad V_B = de, \quad V_0 = V_T + V_B,$$

$$R_T = \ell t^3 + m|c|^3 + rb^3, \quad R_B = de\, k_B, \quad R_0 = R_T + R_B.$$

Then

$$K = V_0^{-1/2}, \qquad C = V_0^{3/2}/G, \qquad o = c - d, \qquad z = o/\sqrt{V_0}.$$

The definition of K gives the necessary polynomial identity

$$\boxed{3(V_0 - R_0) - \sigma o = 0.} \tag{B.16.11}$$

The compact bounds and ratio interval imply

$$25/64 < V_0 < 4, \qquad (49/120)^2 G^2 \le V_0^3 < (227/500)^2 G^2. \tag{B.16.12}$$

At the maximizing pair, each marginal variance lies strictly between 13/200 and 187/200 of the total variance, by (6.2). Hence

$$V_T > \frac{13}{200} V_0, \qquad V_B > \frac{13}{200} V_0. \tag{B.16.13}$$

The actual event probability is $F = \ell + mq$. The density and CDF equations are

$$\boxed{\sigma\sqrt{V_0} - G\phi(z) = 0,} \tag{B.16.14}$$

$$\boxed{G\{\ell + mq - \Phi(z)\} - R_0 = 0.} \tag{B.16.15}$$

Averaging the zero-minimum triple dual (B.16.3) also supplies the necessary positivity test

$$\boxed{B_0 - \frac{3}{2} V_T - R_B > 0.} \tag{B.16.16}$$

For the triple's centered unit-variance direction, the actual CDF derivative at the fixed closed event is

$$L = \frac{(pA-h)V_0}{Ahv}.$$

Combine its first and second variance derivatives and use (B.16.14). After multiplication by the positive factor $GAhvV_0$, the necessary second-order condition is

$$\boxed{12(pA-h)GV_0^2 + 3R_0AhvV_0 + \sigma Ahv\, o(o^2+3V_0) \le 0.} \tag{B.16.17}$$

B.17 Lower-gap collision: elimination of the middle mass

Continue with the supports divided by K in Section B.16.

A positive affine coefficient Retain the previous notation

$$A = t+b, \quad h = t+c, \quad v = b-c, \quad \ell = (b-mv)/A, \quad r = (t-mh)/A,$$

with $H_\ell, H_m \ge 0$, $\sigma = \ell H_\ell + mH_m > 0$, and Bernoulli probabilities $q, p > 0$. The collision support equation is

$$q\ell H_\ell - pmH_m - 3pqh\{1-h(p^2+q^2)\} = 0.$$

Substitution of ℓ makes it affine in m, and gives exactly

$$\boxed{m = \frac{qbH_\ell - 3Apqh\{1-h(p^2+q^2)\}}{qvH_\ell + ApH_m}.} \tag{B.17.1}$$

The denominator $L = qvH_\ell + ApH_m$ is strictly positive: every coefficient is positive, both slopes are nonnegative, and $\sigma > 0$ excludes their simultaneous vanishing. Thus there is no additional zero-denominator collision case to retain.

Let $N = G_c$, $P = G - G_c$, $W = P^2 + N^2$, so $q = N/G$, $p = P/G$. Define

$$\begin{aligned} \mathcal{D} &= G^3(NvH_\ell + APH_m) = G^4 L, \\ M &= NbH_\ell G^3 - 3APNhG^2 + 3APNh^2W. \end{aligned} \tag{B.17.2}$$

Then $m = M/\mathcal{D}$. The four natural mass conditions become polynomial:

$$M > 0, \quad \mathcal{D} - M > 0, \quad b\mathcal{D} - vM > 0, \quad t\mathcal{D} - hM > 0. \tag{B.17.3}$$

A uniform positive denominator bound The canonical mixture-width theorem applies to this triple and its Bernoulli partner without an optimality assumption. Consequently every violating candidate in this system must satisfy

$$\min\{bh, tv\} > V/3 > 25/192. \tag{B.17.4}$$

The Bernoulli variance bound gives

$$pqh^2 > \frac{13}{200}V > \frac{13}{512}.$$

As $h < A = t+b < 6$, this yields $p, q > pq > 13/18432$. Moreover $A \ge 1$.

By (B.3.2), $\sigma > 9/200 > 1/80$. Since $\ell, m < 1$, one has $H_\ell + H_m \ge \sigma$, and hence

$$L > \frac{13}{18432}\frac{25}{672}\frac{1}{80}.$$

Multiplication by $G^4 > 1/16$ proves the explicit bound

$$\boxed{\mathcal{D} > \frac{65}{3170893824} > \frac{1}{50000000}.} \tag{B.17.5}$$

The final comparison has exact numerator slack $65\,50000000 - 3170893824 = 79106176$. Thus the eliminated mass is a smooth rational function on an open neighborhood of every candidate in the compact support charts. In interval arithmetic, (B.17.5) is only a necessary test; division is permitted only if the direct enclosure of $\mathcal{D}$ is positive on the whole current box.

The two three-parameter support charts Delete the independent mass variable from the two charts. For $(s_b, s_t, w) \in [0,1]^3$, set

$$b = 1 + 3s_b/2, \qquad t = 7s_t/2,$$

and take $c = -t(1-w)$ on the negative chart or $c = wb$ on the positive chart. Compute m by (B.17.1) or (B.17.2) and retain all genuine-support, probability, height, slope, ratio, variance, width and global-dual constraints. Every candidate has a preimage in one of these two closed unit cubes, including along the shared boundary $c = 0$.

Four explicit equations without an independent mass variable The following formulas permit denominator-free evaluation before any normal function. Define

$$L_* = b\mathcal{D} - vM, \quad H_* = t\mathcal{D} - hM, \quad S_* = L_*H_\ell + AMH_m, \quad O = cN - tP,$$
$$V_* = (tb\mathcal{D} - Mhv)G^2 + \mathcal{D}PNh^2,$$
$$R_* = (L_*t^3 + AM|c|^3 + H_*b^3)G^4 + A\mathcal{D}PNh^3W, \quad F_* = GL_* + AMN.$$

Then the actual quantities are

$$V = \frac{V_*}{\mathcal{D}G^2}, \quad R = \frac{R_*}{A\mathcal{D}G^4}, \quad \sigma = \frac{S_*}{A\mathcal{D}}, \quad F = \frac{F_*}{A\mathcal{D}G}, \quad z = O\sqrt{\mathcal{D}/V_*}. \tag{B.17.6}$$

The retained Bernoulli height, scale, squared density and CDF equations are, respectively,

$$A\mathcal{D}h(N-P)\{h^2(N-P)^2 - 3hG^2/2\} - S_*hG^3 + AMG^4 = 0,$$
$$3(AG^2V_* - R_*) - S_*OG^3 = 0,$$
$$S_*^2V_* - A^2\mathcal{D}^3G^4\phi(z)^2 = 0,$$
$$G^4\{F_* - A\mathcal{D}G\Phi(z)\} - R_* = 0. \tag{B.17.7}$$

Each clearing factor is positive at every candidate. In particular the two equations preceding the normal functions are exact polynomial equations in the three support parameters on either sign chart.

The ratio test is $(49/120)^2\mathcal{D}^3G^8 \le V_*^3 < (227/500)^2\mathcal{D}^3G^8$. The added width tests are $3bh\mathcal{D}G^2 > V_*$ and $3tv\mathcal{D}G^2 > V_*$.

B.18 Lower-gap collision: direct necessary equations

Use the contact scale of Section B.16.

Independent law coordinates Write the centered triple supports $(-t, c, b)$, set $A = t + b$, $h = t + c$, $v = b - c$, and use its variance $W = V_T$ as the mass coordinate. Exact centering gives

$$\ell = \frac{W + bc}{Ah}, \qquad m = \frac{tb - W}{hv}, \qquad r = \frac{W - tc}{Av}. \tag{B.18.1}$$

These formulas are intersected with normalization, the direct second moment, and the usual middle-mass expression $W = tb - mhv$. For the Bernoulli, low/high probabilities are q, p, supports are $(-d, e)$, and $d = ph$, $e = qh$. Its third absolute moment is evaluated as

$$V_B = de, \qquad R_B = V_B \left(h - \frac{2V_B}{h} \right), \qquad h/2 \le h - 2V_B/h \le h. \tag{B.18.2}$$

$$R_* = \frac{tb(t^2 + b^2)}{A}, \qquad M_* = \frac{tb(b - t)}{A},$$

the exact moments are

$$R_T = R_* + \gamma(W - tb), \qquad \mathbb{E}[T|T|] = M_* + \eta(W - tb). \tag{B.18.3}$$

The direct three-term sum of third absolute moments is also intersected. The bounds $A/2 \le \gamma \le 2A$ and $-1 \le \eta \le 1$ are the established divided- difference bounds. All adjacent and outer gap divisions are performed separately with their positive candidate bounds; a product enclosure that straddles zero is never inverted.

Dual contacts and support pairings The triple majorant is evaluated using the factorization

$$g_b(x) = (x - b)^2(x + 2b - 3/2) + 2(-x)_+^3. \tag{B.18.4}$$

Thus $G = g_b(-t)$, $q = g_b(c)/G$, and $p = 1 - q$. A differently arranged expanded value of G is also intersected. The high minimum gives

$$\sigma = 3\{b(b - 1) - \mathbb{E}[T|T|]\} = \ell H_t + mH_m,$$

where $H_t = 3(t^2 - t + b(b - 1))$ and $H_m = 3(b(b - 1) + c - c|c|)$ are nonnegative. Put

$$Q = 3pqh\{1 - h(p^2 + q^2)\}.$$

The two collision-preserving derivative pairings are evaluated as

$$\ell H_t = p\sigma + Q, \qquad mH_m = q\sigma - Q. \tag{B.18.5}$$

The Bernoulli height equation is

$$mG = \sigma h - (e - d)\{(e - d)^2 - 3h/2\}. \tag{B.18.6}$$

Normalization and inverse identities The independently stored values satisfy the contact equations

$$C = V^{3/2}/G, \quad D = R/G = F - \Phi(z), \quad \sigma\sqrt{V} = G\phi(z), \quad 3(V-R) = \sigma o, \quad o = c-d = e-t = z\sqrt{V}. \tag{B.18.7}$$

They are evaluated both forward and backward, including $V = (CG)^{2/3}$, $\sqrt{V} = G\phi/\sigma$, and the exact inverse density

$$z^2 = -\log(2\pi\phi(z)^2). \tag{B.18.8}$$

Equation (B.18.8) has a known negative square-root branch on the negative chart. On the positive chart a branch is used only when the current candidate interval has proved its sign; otherwise its symmetric hull is retained. A nonnegative interval for z^2 is intersected before taking a square root. The CDF is evaluated through the error function.

The following bounds hold at every candidate: $49/120 \le C \le 227/500$, $49/(120\sqrt{2}) \le D \le 11/20$, $-5/4 \le z \le 3/5$, $\phi \ge 9/50$, $V \ge 2401/5476$ (or $2401/4356$ on the negative chart), and $V \le 4$. It also uses the established coordinate variance fraction bounds $13V/200 \le V_i \le 187V/200$ and Hölder third-moment bounds. References are Section B.3, Section B.4, and Section B.17. Strict bounds are relaxed to closed bounds for enclosure.

The canonical mixture-width inequalities $3bh > V$, $3tv > V$ are also used backward to obtain necessary bounds on each positive factor. The elementary Bernoulli bound $h \ge 2\sqrt{V_B}$ is included. Initial gap and probability floors are exactly the proved floors: $h \ge \sqrt{13/128}$, $v \ge 25/672$, $p, q \ge 13/18432$, and $t \ge 13/1280$. The fixed certificate `collision210` proves that no point of either closed support chart satisfies the necessary conditions in Sections B.16–B.18. The enclosure and partition rules are given in Section 8. This excludes the lower-gap collision.

B.19 Two triples: asymmetric events with a unique representation

Normalize total variance to one and use the dimensionless coordinates of Section B.6. Thus the two coordinate supports are $K(-a_i, c_i, b_i)$, with

$$\frac{1}{2} < a_i < 1 < b_i, \qquad a_i^2 + b_i^2 = a_i + b_i, \qquad h_i = a_i + b_i,$$

and the two nonselected minima have height difference

$$CK^3 D(a_i), \qquad D(a) = \frac{1}{2}(b-a)(a^2 + 4ab + b^2).$$

The decreasing deletion CDF gives these same strict inequalities for a_i, b_i in 221 and 320, because the two nonselected contact levels are distinct. Along the curve $a^2 + b^2 = a + b$,

$$D'(a) = -3(2a-1)(a+b) < 0, \qquad h'(a) = \frac{2(b-a)}{2b-1} > 0. \tag{B.19.1}$$

For shape 221, the first coordinate's three deletion-CDF values are q_2, q_2, l_2, where l_2 is the lowest mass, m_2 the middle mass, and $q_2 = l_2 + m_2$. The second coordinate's three values are $1, q_1, 0$. Therefore

$$CK^3 D(a_1) = m_2 < 1 = CK^3 D(a_2).$$

Equation (B.19.1) forces $a_1 > a_2$ and hence $h_1 > h_2$.

On the other hand, the selected representation is middle-middle. The high-low corner is included, and the low-high corner is excluded. The unique threshold representation makes both relevant comparisons strict:

$$b_1 - a_2 < c_1 + c_2 < -a_1 + b_2.$$

With $u_i = a_i + c_i$, these are exactly

$$h_1 < u_1 + u_2 < h_2,$$

contradicting $h_1 > h_2$. Interchanging the two coordinates transposes 221 to 320, so that shape is excluded as well.

B.20 Two triples: triangular event with a unique representation

Contact geometry and notation Normalize total variance to one. Write the discrepancy as $D = CR$, the selected threshold as z, and put

$$K = \frac{2A}{3C} = R + \frac{z\phi(z)}{3C}.$$

The full-law cubic contacts from Proposition 6.4 give supports $K(-a_i, c_i, b_i)$, with the selected point in the middle of each coordinate. The two nonselected points are smooth minima. In shape 321 their deletion-CDF values are 1 and the other coordinate's lowest mass, so the minimum-height difference is positive. Consequently

$$\frac{1}{2} < a_i < 1 < b_i, \qquad a_i^2 + b_i^2 = a_i + b_i =: h_i \le 2. \tag{B.20.1}$$

Define $u_i = a_i + c_i$ and $v_i = b_i - c_i$. The two cross corners are strictly below the selected middle-middle sum. Thus

$$v_1 < u_2, \qquad v_2 < u_1, \qquad c_1 + c_2 > \max\{b_1 - a_2, b_2 - a_1\} > 0. \tag{B.20.2}$$

In particular $z = K(c_1 + c_2) > 0$ and $K > 0$. Let l_i, m_i, r_i denote the lowest, middle, and highest masses, and put $p_i = m_i + r_i$. The actual selected event probability and its complement are

$$F = 1 - p_1 p_2 + m_1 m_2, \qquad P_{\text{out}} = r_1 p_2 + r_2 p_1 - r_1 r_2. \tag{B.20.3}$$

A divided-difference inequality on the actual triangle For a pair a, b satisfying (B.20.1), let

$$d_a(c) = [-a, c, b]|x|^3.$$

The divided-difference convention is $[x_0, x_1, x_2]x^2 = 1$. Direct interpolation gives

$$d_a(c) = \begin{cases} b + c - a + \dfrac{2a^3}{(a+b)(a+c)}, & c \ge 0, \\ a - b - c + \dfrac{2b^3}{(a+b)(b-c)}, & c \le 0. \end{cases} \tag{B.20.4}$$

This is a strictly convex function of c on $(-a, b)$, with

$$d_a(0) = 1, \qquad d_a'(0) = \frac{b-a}{a+b} > 0. \tag{B.20.5}$$

We prove that (B.20.2) implies

$$\boxed{d_{a_1}(c_1) + d_{a_2}(c_2) > 2.} \tag{B.20.6}$$

Relabel so $a_1 \le a_2$. Then $b_1 \ge b_2$, $h_1 \le h_2$, and the slopes $k_i = (b_i - a_i)/h_i$ satisfy $k_1 \ge k_2$. Put $z_0 = b_2 - a_1 > 0$. We need one derivative comparison:

$$d_{a_2}'(z_0) \ge k_1. \tag{B.20.7}$$

By (B.20.4), this is equivalent to

$$a_2^3 h_1 \le a_1 h_2 (h_2 - a_1)^2.$$

Use $h_1 \le 2$ and $h_2 \ge a_2 + 1$. With $a = a_1$, $b = a_2$, it is sufficient that

$$a(b+1)(b+1-a)^2 \ge 2b^3, \qquad 1/2 \le a \le b \le 1. \tag{B.20.8}$$

For fixed b, the function $a(b+1-a)^2$ has at most one interior critical point, a maximum. Its minimum on $[1/2, b]$ is at an endpoint. At $a = b$, (B.20.8) follows from $b + 1 \ge 2b^2$. At $a = 1/2$, it follows from

$$(b+1)(b+1/2)^2 - 4b^3 = \frac{1}{2} + (1-b)(3b^2 + b - 1/4) > 0.$$

This proves (B.20.7).

If both middle points are nonnegative, (B.20.6) follows immediately from (B.20.5), since their sum is positive. If $c_1 < 0$, then (B.20.2) gives $c_2 > z_0 - c_1$. Along s from c_1 to zero, strict convexity and (B.20.7) give

$$\frac{d}{ds}\{d_{a_1}(s) + d_{a_2}(z_0 - s)\} \le k_1 - d_{a_2}'(z_0) \le 0.$$

The sum is therefore at least $1 + d_{a_2}(z_0) > 2$. The replacement argument stays within the middle-knot intervals because $z_0 - c_1 < c_2 < b_2$. If $c_2 < 0$, the same argument uses $d_{a_1}'(z_0) \ge k_1 \ge k_2$. This proves (B.20.6), including mixed signs.

The actual probability Hessian forces K>1 Use the centered unit-variance signed probability directions on the two three-point supports displayed in Proposition 6.4. Their third-moment derivatives are

$$\gamma_i = K d_{a_i}(c_i).$$

For their two perturbation parameters s, t, the total variance is $1 + s + t$, the sum of third absolute moments is $R + \gamma_1 s + \gamma_2 t$, and the selected CDF is $F + f_1 s + f_2 t + gst$. Both signs of each sufficiently small parameter are admissible. At the maximizing ratio, differentiate

$$J(s,t) = (1+s+t)^{3/2}\{F(s,t) - \Phi(z/\sqrt{1+s+t})\} - C(R + \gamma_1 s + \gamma_2 t).$$

It has a local maximum zero. First stationarity and direct second differentiation give its Hessian entries

$$J_{ii} = 3C\gamma_i - B_* = -b_i^*, \qquad J_{12} = g - \frac{b_1^* + b_2^*}{2},$$

$$B_* = (15CR + \phi'''(z))/4, \qquad b_i^* = B_* - 3C\gamma_i \geq 0. \tag{B.20.9}$$

This is the two-coordinate extension of Section B.1. Negative semidefiniteness implies

$$g \leq \frac{b_1^* + b_2^*}{2} + \sqrt{b_1^* b_2^*} \leq b_1^* + b_2^*. \tag{B.20.10}$$

The actual 321 grid gives, exactly,

$$g = \frac{1 - v_1 v_2/(h_1 h_2)}{K^4 u_1 v_1 u_2 v_2} > \frac{3}{4K^4}. \tag{B.20.11}$$

Indeed (B.20.2) gives $h_1 h_2 > (v_1 + v_2)^2 \geq 4 v_1 v_2$, and $u_i v_i \leq h_i^2/4 \leq 1$.

Since $CR = CK - z\phi(z)/3$ and $\phi'''(z) = (3z - z^3)\phi(z)$, equations (B.20.6), (B.20.9)–(B.20.11) yield

$$\frac{3}{4K^4} < \frac{3}{2} CK - \frac{1}{2} z(z^2 + 2)\phi(z). \tag{B.20.12}$$

Put $N(z) = \Phi(z) - z\phi(z)/3$. It is strictly increasing and exceeds $1/2$ for positive z. The ratio identity is $CK = F - N(z)$, so $CK < 1/2$. Equation (B.20.12) now implies

$$\boxed{K > 1, \qquad K < 1/(2C).} \tag{B.20.13}$$

The Hessian also forces z<1/4 Let

$$L(z) = 1 - N(z) = Q(z) + z\phi(z)/3, \qquad A_*(z) = \frac{3}{2} Q(z) - \frac{1}{2} z(z^2 + 1)\phi(z).$$

We have $CK < L(z)$. Equation (B.20.12), and $C \geq 49/120$, imply

$$\frac{3}{4} < K^4 A_*(z) < \left(\frac{L(z)}{49/120} \right)^4 A_*(z). \tag{B.20.14}$$

The middle expression must be positive, so (B.20.14) is only used when $A_*(z) > 0$. First, $z < 1$: the moment inequality $R \geq 1/\sqrt{2}$ gives $CR > 1/4$, whereas $Q(1) < \phi(1) < 1/4$. A threshold $z \geq 1$ would thus contradict $CR = F - \Phi(z) \leq Q(z)$.

On $[0, 1]$, both L and A_* strictly decrease, since

$$L'(z) = -(2 + z^2)\phi(z)/3, \qquad A_*'(z) = -(2 + z^2 - z^4/2)\phi(z).$$

At $z = 1/4$, use $Q(z) \leq 1/2 - z\phi(z)$ and $\phi(1/4) > 3/8$ to get

$$L(1/4) < 7/16, \qquad A_*(1/4) < 573/1024.$$

If $z \geq 1/4$, (B.20.14) would consequently give

$$\frac{3}{4} < \left(\frac{15}{14} \right)^4 \frac{573}{1024} < \frac{3}{4},$$

a contradiction. The exact slack in the last comparison is $495363/39337984$. We have proved

$$\boxed{0 < z < 1/4.} \tag{B.20.15}$$

The Gaussian comparisons are elementary: $\phi(0) > 39/100$ and $\exp(-1/32) > 31/32$ give $\phi(1/4) > 3/8$; also $\pi e > 8$ gives $\phi(1) < 1/4$.

The necessary outside probability is too large By (B.20.2), (B.20.13), and (B.20.15),

$$b_j - a_i < c_1 + c_2 = z/K < 1/4.$$

Since $b_j > 1$, this forces $a_i > 3/4$ for each coordinate. Its selected middle point also has $c_i > 0$. Indeed its contact value is strictly less than the left-minimum value one, whereas for $c \le 0$ the exact negative branch difference is

$$f(Kc) - f(-Ka) = CK^3(c+a)^2(2a - 3/2 - c) > 0 \quad (a > 3/4).$$

Centering and the variance identity give

$$p_i = \frac{a_i + m_i v_i}{h_i} > \frac{a_i}{h_i} > \frac{39}{100}, \qquad \mathrm{Var}(X_i)/K^2 = a_i c_i + r_i h_i v_i. \tag{B.20.16}$$

For the first estimate, $a_i > 3/4$ and $b_i < (2+\sqrt{7})/4 < 7/6$ give $a_i/h_i > 9/23 > 39/100$. Moreover

$$h_i v_i < h_i b_i \le \frac{17 + 7\sqrt{7}}{16} < \frac{9}{4}. \tag{B.20.17}$$

To check the middle bound, differentiating along (B.20.1) gives $(hb)' = h(3-4a)/(2b-1) \le 0$ for $a \ge 3/4$.

Let $r = r_1 + r_2$. From (B.20.13), (B.20.16)–(B.20.17), $a_i < 1$, and $c_i > 0$,

$$\frac{9}{4} r > \frac{1}{K^2} - \sum_i a_i c_i > \left(\frac{49}{60}\right)^2 - z.$$

Hence

$$r > r_0 - 4z/9 =: r_*(z), \qquad r_0 = \frac{2401}{8100} < 3/10. \tag{B.20.18}$$

Also $0 < r < 1$, because $r_i < a_i/h_i < 1/2$. Equation (B.20.3), (B.20.16), and $r_1 r_2 \le r^2/4$ now imply

$$P_{\mathrm{out}} > f(r) := \frac{39}{100} r - r^2/4 \ge f(r_*(z)). \tag{B.20.19}$$

For the last inequality, f is concave, so its minimum on $[r_*(z), 1]$ is at an endpoint. The second endpoint has $f(1) = 7/50$, whereas $f(r_*(z)) < (39/100)(3/10) < 7/50$.

On the other hand, $\phi(z) > 3/8$ gives $N(z) \ge 1/2 + (2/3)z\phi(z) > 1/2 + z/4$. The ratio identity and (B.20.13) give

$$P_{\mathrm{out}} = 1 - N(z) - CK < \frac{11}{120} - \frac{z}{4}. \tag{B.20.20}$$

Equations (B.20.19)–(B.20.20) are incompatible. At zero their required gap is exactly

$$f(r_0) - \frac{11}{120} = \frac{103447}{52488000} > 0.$$

For $0 \le z \le 1/4$, the derivative of $f(r_0 - 4z/9) - (11/120 - z/4)$ is at least $1/4 - (4/9)(39/100) = 23/300 > 0$. The gap therefore stays positive. This is the final contradiction and proves the 321 exclusion.

B.21 Two triples: upper collisions

Normalize the total variance to one; factors of K in the supports are displayed explicitly below.

$$K = R + z\phi(z)/(3C) > 0, \qquad \phi_0 = \phi(0).$$

For shape 322, the maximal included cells are exactly $(0, 2)$ and $(2, 1)$. A genuine collision at the selected threshold therefore has precisely these two representations. Let the second coordinate's high atom have mass $r_2 > 0$, and put $q = 1 - r_2$.

The first triple forces a substantially negative threshold The first coordinate's deletion levels are $(1, q, q)$. If its successive support gaps are $u, v > 0$, its actual centered unit-variance CDF derivative is

$$L = \frac{r_2}{u(u+v)} > 0.$$

The first and second variance derivatives are valid at collisions because the support and closed threshold are fixed. Their elimination, derived directly in Section B.14, gives

$$12L + 3CR + (z^3 + 3z)\phi(z) \le 0.$$

Consequently $z = -s < 0$, and substitution of $CR = CK + s\phi(s)/3$ yields

$$12L + 3CK \le s(s^2 + 2)\phi(s) \le 2s\phi_0. \tag{B.21.1}$$

The last inequality follows by differentiating $(s^2 + 2)\phi(s)$.

The second triple's equal-level contact prevents that threshold The second coordinate's deletion levels are $(1, 1, \ell_1)$, where ℓ_1 is the positive low mass of the first triple. Its low support point is nonselected, hence a smooth negative minimum of the global cubic dual. Write its first two support points as $-Ka, Kc$.

The contact at Kc has the same value as the negative minimum, and its derivative is nonpositive. Therefore the elementary equal-level cubic calculation gives

$$c \ge -1/2. \tag{B.21.2}$$

For clarity, if $c \le 0$, factor the negative cubic at $-a$: its other equal-height contact is $c = 2a - 3/2$ with $a > 1/2$, so $c > -1/2$. If $c > 0$, inequality (B.21.2) is immediate. No unique-representation assumption is used for this contact; the nonpositive derivative follows from the global majorant at any deletion-CDF jump.

The selected representation $(2, 1)$ uses the first coordinate's highest support point, which is strictly positive by centering and nondegeneracy, and the second coordinate's middle point Kc. Thus

$$z > -K/2, \qquad 0 < s < K/2. \tag{B.21.3}$$

Combining (B.21.1) and (B.21.3) gives

$$3CK < 2s\phi_0 < K\phi_0, \qquad C < \phi_0/3,$$

contrary to $C \geq 49/120 > \phi_0$. This excludes 322. Exchanging the two coordinates transposes its grid to 331 and proves that exclusion too.

B.22 Unit-span divided differences

For $s, \mu \in [0,1]$, put $T = s(1-s)$ and

$$\rho(s,\mu) = (1-s)\mu^3 + s(1-\mu)^3 - |s-\mu|^3.$$

This is exactly the third-moment slope in the three-representation 321 mass-elimination chart. One has the unconditional bounds

$$\boxed{T/2 \leq \rho(s,\mu) \leq T\max\{1+s, 2-s\} \leq 2T.} \tag{B.22.1}$$

At $0 < s < 1$, ρ/T is the second divided difference of $|x|^3$ on $(-\mu, s-\mu, 1-\mu)$. The elementary three-point divided-difference bound in Section B.1 is half the support diameter, giving the lower bound $\rho/T \geq 1/2$. The endpoint cases $s = 0, 1$ satisfy $\rho = T = 0$ directly.

For the upper bound, direct differentiation in μ gives

$$\partial_\mu^2 \rho = 6\{(1-s)\mu + s(1-\mu) - |s-\mu|\} = \begin{cases} 12\mu(1-s), & \mu \leq s, \\ 12s(1-\mu), & \mu \geq s. \end{cases}$$

Thus ρ is convex in μ on $[0,1]$, including its common derivative at $\mu = s$, and its maximum is at an endpoint. The endpoint values are exactly $\rho(s,0) = T(1+s)$ and $\rho(s,1) = T(2-s)$, proving (B.22.1).

In the complementary-gap two-coordinate chart, both third-moment slopes have this same T. Their half difference $a = (\rho_1 - \rho_2)/2$ therefore satisfies

$$\boxed{|a| \leq 3T/4.} \tag{B.22.2}$$

B.23 Three representations: contact and variance equations

Unit-span coordinates Translations and a common positive scale put the raw supports and selected raw threshold into the form

$$X_{\mathrm{raw}} = (0, s, 1), \qquad Y_{\mathrm{raw}} = (0, 1-s, 1), \qquad y_{\mathrm{raw}} = 1, \qquad 0 < s < 1. \tag{B.23.1}$$

Let $p = (p_0, p_1, p_2)$ and $q = (q_0, q_1, q_2)$ be the positive probability vectors of the two laws, with means μ, ν. Put $m = p_1$ and $n = q_1$. Write x, y for their centered support vectors, v_i, r_i for their variances and third absolute moments, and $M_1 = \sum_j p_j x_j |x_j|$, $M_2 = \sum_j q_j y_j |y_j|$. Set

$$V = v_1 + v_2, \qquad R = r_1 + r_2, \qquad o = 1 - \mu - \nu, \qquad z = o/\sqrt{V}. \tag{B.23.2}$$

Event and contact equations The event probability is the actual six-cell sum

$$F = 1 - p_1 q_2 - p_2(q_1 + q_2), \qquad D = F - \Phi(z) > 0, \qquad C = DV^{3/2}/R. \tag{B.23.3}$$

Thus require

$$(49/120)^2 R^2 \le D^2 V^3 < (227/500)^2 R^2. \tag{B.23.4}$$

For the two coordinates the actual deletion levels are respectively

$$H_1 = (1, q_0 + q_1, q_0), \qquad H_2 = (1, p_0 + p_1, p_0). \tag{B.23.5}$$

Write $L = 3D + z\phi(z)$. After multiplication by the positive factor $2VR$, the global dual contact equation at a raw centered support point u of coordinate i is

$$\boxed{2VD(|u|^3 - r_i) - LR(u^2 - v_i) - (6VDM_i + 2\sqrt{V}\phi(z)R)u + 2VR(F - H_i(u)) = 0.} \tag{B.23.6}$$

Indeed the unscaled dual price is D/R, its quadratic coefficient is $L/(2V)$, its linear coefficient is $3DM_i/R + \phi(z)/\sqrt{V}$, and averaging the majorant over the coordinate determines its constant. This proves (B.23.6) directly and preserves the actual centered moments.

Variance Hessian For the two centered signed mass vectors, use

$$w_1 = (1 - s, -1, s), \qquad w_2 = (s, -1, 1 - s), \qquad T = s(1 - s).$$

Both raw variance derivatives equal T. Their raw cubic derivatives are $\rho_i = \sum_j w_{i,j}|x_{i,j}|^3$, where $x_{1,j} = x_j$ and $x_{2,j} = y_j$ are the centered raw supports. The actual 321 mixed event coefficient is

$$M_0 = 1 - s + s^2 = 1 - T.$$

Put

$$\begin{aligned} W &= R\{15D + \phi'''(z)\}/4, \\ N_i &= WT^2 - 3DVT\rho_i, \\ H_{12} &= RV^2 M_0 + \frac{3}{2}DVT(\rho_1 + \rho_2) - WT^2. \end{aligned} \tag{B.23.7}$$

The necessary conditions from the original variance Hessian are

$$\boxed{N_1 \ge 0, \quad N_2 \ge 0, \quad N_1 N_2 - H_{12}^2 \ge 0.} \tag{B.23.8}$$

To verify this normalization, start from the unnormalized Hessian derived at the end of Section B.31, substitute $G = R/D$, and multiply all matrix entries by $D > 0$, with $\phi'''(z) = (3z - z^3)\phi(z)$.

B.24 Three representations: the gap product

Use the unit-span scale of Section B.23.

In the three-representation 321 case, retain the notation

$$T = s(1-s), \quad L = 1 - T, \quad W = \frac{15D + \phi'''(z)}{4}, \quad C = \kappa V^{3/2}.$$

The original variance Hessian has diagonal necessary quantities

$$N_i = WT^2 - 3\kappa VT\rho_i,$$

and its determinant implies

$$N_1 + N_2 \geq V^2 L. \tag{B.24.1}$$

The pointwise third-moment slope bound Section B.22 gives $\rho_1 + \rho_2 \geq T$. Hence

$$V^2 L \leq N_1 + N_2 \leq (2W - 3\kappa V)T^2. \tag{B.24.2}$$

The bound (B.3) gives $W < 79/40$. Also $V \leq 1/2$ and $C \geq 49/120$ yield

$$\kappa V = \frac{C}{\sqrt{V}} > \frac{49\sqrt{2}}{120} > \frac{343}{600},$$

where the final step uses $\sqrt{2} > 7/5$. Thus

$$2W - 3\kappa V < 79/20 - 343/200 = 447/200.$$

Since $L \geq 3/4$, (B.24.2) proves

$$\boxed{T^2 > \frac{50}{149}V^2.} \tag{B.24.3}$$

The unit-span diameter bound is $V > 2401/24964$. Substituting it into (B.24.3) gives

$$T^2 > \frac{144120025}{46428496552} > \frac{1}{324}, \qquad \boxed{s(1-s) > 1/18.} \tag{B.24.4}$$

The exact final slack is $66597887/3760708220712 > 0$, saved in the computational supplement. Equivalently,

$$\frac{3 - \sqrt{7}}{6} < s < \frac{3 + \sqrt{7}}{6}.$$

In particular $1/18 < s < 17/18$.

B.25 Three representations: an upper variance bound

The scale is the unit-span scale of Section B.23.

At an attained violating fixed-two-coordinate maximum with two genuine three-point laws and all three representations of the closed 321 event,

$$\boxed{V < 3/10.} \tag{B.25.1}$$

A scalar lower bound Retain $D > 49/(120\sqrt{2})$ and $-5/4 < z < 3/5$ from Section B.3. Define

$$J = 3D + (z^3 + 3z)\phi(z).$$

Then

$$\boxed{J > -9/50.} \tag{B.25.2}$$

For $z \geq 0$ this is immediate. For $z = -t$, $0 < t < 5/4$, the derivative of $(t^3 + 3t)\phi(t)$ is $(3 - t^4)\phi(t) > 0$. Thus its maximum on this interval is less than $(365/64)\phi(5/4)$. The elementary bound $\phi(0) < 399/1000$, used in Section B.3, and the alternating exponential upper polynomial give

$$\phi(5/4) < \frac{399}{1000}\sum_{j=0}^{6}\frac{(-25/32)^j}{j!} = \frac{9415683872273}{51539607552000} < \frac{183}{1000}.$$

Also $\sqrt{2} < 99/70$, so $3D > 343/396$. Consequently

$$J > \frac{343}{396} - \frac{365}{64}\frac{183}{1000} = -\frac{9}{50} + \frac{631}{253440} > -\frac{9}{50}.$$

Eliminate the cubic slopes from the mass Hessian Write

$$T = s(1-s), \quad L = 1 - T, \quad u = (1-s)\nu, \quad v = s\mu, \quad w = m + n,$$
$$E_0 = \mu(1-\mu) + \nu(1-\nu), \quad V = E_0 - Tw,$$
$$A = \frac{3D + z\phi(z)}{2V}, \quad W = \frac{15D + \phi'''(z)}{4}, \quad \kappa = D/R.$$

Applying the centered mass direction $(1-s, -1, s)$ to the first coordinate's three actual contact equations cancels its constant and linear terms. The deletion levels $(1, q_0 + q_1, q_0)$ give

$$\kappa\rho_1 - AT = (1-s)q_2 - sq_1 = u - Ln =: \eta_1.$$

Similarly, the second coordinate's direction $(s, -1, 1-s)$ gives

$$\kappa\rho_2 - AT = sp_2 - (1-s)p_1 = v - Lm =: \eta_2. \tag{B.25.3}$$

The regularized diagonal Hessian quantities from Section B.23 are therefore

$$H_i = W - 3\kappa V\rho_i/T = -\frac{J}{4} - \frac{3V\eta_i}{T}, \tag{B.25.4}$$

since $W - 3VA = -J/4$. The determinant condition implies $H_1 + H_2 \geq V^2L/T^2$. Set

$$B_0 = LE_0 - T(u+v).$$

As $\eta_1 + \eta_2 = (LV - B_0)/T$, (B.25.4) yields the scalar necessary inequality

$$\boxed{3VB_0 - 4LV^2 - JT^2/2 \geq 0.} \tag{B.25.5}$$

This eliminates both third-moment slopes and their difference from this part of the Hessian.

Maximize the remaining quadratic in the two means For fixed s, independently maximize the two concave quadratics in μ, ν appearing in B_0. This gives

$$\frac{B_0}{L} \le \frac{(L-Ts)^2 + (L-T(1-s))^2}{4L^2} = \frac{(1-2T)(T^2-2T+2)}{4(1-T)^2} =: M(T). \tag{B.25.6}$$

Completing the two squares gives (B.25.6), and $M(T)$ strictly decreases on $0 \le T \le 1/4$. Indeed, with $L = 1-T$, we have

$$M = L/2 - 1/4 + 1/(2L) - 1/(4L^2), \qquad \frac{dM}{dL} = \frac{L^3 - L + 1}{2L^3} > 0$$

on $3/4 \le L \le 1$. Its useful rational value is

$$M(19/100) = \frac{513391}{1312200} = \frac{47}{120} - \frac{277}{656100} < \frac{47}{120}. \tag{B.25.7}$$

Combining (B.25.2), (B.25.5), (B.25.6), and $T^2/L \le 1/12$ gives

$$4V^2 - 3M(T)V - 3/400 < 0. \tag{B.25.8}$$

In particular, whenever $T > 19/100$, (B.25.7) implies

$$f(V) := 4V^2 - (47/40)V - 3/400 < 0. \tag{B.25.9}$$

But $f(3/10) = 0$ exactly and f increases for $V \ge 3/10$.

Two finite bootstrap steps The gap bound is $T^2 > (50/149)V^2$. If $V \ge 1/3$, then

$$T^2 > \frac{50}{1341} > \frac{361}{10000},$$

so (B.25.9) contradicts $V \ge 1/3$. Thus first $V < 1/3$.

It follows that

$$\kappa V = \frac{C}{\sqrt{V}} > \frac{49\sqrt{3}}{120} > \frac{7}{10},$$

using $\sqrt{3} > 12/7$. The original gap-Hessian derivation now improves to

$$V^2 L \le (2W - 3\kappa V)T^2 < \frac{37}{20}T^2, \qquad T^2 > \frac{15}{37}V^2. \tag{B.25.10}$$

If $V \ge 3/10$, (B.25.10) gives

$$T^2 > \frac{27}{740} > \frac{361}{10000}.$$

Hence $T > 19/100$, so (B.25.9) contradicts $V \ge 3/10$. Therefore $V < 3/10$.

As a convenient weaker consequence of (B.25.1), $\kappa V > 37/50$, since $(37/50)^2\,(3/10) < (49/120)^2$. Repeating the same gap inequality gives $T^2 > (75/173)V^2$.

B.26 Three representations: joint mean and variance bounds

Keep the unit-span scale of Section B.23.

In the three-representation 321 case, coupling the two means to the threshold improves the variance bound to

$$\boxed{V < 11/40.} \tag{B.26.1}$$

Couple the quadratic mean bound with the scalar J Retain $T, L, B_0, M(T)$ from Section B.25 and $J = 3D + (z^3 + 3z)\phi(z)$, and write $g = T^2/L$ and $d_{\min} = 49/(120\sqrt{2})$. Set

$$y = z\sqrt{V} = 1 - \mu - \nu, \qquad d = \mu - \nu.$$

Completing the square in d while retaining y gives

$$\frac{B_0}{L} \le N_0(T) + \frac{T}{2L}y - \frac{y^2}{2},$$
$$N_0(T) = \frac{1}{2} - \frac{T}{2L} + \frac{T^2(1-4T)}{8L^2} = M(T) - \frac{T^2}{8L^2} \le M(T). \tag{B.26.2}$$

For $z \le 0$, monotonicity of $(t^2+3)\phi(t)$ for $t \ge 0$ gives

$$(z^3 + 3z)\phi(z) \ge (6/5)z, \qquad J > 3d_{\min} + (6/5)z. \tag{B.26.3}$$

Indeed the derivative of $(t^2+3)\phi(t)$ is $-t(t^2+1)\phi(t) \le 0$, and its value at zero is $3\phi(0) < 6/5$. In the actual contact/Hessian inequality

$$4V \le 3B_0/L - Jg/(2V),$$

substitute (B.26.2)–(B.26.3). The coefficient of $y \le 0$ is

$$\frac{3T}{2L} - \frac{3g}{5V^{3/2}} = \frac{3T}{2L}\left(1 - \frac{2T}{5V^{3/2}}\right) > 0$$

whenever $V \ge 1/4$, because then $T/V^{3/2} \le 2$. Hence that linear term and $-3y^2/2$ are nonpositive. It follows that

$$4V^2 - 3M(T)V + \frac{3d_{\min}}{2}g < 0. \tag{B.26.4}$$

For $z \ge 0$, the same inequality follows directly from $J \ge 3D > 3d_{\min}$ and $B_0/L \le M(T)$. Thus (B.26.4) holds for every candidate with $V \ge 1/4$. In particular, since $3d_{\min}/2 > 3/7$,

$$4V^2 - 3M(T)V + (3/7)g < 0. \tag{B.26.5}$$

Two rational bootstrap steps The already proved $V < 3/10$ gives $\kappa V > 37/50$. Therefore the gap-Hessian inequality gives

$$g > \frac{100}{173}V^2. \tag{B.26.6}$$

If $V \ge 7/25$, (B.26.6) implies $g > 196/4325 > 361/8100 = g(19/100)$. Since $g(T)$ increases, $T > 19/100$, and Section B.25 gives $M(T) < 47/120$. Equation (B.26.5) would then force

$$V < \frac{47/40}{4 + 300/1211} = \frac{56917}{205760} < \frac{7}{25}.$$

This contradiction proves $V < 7/25$. Consequently $\kappa V > 77/100$, so

$$g > \frac{25}{41} V^2. \tag{B.26.7}$$

If $V \geq 11/40$, (B.26.7) gives

$$g > \frac{121}{2624} > \frac{576}{12625} = g(24/125).$$

Thus $T > 24/125$, and exact evaluation gives

$$M(24/125) = \frac{994301}{2550250} = \frac{39}{100} - \frac{593}{5100500} < \frac{39}{100}.$$

Equation (B.26.5) now forces

$$V < \frac{117/100}{4 + 75/287} = \frac{33579}{122300} < \frac{11}{40},$$

again a contradiction. This proves (B.26.1).

B.27 Three representations: scalar exclusion

Keep the unit-span scale of Section B.23.

Use the notation $V, z, T = s(1-s), L = 1 - T, m, n, w = m + n$, and $v_0 = 2401/24964$. The range is

$$v_0 < V < 11/40, \qquad -5/4 < z < 3/5. \tag{B.27.1}$$

The actual bounded sum and its exact slack The raw sum U is in $[0, 2]$, has mean $1 - o$, variance V, and the selected event is $U \leq 1$, where $o = z\sqrt{V}$. Pointwise $U(U-1) \leq 2\mathbf{1}_{\{U>1\}}$. Therefore

$$F \leq 1 - \{V - o + o^2\}/2. \tag{B.27.2}$$

The precise nonnegative slack, using the actual independent 321 event, is

$$S := 2(1-F) - (V - o + o^2) = T(mq_0 + p_0 n) + (2 - s - s^2)mq_2 + (3s - s^2)p_2 n.$$

Both outside coefficients are at least T, so

$$S \geq T\{m(1-n) + n(1-m)\} \geq T\{w - w^2/2\}. \tag{B.27.3}$$

Let $H_\Sigma = H_1 + H_2$, with H_i defined in (B.25.4). The mass-Hessian condition gives

$$H_\Sigma \geq \frac{V^2 L}{T^2} \geq 12V^2, \qquad T = s(1-s) \leq \frac{1}{4}, \quad L = 1 - T.$$

The contact identities yield

$$\eta_{\text{sum}} = u + v - Lw = -\frac{T(H_\Sigma + J/2)}{3V}.$$

Since $J > -9/50$ and $12v_0^2 > 9/100$, this is negative. Therefore

$$w > \frac{u+v}{L}. \tag{B.27.4}$$

If $T > 1/5$, both s and $1-s$ belong to $(1/4, 3/4)$ and $L < 4/5$. Thus (B.27.4) gives $w > 5(\mu+\nu)/16$. The range (B.27.1) gives $\mu+\nu = 1 - z\sqrt{V} > 137/200$, using $\sqrt{11/40} < 21/40$. Hence

$$w > \frac{137}{640} =: a.$$

Let X_1, X_2 denote the raw marginals, and set $a_1 = s$, $a_2 = 1-s$, $m_1 = m$, $m_2 = n$. Since $a_i \in (1/4, 3/4)$, the raw marginal variances satisfy

$$v_i \le \mathbb{E}(X_i - a_i)^2 < 9(1-m_i)/16.$$

Summing gives

$$w \le 2 - 16V/9 < 2 - 16v_0/9 =: b.$$

The concave function $f(w) = w - w^2/2$ has its minimum on $[a, b]$ at an endpoint. Both exact endpoint values exceed $3/20$. Consequently

$$\boxed{T > 1/5 \quad \Longrightarrow \quad S > 3/100.} \tag{B.27.5}$$

Combine with the mass Hessian The mass Hessian and $\rho_1 + \rho_2 \ge T$ give

$$V^2 \le \left\{ \frac{15D + \phi'''(z)}{2} - \frac{3C}{\sqrt{V}} \right\} \frac{T^2}{L}. \tag{B.27.6}$$

If $T \le 1/5$, then $T^2/L \le 1/20$, so $15D + \phi'''(z) \ge 40V^2 + 6C/\sqrt{V}$. If $T > 1/5$, use $T^2/L \le 1/12$ instead, giving $15D + \phi'''(z) \ge 24V^2 + 6C/\sqrt{V}$, together with (B.27.5).

The two cases are therefore enclosed by $(A, S_0) = (40, 0)$ and $(A, S_0) = (24, 3/100)$. Put $c_0 = 49/120$ and

$$B(V, z; A, S_0) = 15(1-\Phi(z)) - \frac{15}{2}V(1+z^2) + \frac{15}{2}z\sqrt{V} - \frac{15}{2}S_0 + (3z - z^3)\phi(z) - AV^2 - \frac{6c_0}{\sqrt{V}}. \tag{B.27.7}$$

The plain probability bound also gives

$$P(V, z; A) = 15(1 - \Phi(z)) + (3z - z^3)\phi(z) - AV^2 - 6c_0/\sqrt{V}.$$

When $z < 0$, Cantelli gives the additional bound

$$Q(V, z; A) = 15\{(1+z^2)^{-1} - \Phi(z)\} + (3z - z^3)\phi(z) - AV^2 - 6c_0/\sqrt{V}. \tag{B.27.8}$$

Every genuine candidate requires each applicable upper expression to be strictly positive, including when $C = c_0$. The proper selected downset has probability less than one. The bounded-sum comparison has positive slack at an interior support point; when the prescribed slack is positive, (B.27.5) gives a strict improvement over it. For negative thresholds, equality in Cantelli's inequality would force a two-point sum law, whereas the sum of two genuine triples has at least five support points. At $z = 0$ the Cantelli expression equals the probability expression. Thus a nonpositive enclosure excludes the candidate also at $C = c_0$. It suffices to cover (B.27.1), separately for both (A, S_0) cases, by boxes on which at least one of B, P, or Q (when the whole box has $z \le 0$) is nonpositive.

Whole-box derivative enclosures For (B.27.7), the exact first derivatives are

$$B_V = -\frac{15}{2}(1+z^2) + \frac{15z}{4\sqrt{V}} - 2AV + 3c_0/V^{3/2},$$

$$B_z = (z^4 - 6z^2 - 12)\phi(z) - 15Vz + \frac{15}{2}\sqrt{V}.$$

For P, Q, their V derivative is $-2AV + 3c_0/V^{3/2}$; their z derivatives are respectively $(z^4 - 6z^2 - 12)\phi(z)$ and that quantity minus $30z/(1+z^2)^2$. These formulas apply throughout the original rectangle, independently of candidate constraints. A midpoint value plus these whole-box gradient enclosures gives a valid enclosure by the mean value integral. The numerical domain has $V \geq v_0 > 0$. The fixed certificate `three_owner321` covers both cases in (B.27.1) and proves that every point has a nonpositive applicable exclusion expression. Therefore no maximizing pair can have all three representations of the 321 threshold. Section 8 gives the enclosure and partition rules.

B.28 Two representations: contact geometry

Use the contact scale and the strict bounds in (6.1): $K = R_{\text{normalized}} + z\phi(z)/(3C)$, scaled support coordinates equal the normalized coordinates divided by K, and $\tau = CK^3$. Raw moments in this calculation refer to those scaled supports. Thus

$$\frac{1}{20} < \tau < 2, \qquad \frac{25}{64} < V < 4, \qquad C = \tau V^{3/2}. \tag{B.28.1}$$

The two coordinate variance fractions exceed 13/200. Define the piecewise cubic function

$$P_\beta(x) = |x|^3 - \frac{3}{2}x^2 - 3\beta x.$$

Every global majorant in this scale is a constant plus τP_β. A nonselected positive contact b has $\beta = b(b-1)$; a nonselected negative contact $-a$ has $\beta = a(1-a)$.

Adjacent representations Write the supports and the selected threshold as

$$X = (-t, c, b), \quad Y = (-a, d, e), \quad o = e - t = c + d, \qquad d = e - t - c. \tag{B.28.2}$$

Here the representations are $(0,2), (1,1)$. The unused maximal cell $(2,0)$ is strictly lower: $b - a < o$. The high point b and the low point $-a$ are nonselected smooth minima, so $b > 1/2$ and $a > 1/2$. In fact $b \geq 1$. If $1/2 < b < 1$, then $\beta = b(b-1) < 0$ and P_β has a negative minimum at $-a_*$ with $a_* > 1$. Its value is less than $-1/2$, whereas $P_\beta(b) = b^2(3/2 - 2b) > -1/2$. The deletion CDF at $-a_*$ is at least its value at b; global majorization therefore rules out this lower minimum.

Also $a < 1$. Otherwise $-a$ would be a global minimum of $P_{a(1-a)}$. Indeed, for $x \geq 0$,

$$P_{a(1-a)}(x) = x^3 - \frac{3}{2}x^2 + 3a(a-1)x \geq -\frac{1}{2} \geq P_{a(1-a)}(-a),$$

and for $x = -t \leq 0$ the difference from the value at $-a$ is $(t-a)^2(t + 2a - 3/2) \geq 0$. The contact level at $-a$ is one, so a global minimum there would preclude the smaller contact levels at the other support points.

The nonpositive derivative at the selected high point gives $e \leq B(a)$, where B is defined below. The unused-corner inequality now gives

$$t + b < a + e \leq a + B(a) \leq 2.$$

Since $b \geq 1$, we have $t < 1$. The nonpositive derivative at $-t$ gives $b(b-1) \geq t(1-t)$, hence $b \geq B(t)$. The function $x + B(x)$ is strictly increasing on $(0,1)$, because its derivative is $1 + (1-2x)/\sqrt{1+4x(1-x)} > 0$. Thus $t \geq a$ would contradict the strict span inequality. Finally $e > t + b - a > t$, so $o = e - t > 0$. We have proved

$$\frac{1}{2} < a < 1, \quad 0 < t < a, \quad B(t) \leq b < 2 - t, \quad 0 < e \leq B(a), \quad -t < c < b,$$

$$B(v) = \frac{1 + \sqrt{1 + 4v(1-v)}}{2}, \qquad A_2 := a + e > A_1 := t + b, \qquad o > 0. \tag{B.28.3}$$

The strict span ordering also implies $-a < d < e$. Set $h = t + c = e - d > 0$ and $v = b - c > 0$.

Endpoint representations Now the representations are $(0,2), (2,0)$ and the unused middle-middle cell lies strictly below their common threshold. Both nonselected middle contacts are negative smooth minima. A positive smooth minimum would force a positive derivative at the larger contact, contrary to Lemma 6.2; zero cannot be a minimum because the quadratic coefficient is negative. Write

$$X = (-t_1, -a_1, b_1), \qquad Y = (-t_2, -a_2, b_2),$$

$$\frac{1}{2} < a_i < 1, \quad a_i < t_i < 4, \quad 0 < b_i \leq B(a_i), \quad A = t_1 + b_1 = t_2 + b_2,$$

$$o = b_2 - t_1 = b_1 - t_2 > -a_1 - a_2. \tag{B.28.4}$$

Variance Hessian For either threshold geometry, take arbitrary positive mass vectors (ℓ_i, m_i, r_i) on its centered supports. Let v_i and R_i be their variances and third absolute moments, and put

$$V = v_1 + v_2, \qquad R = R_1 + R_2, \qquad z = o/\sqrt{V}, \qquad F = 1 - m_1 r_2 - r_1(m_2 + r_2).$$

At the attained ratio, $\tau = C/V^{3/2}$ and $D = F - \Phi(z) = \tau R$.

The full variance Hessian can be evaluated in the original signed-mass basis. For each support vector (x_0, x_1, x_2), set

$$w_i = \left(\frac{x_2 - x_1}{x_2 - x_0}, -1, \frac{x_1 - x_0}{x_2 - x_0}\right), \quad T_i = (x_1 - x_0)(x_2 - x_1), \quad \rho_i = \sum_j w_{i,j}|x_j|^3.$$

Let $M_0 = \sum_{j+k\leq 2} w_{1,j} w_{2,k}$, the actual 321 mixed event coefficient, and put

$$W = \{15\tau R + \phi'''(z)\}/4, \quad N_i = WT_i^2 - 3\tau V T_i \rho_i,$$

$$Q = V^2 M_0 + \frac{3}{2}\tau V(T_1\rho_2 + T_2\rho_1) - WT_1T_2.$$

Necessary second-order conditions are

$$N_1 \geq 0, \qquad N_2 \geq 0, \qquad N_1 N_2 - Q^2 \geq 0. \tag{B.28.5}$$

These are the already derived original Hessian entries multiplied by the positive scalar τ, with the actual normal derivative $\phi'''(z) = (3z - z^3)\phi(z)$.

B.29 Adjacent representations: scalar exclusion

Normalize by the larger span Translate each support to start at zero and divide by the second span. The supports and selected threshold become

$$X \in \{0, a, r\}, \quad Y \in \{0, 1-a, 1\}, \quad 0 < a < r < 1, \quad X + Y \le 1. \tag{B.29.1}$$

Let V denote total variance in this raw scale, z the standardized selected threshold, $D = F - \Phi(z)$, and $C = DV^{3/2}/R$. The uniform diameter and threshold bounds give

$$v_0 := 2401/24964 < V \le (1 + r^2)/4 \le 1/2, \qquad 0 < z < 3/5. \tag{B.29.2}$$

Put $s = a/r$, $T = s(1-s)$, and use the centered mass directions $(1-s, -1, s)$ and $(a, -1, 1-a)$. Their raw variance derivatives are $T_1 = a(r-a)$ and $T_2 = a(1-a)$. Summation over the actual 321 cells gives the mixed event derivative

$$M = 1 - a + a^2/r = 1 - rT. \tag{B.29.3}$$

The cubic third-moment derivatives divided by T_i are the centered second divided differences γ_i. The elementary divided-difference diameter bound gives $\gamma_1 \ge r/2$ and $\gamma_2 \ge 1/2$.

Retain the dependence between the mixed coefficient and span Write $W = (15D + \phi'''(z))/4$ and $H_i = W - 3C\gamma_i/\sqrt{V}$. The mass-Hessian condition (B.28.5), divided by the positive variance-direction factors, implies

$$H_1 + H_2 \ge V^2\Lambda, \qquad \Lambda = M/(T_1T_2). \tag{B.29.4}$$

This follows directly from its determinant: the off-diagonal entry is $V^2\Lambda - (H_1 + H_2)/2$, whose absolute value is at most $\sqrt{H_1H_2} \le (H_1 + H_2)/2$.

Now $a(1-a) \le 1/4$, and $T_1\,T_2 = r^2T\,a(1-a)$. Therefore

$$\Lambda \ge \frac{4(1-rT)}{r^2T} \ge \frac{16 - 4r}{r^2} \ge \frac{12}{r^2}. \tag{B.29.5}$$

Combining (B.29.4), (B.29.5), and the two divided-difference bounds proves

$$\boxed{15D + \phi'''(z) \ge \frac{32 - 8r}{r^2}V^2 + \frac{3C(1+r)}{\sqrt{V}}.} \tag{B.29.6}$$

The universal bound $15D + \phi'''(z) < 79/10$ also gives $V^2\Lambda < 79/20$. Hence $r^2 > (240/79)V^2 > (240/79)v_0^2 > 1/36$, proving the convenient strict bound $r > 1/6$. The final comparison is exact rational arithmetic.

The bounded sum supplies the scalar upper bounds Set $U = X + Y$. It belongs to $[0, 1+r]$, has mean $1 - z\sqrt{V}$, and its selected event is $U \le 1$. Pointwise,

$$U(U-1) \le r(1+r)\mathbf{1}_{\{U>1\}}.$$

Consequently

$$F \le 1 - \frac{V(1+z^2) - z\sqrt{V}}{r(1+r)}, \qquad F \le 1. \tag{B.29.7}$$

Put $c_0 = 49/120$, $A(r) = 32/r^2 - 8/r$, and

$$P(r,V,z) = 15(1-\Phi(z)) + (3z - z^3)\phi(z) - A(r)V^2 - \frac{3c_0(1+r)}{\sqrt{V}},$$

$$B(r,V,z) = P(r,V,z) - \frac{15\{V(1+z^2) - z\sqrt{V}\}}{r(1+r)}. \tag{B.29.8}$$

Every genuine candidate requires both $P > 0$ and $B > 0$, including when $C = c_0$. The selected downset is proper, so $F < 1$. The bounded-sum comparison is also strict: the sum has an interior support point at which its pointwise quadratic bound has positive slack. Together with (B.29.6), these strict comparisons show that a nonpositive enclosure excludes the candidate at the equality floor too. It suffices to cover the closed box

$$[1/6, 1] \times [v_0, 1/2] \times [0, 3/5] \tag{B.29.9}$$

by boxes on each of which one expression in (B.29.8) is nonpositive.

Whole-box derivatives for a fixed outward cover Let $N = V(1+z^2) - z\sqrt{V}$. On the whole domain (B.29.9), the exact derivatives are

$$P_r = (64/r^3 - 8/r^2)V^2 - 3c_0/\sqrt{V},$$

$$P_V = -2A(r)V + \frac{3c_0(1+r)}{2V^{3/2}}, \qquad P_z = (z^4 - 6z^2 - 12)\phi(z),$$

$$B_r = P_r + \frac{15N(1+2r)}{r^2(1+r)^2},$$

$$B_V = P_V - \frac{15\{1 + z^2 - z/(2\sqrt{V})\}}{r(1+r)}, \qquad B_z = P_z - \frac{15(2Vz - \sqrt{V})}{r(1+r)}.$$

Thus midpoint values plus whole-box derivative enclosures give a valid mean-value enclosure. Both denominators r and V are uniformly positive. The fixed records `adjacent_owner321` and `adjacent_owner321_completion` cover (B.29.9); at every point they prove $P \le 0$ or $B \le 0$. Thus the adjacent-representation case has no maximizing pair. Section 8 gives the enclosure and partition rules, including the subdivision of the four cells treated by the completion record.

B.30 Endpoint representations: scalar exclusion

Geometry, variance directions, and the first scalar inequality Translate both supports to start at zero and divide by their common span:

$$X \in \{0, a, 1\}, \quad Y \in \{0, b, 1\}, \quad a, b > 0, \quad s := a + b < 1, \qquad U = X + Y \le 1. \tag{B.30.1}$$

Let the respective means be μ, ν, middle masses m, n, and total raw variance and sum of third absolute moments be V, R. Write

$$D = F - \Phi(z), \quad C = DV^{3/2}/R, \quad T_1 = a(1-a), \quad T_2 = b(1-b), \quad M = a+b-ab, \quad g = T_1T_2/M.$$

The diameter and threshold results give

$$v_0 := 2401/24964 < V \le 1/2, \qquad -5/4 < z < 3/5, \qquad D > \frac{49}{120\sqrt{2}}, \quad C \ge c_0 := 49/120. \tag{B.30.2}$$

The actual mixed derivative of the 321 event in the centered mass directions $(1-a,-1,a)$ and $(1-b,-1,b)$ is M. The original mass Hessian, as in the two-representation reduction, implies

$$15D + \phi'''(z) \ge \frac{2V^2}{g} + \frac{6C}{\sqrt{V}}, \tag{B.30.3}$$

using the lower bound $1/2$ for each centered cubic divided difference.

We will use the uniform bound

$$\boxed{g < 1/11.} \tag{B.30.4}$$

Indeed put $p = ab$, $d = (1-a)(1-b) = 1-s+p$; then $g = pd/(s-p)$. For fixed s, its derivative in p has positive numerator $s(1-s+2p)-p^2$. Thus its maximum occurs at $p = s^2/4$, or $a = b = x$. The resulting function is $x(1-x)^2/(2-x)$, $0 < x \le 1/2$, whose derivative has the sign of x^2-3x+1. Its maximum is $(5\sqrt{5}-11)/2 < 1/11$; the last comparison follows from $\sqrt{5} < 123/55$, since $123^2 - 5\,55^2 = 4 > 0$.

At moderately negative thresholds the actual contacts give mass bounds For $z \ge -3/4$, define $J = 3D + (z^3+3z)\phi(z)$. Then

$$J > \frac{343}{396} - \frac{171}{64}\frac{31}{100} = \frac{24001}{633600} > 0. \tag{B.30.5}$$

Here $\sqrt{2} < 99/70$, and $\phi(3/4) < 31/100$. The latter follows from $\phi(0) < 399/1000$ and the degree-four upper Taylor polynomial for $\exp(-9/32)$. On $[0,3/4]$, both $t(t^2+3)\phi(t)$ and $t(t^2+2)\phi(t)$ increase; their derivatives are respectively $(3-t^4)\phi(t)$ and $(2+t^2-t^4)\phi(t)$.

Let $\kappa = D/R$, $A = (3D + z\phi(z))/(2V)$, and $W = (15D + \phi'''(z))/4$. Applying each centered mass direction to its three actual contact equations gives

$$\eta_1 := \kappa\rho_1 - AT_1 = (1-a)\nu - Mn, \qquad \eta_2 := \kappa\rho_2 - AT_2 = (1-b)\mu - Mm. \tag{B.30.6}$$

The diagonal Hessian quantities satisfy $H_i = W - 3\kappa V\rho_i/T_i = -J/4 - 3V\eta_i/T_i \ge 0$. Thus (B.30.5) gives $\eta_i < 0$. Both middle contacts are negative after centering, so $\mu > a$ and $\nu > b$. Consequently

$$\boxed{m > \frac{a(1-b)}{M}, \qquad n > \frac{b(1-a)}{M}.} \tag{B.30.7}$$

Define the cubic contact scale in the present unit-span coordinates by

$$k = \frac{\sqrt{V}\{D + z\phi(z)/3\}}{C}. \tag{B.30.8}$$

A nonselected negative minimum of the cubic majorant has distance strictly larger than $k/2$ from zero: its second derivative is positive, and the quadratic coefficient is $-3k/2$ relative to the cubic coefficient. Hence $\mu - a > k/2$ and $\nu - b > k/2$. Positivity of the low masses now gives

$$\boxed{1-m > \frac{k}{2(1-a)}, \qquad 1-n > \frac{k}{2(1-b)}.} \tag{B.30.9}$$

A uniform positive lower bound on this raw contact scale Combining (B.30.3), (B.30.4), and (B.30.8) yields

$$k \geq \frac{2}{5} + \frac{\sqrt{V}}{15C}\{22V^2 + (z^3 + 2z)\phi(z)\}. \tag{B.30.10}$$

For $z \geq 0$, the braces are positive. For $-3/4 \leq z \leq 0$, their value exceeds $1/5 - (123/64)(31/100) = -2533/6400$, because $22v_0^2 > 1/5$. Moreover $\sqrt{V}/(15C) < (5/7)/(49/8) = 40/343$. In either sign case,

$$\boxed{k > \frac{7}{20}.} \tag{B.30.11}$$

The exact margin in the negative sign case is $2/5 - (40/343)(2533/6400) - 7/20 = 211/54880 > 0$.

Exact event slack, retaining the independent probabilities Let p_0, m, p_2 and q_0, n, q_2 be the actual probabilities. The nonnegative bounded-sum slack is

$$\begin{aligned} S &:= 2(1-F) - \{V - z\sqrt{V} + z^2 V\} \\ &= T_1 m q_0 + T_2 p_0 n + s(1-s)mn + (1-a)(2+a)mq_2 + (1-b)(2+b)p_2 n. \end{aligned} \tag{B.30.12}$$

Since $\mu > a$, $\nu > b$, the high masses satisfy $p_2 > a(1-m)$ and $q_2 > b(1-n)$. After grouping each low/high pair in (B.30.12),

$$S > A_0 m(1-n) + B_0 n(1-m) + s(1-s)mn,$$

where $A_0 = (1-a)(a+2b)$ and $B_0 = (1-b)(b+2a)$. Inserting (B.30.7), (B.30.9) into the first two positive terms proves

$$S > \frac{k}{2M}\{aA_0 + bB_0\} = \frac{k}{2M}\{s^2(1-s) + p(s+2)\}. \tag{B.30.13}$$

The expression in braces is at least $4pd$. To see this, subtract $4pd$: $s^2(1-s) + p(5s-2) - 4p^2$ is concave in $p \in [0, s^2/4]$. Its endpoint values are $s^2(1-s) \geq 0$ and $s^2(2+s-s^2)/4 > 0$. Using $g = pd/M$, (B.30.13) therefore yields the simple exact inequality

$$\boxed{S > 2kg \qquad (z \geq -3/4).} \tag{B.30.14}$$

In particular, (B.30.11) proves

$$\boxed{g > 1/20,\ z \geq -3/4 \quad \Longrightarrow \quad S > 7/200.} \tag{B.30.15}$$

Three fixed scalar rectangles suffice The expressions P, B, Q below are upper bounds obtained respectively from $F \leq 1$, the bounded sum with slack S_0, and Cantelli for $z \leq 0$:

$$\begin{aligned} P &= 15(1 - \Phi(z)) + (3z - z^3)\phi(z) - A_0 V^2 - 6c_0/\sqrt{V}, \\ B &= P - \frac{15}{2}\{V(1+z^2) - z\sqrt{V} + S_0\}, \qquad Q = P + 15\{(1+z^2)^{-1} - 1\}. \end{aligned} \tag{B.30.16}$$

Every candidate requires each applicable expression to be strictly positive, including at $C = c_0$, by the strict probability and bounded-sum comparisons used in Section B.27. For the positive prescribed slack use (B.30.15); for negative thresholds Cantelli equality is impossible for a sum with at least five support points. The following three closed rectangles cover all possibilities:

Case	Coefficient A_0	Slack S_0	z interval
$g \le 1/20$	40	0	$[-5/4, 3/5]$
$g > 1/20$, negative tail	22	0	$[-5/4, -3/4]$
$g > 1/20$, remaining thresholds	22	7/200	$[-3/4, 3/5]$

Every row uses the V interval $[2401/24964, 1/2]$. The first row uses (B.30.3) directly; the others use (B.30.4), and the last also uses (B.30.15).

The whole-box derivative bounds are those of Section B.27 with the displayed coefficient and slack. The fixed certificate `endpoint_owner321` covers all three rectangles and certifies that every point has a nonpositive applicable expression in (B.30.16), excluding the endpoint-representation case.

B.31 Two triples: the collided asymmetric event

$$K = R + z\phi(z)/(3C), \qquad \tau = CK^3, \qquad \sigma = \phi(z)/(CK^2).$$

The elementary fixed-two-coordinate bounds give $1/2 < K < 8/5$, $1/20 < \tau < 2$, and $49/120 \le C < 227/500$. All support quantities below are divided by K.

Supports, actual levels, and exact compact charts The maximal included cells are $(1, 1)$ and $(2, 0)$. A genuine selected collision therefore has these two representations. Write

$$T_1 = (-a, c, b), \qquad T_2 = (-t, d, e), \qquad h = b - c = t + d > 0,$$
$$u = a + c > 0, \qquad A_1 = a + b = u + h, \qquad v = e - d > 0, \qquad A_2 = t + e = h + v. \tag{B.31.1}$$

The unused corner $(0, 2)$ is excluded, giving the strict geometric condition

$$A_2 > A_1, \qquad \text{equivalently } v > u. \tag{B.31.2}$$

Denote low/middle/high probabilities by ℓ_i, m_i, r_i, and put $q_i = \ell_i + m_i = 1 - r_i$. The first triple's deletion levels are (q_2, q_2, ℓ_2); the second triple's levels are $(1, q_1, 0)$. The first low support and the second high support are nonselected. They are smooth minima of their global cubic majorants.

In the first triple the negative minimum $-a$ and the selected middle contact c have equal height and nonpositive derivatives. The complete equal-height charts from the contact formulas above are

	$a(k)$	$c(k)$
negative	$\frac{1}{2} + \frac{k}{4}$	$\frac{k-1}{2}$
positive	$\frac{3(1+k)^2}{2(k^3+3k+2)}$	$ka(k)$

$0 \le k \le 1.$ (B.31.3)

The next positive minimum is

$$B(a) = \frac{1 + \sqrt{1 + 4a(1-a)}}{2}.$$

The selected high contact satisfies $0 < b \le B(a)$. Set $b = wB(a)$, $0 \le w \le 1$, retaining $b > c$ as a strict necessary condition. The genuine level drop below the negative minimum ensures $a < 1$; including the boundary $a = 1$ in (B.31.3) is harmless because the level drop is tested separately.

For the second triple its nonselected high contact has level zero. Its scaled cubic majorant is exactly

$$g_e(x) = |x|^3 - \frac{3}{2}x^2 - 3e(e-1)x + B_0, \qquad B_0 = e^2(2e - 3/2). \tag{B.31.4}$$

Necessarily $e \geq 1$. Indeed a smooth positive minimum has $e > 1/2$; if $1/2 < e < 1$, the corresponding negative minimum has smaller potential value. That contradicts global majorization of a nonnegative deletion CDF when $g_e(e) = 0$. More generally the same argument applies to a nonselected positive minimum at any deletion level, since the deletion CDF is nonincreasing.

At the low contact set

$$G = g_e(-t) = t^3 - \frac{3}{2}t^2 + 3e(e-1)t + B_0 = 1/\tau. \tag{B.31.5}$$

Thus $1/2 < G < 20$. It follows that $1 \leq e < 5/2$ and $0 < t < 7/2$. For $e \geq 5/2$, the expression in (B.31.5) increases with $t \geq 0$ and is at least $B_0 \geq 175/8 > 20$. For $t \geq 7/2$, it is at least $t^3 - (3/2)t^2 + 1/2 \geq 25 > 20$.

Use $e = 1 + (3/2)E$, $t = (7/2)T$, and $d = h - t$. Together with (B.31.3), the two parameter cubes are $(k, w, E, T, m_1) \in [0,1]^5$.

Contact values determine the remaining masses The first triple's scaled level difference is

$$U = a^2(3/2 - 2a) - b^3 + \frac{3}{2}b^2 + 3a(1-a)b > 0. \tag{B.31.6}$$

Actual deletion levels give

$$\boxed{m_2 = U/G.} \tag{B.31.7}$$

Centering determines all remaining probabilities:

$$\ell_1 = \frac{b - m_1 h}{A_1}, \quad r_1 = \frac{a - m_1 u}{A_1}, \qquad \ell_2 = \frac{e - m_2 v}{A_2}, \quad r_2 = \frac{t - m_2 h}{A_2}. \tag{B.31.8}$$

Require all six masses strictly positive. The second middle contact supplies the exact equation

$$\boxed{(1 - r_1)G - g_e(d) = 0.} \tag{B.31.9}$$

In the open region $u > 0$, equation (B.31.9) can eliminate the last free mass: $m_1 = (A_1 g_e(d)/G - b)/u$. The five-parameter box retains m_1 and (B.31.9), avoiding a division whose denominator vanishes on the coalescing-support face. This is a numerical choice, not an extra mathematical degree of freedom in an actual solution.

Collision-preserving gradients and actual moments Define the nonnegative contact slopes

$$\begin{aligned} H_c &= 3[a(1-a) + c - c|c|], \quad H_b = 3[a(1-a) + b - b^2], \\ H_t &= 3[t^2 - t + e(e-1)], \quad H_d = 3[e(e-1) + d - d|d|]. \end{aligned} \tag{B.31.10}$$

Moving the two support points in either selected representation by opposite amounts, with the usual recentering of the two marginals, preserves both selected raw sums. These two admissible variations give

$$\boxed{m_1 H_c = m_2 H_d, \qquad r_1 H_b = \ell_2 H_t.} \tag{B.31.11}$$

Consequently

$$\sigma = m_1 H_c + r_1 H_b = \ell_2 H_t + m_2 H_d > 0. \tag{B.31.12}$$

The scaled variances and third absolute moments are

$$\begin{gathered} V_1 = ab - m_1 uh, \quad V_2 = te - m_2 hv, \quad V = V_1 + V_2, \\ R_1 = \ell_1 a^3 + m_1 |c|^3 + r_1 b^3, \quad R_2 = \ell_2 t^3 + m_2 |d|^3 + r_2 e^3, \quad R_0 = R_1 + R_2. \end{gathered} \tag{B.31.13}$$

The selected scaled sum is $o = b - t = c + d$. Thus

$$\begin{gathered} K = V^{-1/2}, \quad z = o/\sqrt{V}, \quad C = V^{3/2}/G, \\ \boxed{3(V - R_0) - \sigma o = 0, \qquad \sigma\sqrt{V} - G\phi(o/\sqrt{V}) = 0.} \end{gathered} \tag{B.31.14}$$

The actual closed CDF is

$$F = q_2 - m_2 r_1.$$

Its exact normal equation and a useful necessary positive expression are

$$\boxed{G[F - \Phi(o/\sqrt{V})] - R_0 = 0,} \tag{B.31.15}$$

$$\boxed{B_0 - \tfrac{3}{2} V_2 - R_1 > 0.} \tag{B.31.16}$$

Equation (B.31.16) follows by averaging the second dual (B.31.4), giving $GF = R_2 - (3/2)V_2 + B_0$, and using $\Phi(z) = F - R_0/G > 0$.

The fixed-two-coordinate and dominant-coordinate results impose

$$\begin{gathered} \frac{25}{64} < V < 4, \qquad V_i > \frac{13}{200} V \quad (i = 1, 2), \\ (49/120)^2 G^2 \le V^3 < (227/500)^2 G^2. \end{gathered} \tag{B.31.17}$$

The unconditional two-triple mixture-window result additionally requires

$$3(W_1 + W_2) > V, \tag{B.31.18}$$

where the actual canonical variance widths are $W_1 = \min(bu, ah)$ and $W_2 = \min(eh, tv)$. Equivalently, $W_1 = bu$ for $c \le 0$ and $W_1 = ah$ for $c \ge 0$, while $W_2 = (eh + tv - |d|A_2)/2$. The adjacent-gap products uh, hv only give upper bounds for these widths; they are not the canonical widths.

The full actual variance Hessian, with cleared denominators For the centered unit-variance directions of the two triples, the actual CDF first and mixed derivatives are

$$L_1 = -\frac{m_2 V}{hA_1}, \qquad L_2 = \frac{(r_1 A_2 - h)V}{hvA_2}, \qquad g = \frac{V^2}{A_1 h^2 v} > 0. \tag{B.31.19}$$

These follow directly from the deletion levels and signed weights in Proposition 6.4. In particular the mixed coefficient is that of the actual 221 event.

Use centered mass parameters whose individual variations have raw total-variance derivative V; these are unit-variance directions after standardization. Write

$$J = F - \Phi(o/\sqrt{V}) - \frac{CR}{V^{3/2}},$$

where C is fixed, the raw quantities F, V, R vary, and the raw threshold o stays fixed. At the attained maximum, both first derivatives vanish and the Hessian is negative semidefinite. If γ_i is the scaled second divided difference of $|x|^3$, put

$$B_i = \frac{15R_0 + G\phi'''(z)}{4} - 3V\gamma_i = -GJ_{ii}.$$

The full condition is

$$B_1 \geq 0, \quad B_2 \geq 0, \quad B_1B_2 - \left[Gg - \frac{B_1 + B_2}{2}\right]^2 \geq 0. \tag{B.31.20}$$

Eliminating both divided differences and the normal derivative gives polynomial necessary inequalities. The variance identity in Section B.14 is

$$4J_{ii} = 12L_i + 3D + (z^3 + 3z)\phi(z), \qquad D = \frac{R_0}{G} = \frac{CR_0}{V^{3/2}}.$$

For completeness, the raw third-moment derivative in this direction is $V\gamma_i$. The first derivative gives $C\gamma_i/\sqrt{V} = L_i + 3D/2 + z\phi(z)/2$, while the second gives $J_{ii} = 3C\gamma_i/\sqrt{V} - (15D + \phi'''(z))/4$. Eliminating γ_i proves the displayed identity.

Using (B.31.14), define

$$\begin{aligned} M &= 3R_0V + \sigma o(o^2 + 3V), \qquad D_* = 4Vh^2A_1vA_2 > 0, \\ P_1 &= 12V^2UhvA_2 - Mh^2A_1vA_2, \\ P_2 &= -12GV^2(r_1A_2 - h)hA_1 - Mh^2A_1vA_2, \qquad Q = 4GV^3A_2. \end{aligned} \tag{B.31.21}$$

Direct substitution gives $P_i = D_*B_i$ and $Q = D_*Gg$. Therefore the equivalent denominator-cleared necessary conditions are

$$\boxed{P_1 \geq 0, \qquad P_2 \geq 0, \qquad 4P_1P_2 - (2Q - P_1 - P_2)^2 \geq 0.} \tag{B.31.22}$$

Only division by $G > 1/2$ and the positive outer spans is needed to evaluate the necessary system. The Hessian tests remain finite when one of the adjacent gaps approaches zero.

Additional global-dual and threshold-gap tests At a maximizing threshold, every excluded sum atom at $z + \delta$ with $\delta > 0$ has mass at most $\Phi(z + \delta) - \Phi(z) \leq \phi(0)\delta$. Since $\phi(0) < 2/5$, a scaled gap $\Delta = \delta/K$ and atom mass w_0 must satisfy

$$25w_0^2V < 4\Delta^2. \tag{B.31.23}$$

Four available excluded atoms give (w_0, Δ) respectively

$$(\ell_1r_2, A_2 - A_1), \quad (m_1r_2, v), \quad (r_1m_2, h), \quad (r_1r_2, A_2).$$

For the first pair, the strict sum-gap (B.31.2) cannot approach zero with a fixed positive atom mass. This controls the transition from 221 to 321 at positive masses, rather than treating it as a missing boundary face.

Two further tests follow by evaluating the global majorants at unused jumps. For the first dual, the point $o - e = b - A_2 < -a$ has deletion level one. The negative cubic factors at $-a$, giving

$$\boxed{(A_2 - A_1)^2(A_2 - b + 2a - 3/2) - r_2G \geq 0.} \tag{B.31.24}$$

For the second dual, $o + a = A_1 - t$ lies strictly between d and e, and its deletion level is ℓ_1. Hence

$$\boxed{g_e(A_1 - t) - \ell_1 G \geq 0.} \tag{B.31.25}$$

These tests use only global majorization and the actual convolution levels.

B.32 Two triples: eliminating the masses

Keep the contact scale of Section B.31.

Use the notation and equal-height charts of Section B.31:

$$T_1 = (-a, c, b), \quad T_2 = (-t, d, e), \quad u = a + c,\ h = b - c = t + d,\ v = e - d,$$
$$A_1 = a + b = u + h, \quad A_2 = t + e = h + v, \quad A_2 > A_1.$$

The supports depend on only $(k, w, E, T) \in [0, 1]^4$ in each chart: a, c are the equal-height functions, $b = wB(a)$, $e = 1 + 3E/2$, $t = 7T/2$, and $d = b - c - t$. The eliminated variable was the first middle mass.

Contact height determines the useful mass products Let $G = g_e(-t)$, $g = g_e(d)$, and U be the first triple's positive level drop, as in the reduction. Put

$$N = A_1 g - bG, \qquad N_0 = bG - hg, \qquad Z = N/G. \tag{B.32.1}$$

The second middle deletion level gives $q_1 = 1 - r_1 = g/G$. Centering the first coordinate then yields

$$\boxed{r_1 = 1 - g/G, \quad m_1 u = Z, \quad m_1 = \frac{N}{uG}, \quad \ell_1 = \frac{N_0}{uG}.} \tag{B.32.2}$$

Thus all three masses are positive exactly when $N > 0$, $N_0 > 0$, $G - g > 0$ (with $G, u > 0$). They sum to one and have mean zero by direct substitution. Use $Z = m_1 u$; the formulas below then require no division by u. The second coordinate is as follows:

$$m_2 = U/G, \qquad \ell_2 = (e - m_2 v)/A_2, \qquad r_2 = (t - m_2 h)/A_2. \tag{B.32.3}$$

Equal height removes the apparent cubic-moment singularity The first low and middle contacts have equal cubic-majorant height. Subtracting those values gives the exact identity

$$|c|^3 - a^3 = (a + c)\left[\frac{3}{2}(c - a) + 3a(1 - a)\right].$$

Define its bracket as W_c. Since $\ell_1 + m_1 = q_1$, the first variance and sum of third absolute moments become

$$\boxed{V_1 = ab - hZ, \qquad R_1 = q_1 a^3 + ZW_c + r_1 b^3.} \tag{B.32.4}$$

These are exactly the centered raw moments. Neither contains a small-gap denominator. The same holds for the selected derivative. The quotient $J_c = H_c/u$ has the explicit closed-chart extensions

$$\boxed{J_c = \begin{cases} u = 3k/4, & \text{negative chart,} \\ \dfrac{3(1-k)(k^2+4k+1)}{2(k^3+3k+2)}, & \text{positive chart.} \end{cases}} \tag{B.32.5}$$

For the negative chart, $H_c = u^2$. For the positive chart, substitute $c = ka$ into $H_c/u = 3[1 - a(1+k^2)/(1+k)]$, and then the exact rational formula for $a(k)$. Hence

$$\boxed{\sigma = ZJ_c + r_1 H_b.} \tag{B.32.6}$$

The two retained representation-gradient equations can be written

$$\boxed{NJ_c - UH_d = 0, \qquad A_2(G-g)H_b - (eG - Uv)H_t = 0.} \tag{B.32.7}$$

Remaining equations, probability signs and curvature Set $V_2 = te - m_2 hv$ and compute R_2 from the second raw support and masses. Then $V = V_1 + V_2$, $R = R_1 + R_2$, $o = b - t$, $z = o/\sqrt{V}$. The actual event is

$$F = \ell_2 + m_2 q_1.$$

The remaining three zero equations are as follows:

$$3(V - R) - \sigma o = 0, \qquad \sigma^2 V - G^2 \phi(z)^2 = 0, \qquad G[F - \Phi(z)] - R = 0. \tag{B.32.8}$$

The positive slope requirement retains the sign of the unsquared density equation. All ratio, raw moment, variance-fraction, width, diameter and majorant tests remain necessary. In particular the useful CDF expression remains $e^2(2e - 3/2) - (3/2)V_2 - R_1 > 0$.

The full variance Hessian in (B.31.21)–(B.31.22) uses only U, V, R, σ, r_1, h, v, A_1, A_2, G and o. All are now functions of the four retained parameters with no division by u, so the identical three cleared Hessian inequalities can be retained. For an independent original-Hessian formulation, its first divided difference also has the regular expression

$$\gamma_1 = \frac{b^3 - a^3 - A_1 W_c}{hA_1}. \tag{B.32.9}$$

This follows by subtracting the two secant slopes and using the equal-height identity. It likewise has no u denominator.

Uniform bounds for the denominators that remain The canonical widths satisfy

$$W_1 = \min(bu, ah), \quad W_2 = \min(eh, tv), \quad W_1 + W_2 > V/3 > 25/192.$$

Since $a < 1$ and $e < 5/2$, their sum is at most $(a+e)h < (7/2)h$. Since $v > u$, $b < B(a) < 5/4$, and $t < 7/2$, it is also smaller than $(b+t)v < (19/4)v$. Therefore

$$\boxed{h > 25/672, \qquad v > 25/912.} \tag{B.32.10}$$

Also $V_1 > 13/512$ and $V_1 \le ab < b$ imply $b > 13/512$. Similarly $t > 13/1280$ follows from $V_2 > 13/512$ and $e < 5/2$. The outer spans obey $A_1 > 1/2$, $A_2 \ge 1$, and $G > 1/2$ was already proved. Thus every remaining rational moment/curvature denominator is uniformly bounded away from zero at a genuine candidate. The first gap u may tend to zero, but no evaluation denominator uses it.

Global-dual and atom-gap tests without individual first masses The first unused dual jump is the inequality already stated for g_a. For the second unused jump, multiply its necessary inequality by $u > 0$ and use (B.32.2):

$$\boxed{u\, g_e(A_1 - t) - N_0 \geq 0.} \tag{B.32.11}$$

For the excluded low/high and middle/high atoms, multiply the squared atom-gap inequalities by $(uG)^2 > 0$:

$$\begin{aligned} 4(A_2 - A_1)^2 u^2 G^2 - 25 N_0^2 r_2^2 V > 0, \\ 4v^2 u^2 G^2 - 25 N^2 r_2^2 V > 0. \end{aligned} \tag{B.32.12}$$

The other two atom-gap inequalities need only r_1, m_2, r_2, already available without u division. Two aggregate tests remain nontrivial on the coalescing face. At the middle/high sum, both first-low/second-high and first-middle/second-high atoms have entered the CDF, with total mass $q_1 r_2$ and maximal raw gap v. At the largest sum, the entire excluded mass $1 - F$ has entered, with raw gap A_2. Therefore

$$\boxed{4v^2 - 25(q_1 r_2)^2 V > 0, \qquad 4A_2^2 - 25(1 - F)^2 V > 0.} \tag{B.32.13}$$

Both follow from threshold maximality and $\phi(0) < 2/5$, exactly as the individual atom-gap tests. Coincidences among later atoms cause no problem.

Each genuine collided maximum is covered by the support chart and maps into one of the two original closed four-cubes. Conversely, on $u, G > 0$, the retained probability signs reconstruct a normalized centered first triple through (B.32.2). The closed face $u = 0$ is included by continuity.

B.33 Support geometry inequalities

The following identities are used in the contact-scale charts of Sections B.16 and B.31.

If the candidate values satisfy an affine equation $\sum_j a_j x_j = b$, with fixed nonzero exact coefficients, then for every i they satisfy

$$x_i = \frac{b - \sum_{j \neq i} a_j x_j}{a_i}. \tag{B.33.1}$$

Evaluate the right side on current candidate intervals and intersect with the interval for x_i. This preserves every candidate. Any fixed sequence of such operations is valid by induction; stopping before a fixed point only weakens the resulting test. No derivative is involved. The coefficients used below are integers or exact half-integers.

For collided 210, use all support equations

$$A = t + b = h + v, \quad h = t + c = d + e, \quad v = b - c, \quad o = c - d = e - t,$$

and the probability, variance, and third-moment sums. For collided 221, use

$$A_1 = a + b = u + h, \quad A_2 = t + e = h + v, \quad u = a + c, \quad h = b - c = t + d, \quad v = e - d,$$

$$\delta = A_2 - A_1 = v - u, \qquad o = b - t = c + d.$$

On its chart with nonpositive first middle contact also use $3a - u = 3/2$, $3c - 2u = -3/2$, and $4u = 3k$, where k is that chart's parameter (the contact scale remains one). These are the exact

chart identities, including the retained $u = 0$ face. On the positive chart $a > 0$, so $k = c/a$ may be intersected before re-evaluating its rational chart formulas.

After a support interval narrows, re-evaluate the original cubic duals and their slopes on those narrower candidate values. For example in 221, $G = g_e(-t)$, $g = g_e(d)$, and $m_2 = U/G$ remain exact candidate equalities. These are not ambient derivatives of the support-chart residuals.

Two existing unused-jump majorizations are also useful in inverse form:

$$r_2 \le \frac{\delta^2(A_2 - b + 2a - 3/2)}{G}, \qquad Y \le \frac{u\, g_e(A_1 - t)}{G}. \tag{B.33.2}$$

Here $Y = u\ell_1$; (B.33.2) never divides by u. The factor in the first bound is nonnegative at every candidate since $a \ge 1/2$ and $A_2 - b + 2a - 3/2 = \delta + 3a - 3/2$. On the negative chart it is exactly $\delta + u = v$, so the first bound simplifies to

$$\boxed{r_2 \le \delta^2 v/G.} \tag{B.33.3}$$

The potential $g_e(A_1 - t)$ in the second bound is nonnegative because the global cubic majorant dominates a nonnegative deletion CDF. Its factorized form is $\delta^2(3e - \delta - 3/2)$ when $A_1 - t \ge 0$, and its absolute-cubic form works without a sign restriction. Divisions by G, outer spans and a use their already proved positive lower bounds.

B.34 The support Hessian with fixed moments

Keep the contact scale of Section B.31.

Keep the scaled supports $(-a, c, b)$, $(-t, d, e)$, gaps $u = a + c$, $h = b - c = t + d$, $v = e - d$, spans $A_1 = u + h$, $A_2 = h + v$, and $A_2 > A_1$. Let G, the six masses, and the contact slopes have their previous meanings. Write $Z = um_1$, $J_c = H_c/u$, and $L(x) = 6|x| - 3$. The selected-support equations are

$$m_1 H_c = m_2 H_d, \qquad r_1 H_b = \ell_2 H_t. \tag{B.34.1}$$

An actual two-parameter family Vary the supports as

$$(-a, c + \epsilon, b + \eta), \qquad (-t - \eta, d - \epsilon, e). \tag{B.34.2}$$

For each marginal separately choose its three masses to keep its total mass, mean zero, and original variance fixed. The two 3×3 Vandermonde systems have unique smooth solutions near every genuine candidate. The base masses are positive, so both signs of small ϵ and η are admissible. Both selected raw sums remain exactly $o = c+d = b-t$ and all other event inequalities remain strict. Thus the event matrix is the actual 221 matrix throughout this family. Its CDF is $F = \ell_2 + m_2(1 - r_1)$.

Both the centered threshold and the total variance are fixed. The normal CDF term is therefore constant, and the relevant objective is $F - R/G$. Let P_1, P_2 be the scaled cubic dual potentials. Their values at the three supports equal G times the actual deletion probabilities. Their first derivatives are respectively $(0, -H_c, -H_b)$ and $(-H_t, -H_d, 0)$, and their second derivatives are $L(x)$. Adding the fixed mean, variance and mass constraints to the objective changes it only by a constant on (B.34.2). The resulting Lagrangian has zero derivatives in all mass variables at the base point.

Consequently, for two tangent support motions s, t, its Hessian multiplied by G is exactly

$$G\sum_{ij} I_{ij}(w^s_{1i}w^t_{2j} + w^t_{1i}w^s_{2j}) - \sum_{ij} p_{ij}L(x_{ij})\dot{x}^s_{ij}\dot{x}^t_{ij} - \sum_{ij} P_i'(x_{ij})(w^s_{ij}\dot{x}^t_{ij} + w^t_{ij}\dot{x}^s_{ij}). \tag{B.34.3}$$

There are no second derivatives of the masses in (B.34.3): their coefficients are the zero first mass derivatives of this Lagrangian. The support paths are affine. Equation (B.34.1) makes both first support derivatives zero.

Exact weights and regular Hessian For a marginal with support points x_0, x_1, x_2, let L_i be its quadratic Lagrange basis, defined by $L_i(x_j) = \mathbf{1}_{\{i=j\}}$. If x_j moves with velocity s, preserving mass, mean, and variance requires the probability derivative $w_i = -p_j s L_i'(x_j)$. The entries needed in the actual event are

	ϵ	η
w_{1r}	$-Z/(A_1 h)$	$-r_1(A_1+h)/(A_1 h)$
w_{1m}	$-m_1(h-u)/(uh)$	$r_1 A_1/(uh)$
w_{2m}	$m_2(v-h)/(hv)$	$\ell_2 A_2/(hv)$
$w_{2\ell}$	$-m_2 v/(hA_2)$	$-\ell_2(h+A_2)/(hA_2)$.

(B.34.4)

The mixed event form is exactly $-w_{1r}w_{2m}$. Substitution in (B.34.3), followed only by (B.34.1), gives the scaled Hessian

$$\begin{pmatrix} -C_m & B \\ B & -C_e \end{pmatrix} \preceq 0, \tag{B.34.5}$$

where

$$C_m = m_2 L(d) + Z\left(\frac{L(c)+2J_c}{u} - \frac{2J_c}{v}\right) - \frac{2GZm_2(v-h)}{A_1 h^2 v}, \tag{B.34.6}$$

$$C_e = r_1 L(b) + \ell_2 L(-t) + 2r_1 H_b\left(\frac{1}{A_1} - \frac{1}{A_2}\right) - \frac{2Gr_1\ell_2 A_2(A_1+h)}{A_1 h^2 v}, \tag{B.34.7}$$

$$B = \frac{G\{Z\ell_2 A_2 + r_1 m_2(A_1+h)(v-h)\}}{A_1 h^2 v} - \frac{ZH_b}{A_1 h} + \frac{r_1 A_1 J_c}{h} + \frac{m_2 v H_t}{hA_2} - \frac{\ell_2 A_2 H_d}{hv}. \tag{B.34.8}$$

Thus every genuine candidate satisfies $C_m \geq 0$, $C_e \geq 0$, and $C_m C_e - B^2 \geq 0$. On the negative first-middle chart, $J_c = u$, $L(c) = -4u$, so (B.34.6) has the regular form

$$\boxed{C_m = m_2 L(d) - 2Z(1+u/v) - \frac{2GZm_2(v-h)}{A_1 h^2 v}.} \tag{B.34.9}$$

Analytic restriction when $c \leq 0$ and $v \geq h$ In this genuine subcase $Z > 0$, $m_2 > 0$, and $0 < u \leq 3/4$. Equation (B.34.9) first gives $L(d) > 0$, hence $|d| > 1/2$. If $d > 1/2$, then $h = t + d > d$ and $e = d + v \geq d + h > 2d \geq 1$. Since $x(x-1)$ increases for $x \geq 1$,

$$H_d = 3\{e(e-1) + d - d^2\} > 3d(3d-1).$$

But (B.34.1) and (B.34.9) also give $L(d) \geq 2Z/m_2 = 2H_d/u$, so

$$6d - 3 > 8d(3d-1).$$

This is impossible because

$$24d^2 - 14d + 3 = 24(d - 7/24)^2 + 23/24 > 0.$$

Therefore

$$\boxed{c \le 0, \quad v \ge h \quad \Longrightarrow \quad d < -1/2.} \tag{B.34.10}$$

There is also a useful upper-support bound. Write $w = -d > 1/2$. Equation (B.34.9) is strict in the weaker form $m_2 L(d) > 2Z$, since $u/v > 0$. Using (B.34.1) yields

$$e(e-1) + w^2 - w < u(w - 1/2).$$

Complete the square in w to obtain

$$\boxed{e(e-1) < \frac{1+u^2}{4}, \qquad e < \frac{1+\sqrt{2+u^2}}{2} \le \frac{1}{2} + \frac{\sqrt{41}}{8}.} \tag{B.34.11}$$

The last two inequalities hold when $v \ge h$; the next calculation proves this condition on the negative chart.

B.35 The negative middle contact

Keep the contact scale of Section B.31.

On the chart with nonpositive first middle contact,

$$c = 2a - 3/2, \quad u = a + c \in (0, 3/4], \quad h = b - c \le 7/4, \quad v = e - d > u, \quad e \ge 1.$$

The actual first dual height and selected-flow identity give

$$U = A_1^2 h - 2b^3 < A_1^2 h, \qquad (A_1 g - bG)u = UH_d, \quad g = g_e(d), \quad G > 0. \tag{B.35.1}$$

In particular $H_d > 0$ and $U/A_1 < ug/H_d$. The middle support curvature, after using $Zu = m_2 H_d$, is

$$\frac{C_m}{m_2} = L(d) - \frac{2H_d}{u} Q, \quad Q = 1 + \frac{u}{v} - \frac{U(h-v)}{A_1 h^2 v}, \quad L(d) = 6|d| - 3. \tag{B.35.2}$$

A maximum requires $C_m \ge 0$.

First suppose $-1/2 \le d \le 1/8$. If $v \ge h$, then $Q \ge 1 + u/v > 0$. If $v < h$, use the first upper bound in (B.35.1):

$$Q > 2 - h/v + u/h.$$

Since $v = e - d \ge 7/8$ and $h \le 7/4$, the right side is at least $u/h > 0$. But $L(d) \le 0$, so (B.35.2) is strictly negative. This interval is impossible.

Now suppose $d \ge 1/8$. If $v \ge h$, (B.35.2) and $v > u$ give

$$C_m/m_2 < L(d) - 2H_d/v = 3 - 6e < 0.$$

If $v < h$, write $r = v/h$ and $\ell = e + d - 1 \ge d \ge 1/8$. The positive-side cubic potential factors exactly as

$$\frac{g}{H_d} = \frac{v}{2} + \frac{v^2}{6\ell}.$$

Hence, using $0 < r < 1$, $h \le 7/4$,

$$\frac{g(h-v)}{H_d h^2} = \frac{r(1-r)}{2} + \frac{h}{6\ell} r^2(1-r) \le \frac{1}{8} + \frac{7}{3}\frac{4}{27} = \frac{305}{648} < 1. \tag{B.35.3}$$

The second upper bound in (B.35.1) now implies

$$Q > 1 + \frac{u}{v} - \frac{u}{v}\frac{305}{648} > 1 > \frac{u}{v}.$$

Again (B.35.2) is strictly below $L(d) - 2H_d/v = 3 - 6e < 0$. This excludes every $d \ge 1/8$. Combining the two ranges proves

$$\boxed{d < -1/2 \quad \text{throughout the chart with nonpositive first middle contact.}} \tag{B.35.4}$$

In fact the opposite-gap sector is impossible. Put $w = -d > 1/2$. The first high support satisfies the exact bound

$$b < B(a) = \frac{1 + \sqrt{1 + 4a(1-a)}}{2} \le \frac{1+\sqrt{2}}{2}.$$

Consequently

$$h = b - c < 1 + 1/\sqrt{2} - 2u/3 < 1 + 1/\sqrt{2}. \tag{B.35.5}$$

If $v \le h$, then $v = e + w$ lies in $(3/2, 1 + 1/\sqrt{2})$. For fixed v, convexity in w on $[1/2, v-1]$ gives

$$\frac{H_d}{3} = e^2 - e + w^2 - w \le \max\{v^2 - 2v + 1/2,\ (v-1)(v-2)\} < 0.$$

Both endpoint expressions are strictly negative on the stated interval. This contradicts $H_d > 0$ from the selected flow. Thus

$$\boxed{v > h \quad \text{throughout the chart with nonpositive first middle contact.}} \tag{B.35.6}$$

The conditional negative-chart support bound therefore also holds unconditionally on this chart: $e(e-1) < (1+u^2)/4$, hence $e < 1/2 + \sqrt{41}/8$.

B.36 The signed divided-difference inequality

In the two-triple 221 contact scale let $r = \sqrt{V}$. The identities $C = r^3/G$, $\phi(z) = \sigma r/G$, $z = o/r$, and $3(V - R) = \sigma o$ give

$$\frac{C}{r} = D + \frac{z\phi(z)}{3}, \qquad L = \frac{1}{4} + \frac{\sigma o(o^2 + 2V)}{12V^2} = \frac{1}{4} + \frac{rz\phi(z)(z^2+2)}{12C} = \frac{1}{4} + \frac{z\phi(z)(z^2+2)}{12(D + z\phi(z)/3)}.$$

The own-variance curvature inequality is

$$\gamma_i \le \frac{r\{15D + (3z - z^3)\phi(z)\}}{12C} = \frac{3}{2} - L.$$

For $z \ge 0$ one has $L \ge 1/4$. For $z = -y \in [-5/4, 0]$, the function $y(y^2+2)\phi(y)$ increases: its derivative is $(2-y^2)(1+y^2)\phi(y) > 0$. Since $\phi(0) < 2/5$ and $\sum_{j=0}^{5}(25/32)^j/j! > 24/11$, $\phi(5/4) < 11/60$. Using $r < 2$ and $C \ge 49/120$ gives

$$y(y^2+2)\phi(y) < \frac{285}{64}\frac{11}{60} = \frac{209}{256}, \qquad L > \frac{1}{4} - \frac{20}{49}\frac{209}{256} = -\frac{261}{3136} > -\frac{1}{12}.$$

The height identities, with q allowed either sign, are

$$p = \frac{3}{2} - \gamma_1 = \frac{U}{A_1 h} > 0, \qquad q = \frac{3}{2} - \gamma_2 = \frac{G(h - A_2 r_1)}{A_2 h v}, \qquad p, q \geq L.$$

Thus $\gamma_2 < 4965/3136$ and $A_2 < 4965/1568$ by (B.2). The exact mass and event identities read

$$m_2 = \frac{pA_1 h}{G}, \quad r_1 = \frac{h}{A_2} - \frac{qhv}{G}, \quad q_1 = \frac{v}{A_2} + \frac{qhv}{G}, \quad F = \frac{e}{A_2} + \frac{m_2 qhv}{G}.$$

Put $x = p - L$, $y = q - L$, and $b_0 = GV/(3A_1 h^2 v)$. The original probability Hessian is negative semidefinite only if

$$x \geq 0, \quad y \geq 0, \quad x + y \geq b_0, \qquad 4xy - (2b_0 - x - y)^2 \geq 0.$$

These are the signed inequalities used in the supplied positive-chart calculation. They impose no positivity assumption on q or $F - e/A_2$.

B.37 A parameter bound for a positive middle contact

Keep the contact scale of Section B.31.

In the positive chart for the collided two-triple 221 event, $k < 8/9$. Indeed, with $u = a(1 + k)$ and $A_1 = u + h$, the centered marginal and selected contact equations give

$$\beta_1 = 1 - \frac{2a^2}{A_1 u}, \quad \sigma = \frac{aH_b}{A_1} + \left(J_c - \frac{H_b}{A_1}\right) Z, \quad J_c - \frac{H_b}{A_1} = 3h\beta_1, \quad Z = a - A_1 r_1.$$

Thus $0 < Z < a$. If $k \geq 8/9$, then $A_1 u > u^2 > 2a^2$, so $\beta_1 > 0$ and $\sigma \leq aJ_c \leq J_c$. But

$$J_c = \frac{3(1 - k)(k^2 + 4k + 1)}{2(k^3 + 3k + 2)}$$

decreases on $[1/2, 1]$: its derivative has the sign of $k^4 - 4k^3 - 6k^2 - 4k + 1 < 0$, as follows from $k^4 \leq k^3$ and evaluation of the resulting decreasing polynomial at $1/2$. Consequently $\sigma \leq J_c(8/9) = 1299/7828$. On the other hand $\sigma = V\phi(z)/C$, and the bounds $V > 2401/5476$, $\phi(z) > 9/50$, $C < 227/500$ imply

$$\sigma > \frac{108045}{621526} = \frac{1299}{7828} + \frac{19206993}{2432652764}.$$

This contradiction proves the parameter bound used in the positive-chart certificate.

The fixed certificates `tt221_negative` and `tt221_positive` prove that neither compact support chart contains a point satisfying the necessary conditions in Sections B.31–B.37. Thus the collided 221 event and its transpose 320 cannot occur at a maximizing pair. Section 8 gives the enclosure and partition rules.

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